



STRUCTURAL PLANS
PROPOSED DETACHED DUAL
OCCUPANCY
42 DARTMOOR CIRCUIT, EMU
HEIGHTS



ISSUED FOR CDC.

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GENERAL NOTES

1. THESE ENGINEERING DRAWINGS ARE TO BE READ IN CONJUNCTION WITH PROJECT SPECIFICATIONS AND OTHER CONSULTANTS DRAWINGS ON THE PROJECT.
2. THESE ENGINEERING DRAWINGS HAVE BEEN PREPARED FROM INFORMATION AVAILABLE AT THE TIME OF ISSUE. AS THIS INFORMATION MAY BE THE SUBJECT OF CHANGE PRIOR TO OR DURING CONSTRUCTION THE CONTRACTOR IS TO ADVISE THE ENGINEER WHERE DISCREPANCIES OCCUR.
3. THESE DRAWINGS SHALL NOT BE USED FOR FINAL SETOUT OF THE PROJECT UNLESS SPECIFICALLY STATED.
4. ALL INSPECTIONS ARE TO BE CARRIED BY 'NANO CONSULTING ENGINEERS' AT THE TIME OF CONSTRUCTION. ALL INSPECTIONS TO BE CARRIED AS PER THE RELEVANT SCOPE PROVIDED BY 'NANO CONSULTING ENGINEERS'. A MINIMUM OF 48 HOURS NOTICE IS TO BE PROVIDED PRIOR TO 'NANO CONSULTING ENGINEERS' COMPLETING AN INSPECTION.
5. PRIOR TO THE COMMENCEMENT OF WORKS THE CONTRACTOR IS TO IDENTIFY ALL EXISTING SERVICES. ANY DAMAGE TO EXISTING SERVICES IS TO BE RECTIFIED AT THE CONTRACTORS EXPENSE. SERVICES SHOWN ON 'NANO CONSULTING ENGINEERS' DRAWINGS ARE INDICATIVE ONLY.
6. DURING CONSTRUCTION THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING THE STABILITY OF THE WORKS AND ENSURE NO PART IS OVERSTRESSED.
7. ALL MATERIALS AND WORKMANSHIP SHALL BE IN ACCORDANCE WITH THE CURRENT AUSTRALIAN STANDARDS AND BCA STATUTORY REQUIREMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
8. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ENSURING THAT SUFFICIENT TOLERANCES ARE PROVIDED AND INTEGRATED THROUGHOUT ALL THE ELEMENTS OF THE WORKS.
9. ALL NON-LOAD BEARING ELEMENTS SHALL BE KEPT CLEAR OF THE STRUCTURE SOFFIT BY AN ALLOWANCE DETERMINED FROM SPAN/250 OR CANTILEVER/125 BUT NOT LESS THAN 20mm, UNLESS NOTED OTHERWISE ON THE DRAWINGS.
10. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS STATED OTHERWISE.
11. WIND LOADS HAVE BEEN DETERMINED IN ACCORDANCE WITH AS1170.2 AND AS4055.
12. SUPERIMPOSED DEAD LOADS AND LIVE LOADS HAVE BEEN DETERMINED IN ACCORDANCE WITH AS1170.0 AND AS1170.1 AND ARE SHOWN ON THE GENERAL ARRANGEMENT DRAWINGS.
13. THE GEOTECHNICAL INVESTIGATION(S) THAT WERE CARRIED OUT AND IMPLEMENTED IN OUR DESIGN ARE INCLUDED IN THE TABLE BELOW :-

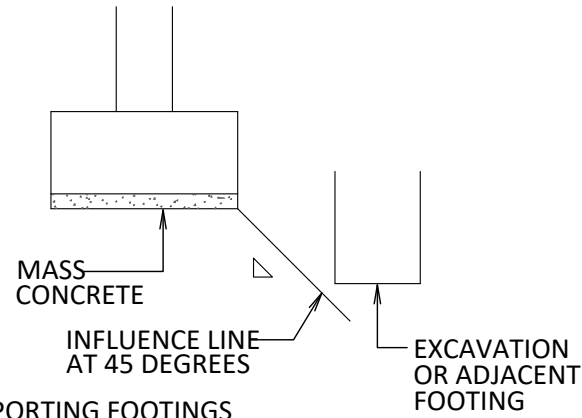
GEOTECHNICAL REPORT		
CONSULTANT	REFERENCE	DATE
GEOSOL CONSULTING ENGINEERS PTY LTD	GS-26024	26/02/2026

14. ALL STRUCTURAL ELEMENTS DESIGNED WITHIN THESE SET OF DRAWINGS COMPLY WITH THE PROVISIONS OF THE RELEVANT BUILDING CODE OF AUSTRALIA AND THE CURRENT AUSTRALIAN STANDARDS NAMELY:
- NCC 2019 - BUILDING CODE OF AUSTRALIA
 - AS1170.0-2002 - GENERAL PRINCIPLES
 - AS1170.1-2002 - PART 1 PERMANENT, IMPOSED AND OTHER ACTIONS
 - AS1170.2-2021 - PART 2 WIND ACTIONS
 - AS2159-2009 - PILING DESIGN AND INSTALLATION
 - AS2870-2011 - RESIDENTIAL SLABS AND FOOTINGS
 - AS3600-2018 - CONCRETE STRUCTURES CODE
 - AS3700-2018 - SAA MASONRY CODE
 - AS1684.2-2021 - RESIDENTIAL TIMBER-FRAMED CONSTRUCTION, NON CYCLONIC AREAS
 - AS4100-2020 - STEEL STRUCTURES CODE

FOUNDATION NOTES

1. REFER TO THE GEOTECHNICAL ENGINEERING REPORT SPECIFIED IN THE GENERAL NOTES FOR SITE SPECIFIC GEOTECHNICAL INFORMATION.
2. REQUIRED MINIMUM SAFE BEARING CAPACITY OF MATERIAL SHALL BE 100 kPa FOR SHALLOW FOOTING/ SLABS AND SCREW PILES DESIGNED BY OTHERS TO BE CONFIRMED BY GEOTECHNICAL ENGINEER. WHERE BEARING CAPACITY IS NOT ACHIEVED OR IS DIFFICULT DUE TO SITE CONSTRAINTS, 'NANO CONSULTING ENGINEERS' IS TO BE CONTACTED TO RE-ASSESS THE FOOTING DESIGN.
3. THE CONTRACTOR IS TO ENGAGE AND PAY A GEOTECHNICAL ENGINEER TO VERIFY THE BEARING CAPACITY OF THE FOUNDATIONS PRIOR TO PLACEMENT OF THE BLINDING LAYER.
4. ALL LOOSE MATERIAL AND WATER TO BE CLEANED OUT OF THE FOUNDATION. FORMWORK TO BE USED WHERE THE SIDES OF THE FOUNDATION ARE NOT STABLE.
5. A 50mm MINIMUM BLINDING LAYER SHOULD BE APPLIED TO THE BASE OF ALL FOUNDATIONS IMMEDIATELY AFTER VERIFICATION OF THE BEARING CAPACITY BY THE GEOTECHNICAL ENGINEER. WHERE THE FOUNDING MATERIAL IS DEEPER THAN REQUIRED FOR THE FOOTING THE EXCAVATION IS TO BE BACKFILLED WITH A WEAK MIX CONCRETE (N10) TO THE UNDERSIDE OF THE FOOTING.

6. WHERE EXCAVATION IS REQUIRED WHICH EXISTS BELOW THE BASE OF THE NEIGHBOURING FOOTING AND/OR ADJACENT TO THE SIDE OF THE FOOTING, THE NEW FOOTING IS TO BE LOCATED 'X' DISTANCE AWAY TO ENSURE IT IS OUTSIDE THE EXISTING FOOTINGS ZONE OF INFLUENCE. IF THE FOOTING CANNOT BE PUSHED AWAY, MASS CONCRETE IS TO BE PLACED UNDER THE FOOTING AS PER THE DIAGRAM ON THE RIGHT. ALL MASS CONCRETE TO BE POURED AT MIN 32MPA. IF THE MASS CONCRETE DEPTH EXCEED 2M, PLEASE NOTIFY 'NANO CONSULTING ENGINEERS', FOR FURTHER ADVICE.



7. ALL WALLS AND COLUMNS SHALL BE CONCENTRIC WITH THE SUPPORTING FOOTINGS UNLESS NOTED OTHERWISE ON THE DRAWINGS.

CONCRETE NOTES

- GENERAL
1. CONCRETE WORK SHALL BE IN ACCORDANCE WITH AS3600 AND WITH THE PROJECT SPECIFICATIONS.
2. ANY JOINTS, I.E. (PERMANENT OR TEMPORARY), SHALL BE PROPERLY FORMED AND USED ONLY WHERE SHOWN ON 'NANO CONSULTING ENGINEERS' DRAWINGS OR SPECIFICALLY APPROVED BY 'NANO CONSULTING ENGINEERS'
3. ALL THICKNESSES SHOWN ARE MINIMUM STRUCTURAL REQUIREMENTS, NO REDUCTION IN THICKNESS DUE TO FALLS OR TOPPING IS PERMITTED. REFER ARCHITECT DRAWINGS FOR ALL SLAB FALLS AND CONFIRMATION OF SLAB STEPS.
5. ANY NEW SLAB CAST AGAINST OLD SLABS TO BE THOROUGHLY MECHANICALLY SCABBLED, FULLY EXPOSING THE AGGREGATE MATRIX.
6. NO PENETRATIONS GREATER THAN 150mm DIAMETER, OR EMBEDMENT OF PIPES GREATER THAN 40mm DIAMETER OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN CONCRETE SLABS. FOR ALL OTHER CONCRETE MEMBERS NO PENETRATIONS, CHASES OR EMBEDMENTS SHALL BE MADE WITHOUT PRIOR APPROVAL BY "NANO CONSULTING ENGINEERS". TYPICALLY ALL APPROVED PIPES IN SLABS TO BE PLACED IN THE MIDDLE THIRD.
7. CONDUITS GREATER THAN 25mm DIAMETER CAST INTO CONCRETE MEMBERS SHALL BE SPACED AT A MAXIMUM DISTANCE POSSIBLE AND UNDER NO CIRCUMSTANCES CLOSER THAN A CLEAR SPACING OF TWICE THE LARGER CONDUIT DIAMETER FROM PARALLEL REINFORCEMENT OR ANY OTHER CONDUIT.
8. CONCRETE TO BE KEPT FREE OF SUPPORTING BRICKWORK BY 'HERCUSLIP COMPOSITE (HSC)' SLIDING JOINT OR EQUIVALENT U.N.O.
9. NO ADMIXTURES TO BE USED IN CONCRETE UNLESS APPROVED IN WRITING.
10. AGGREGATE SIZE, SLUMP AND CONCRETE STRENGTH TO BE FOLLOWED AS PER DRAWINGS.
11. ALL CONCRETE SHALL BE VIBRATED USING MECHANICAL VIBRATORS.
12. ALL CONCRETE SHALL BE CURED IN ACCORDANCE WITH THE SPECIFICATION.
13. ALL CONCRETE SHALL BE SAMPLED AND TESTED IN ACCORDANCE WITH AS1012 AND THE PROJECT SPECIFICATION.
14. ALL FORMWORK SHALL COMPLY WITH AS3610
15. EACH FLOOR SHALL BE FULLY PROPPED TO THE FLOOR BELOW IN ACCORDANCE WITH AS3610 (FORMWORK CODE).

16. THE FLOOR BELOW SHALL BE BACKPROPPED THROUGH A MINIMUM OF TWO STOREYS BELOW. THIS RESULTS IN A MINIMUM OF THREE STOREYS PROPPED AT ALL TIMES. (REFER TO TYPICAL BACK PROPPING SEQUENCE DIAGRAM)

REINFORCEMENT

1. REINFORCEMENT IS TO BE MANUFACTURED IN ACCORDANCE WITH AS1302 AND SHALL BE FIXED AS SHOWN ON DRAWINGS.
2. ALL REINFORCEMENT TO BE AS FOLLOWS:
- | SYMBOL | TYPE |
|--------|--|
| R | STRUCTURAL GRADE PLAIN BARS TO AS/NZS 4671 (250 MPa) |
| SL, RL | FABRIC MESH TO AS/NZS 4671 (500 MPa) |
| N | HOT ROLLED DEFORMED BARS TO AS/NZS 4671 (500 MPa) |
- NOTE: THE NUMBER FOLLOWING R OR N INDICATES THE BAR DIAMETER IN MILLIMETRES.
3. THE BAR SIZE IS INDICATED BY A NUMBER AFTER THE SYMBOL, WHICH INDICATES THE BAR DIAMETER IN MILLIMETERS.
4. REINFORCEMENT SPACING NOMINATED ON DRAWINGS IS TO ASSIST SCHEDULER AND STEELFIXER TO ASSESS TOTAL NUMBER OF BARS REQUIRED. WHERE BARS PLACED IN ACCORDANCE WITH SPACING NOMINATED FOUL WITH OTHER STRUCTURAL REQUIREMENTS, PREFERENCE IS TO BE GIVEN TO RELOCATING BARS BY LOCALLY ADJUSTING SPACING TO ENABLE ASSEMBLY OF REINFORCEMENT TO BE COMPLETED. ENGINEER IS TO BE CONTACTED IN THE EVENT THAT REINFORCEMENT IS NEEDED TO BE CUT ON SITE PRIOR TO CONTINUING.
5. LAP LENGTHS TO REINFORCEMENT BARS TO BE AS NOTED ON THE RELEVANT DRAWINGS.
6. WELDING OF REINFORCEMENT BARS IS NOT PERMITTED UNLESS APPROVED.
7. COVER SHALL BE AS NOTED ON THE RELEVANT DRAWINGS.
8. CONCRETE COVERS NOTED ARE MEASURED FROM THE FORMWORK OR GROUND FACE TO THE OUTERMOST REINFORCEMENT COMPONENT. ie. IN COLUMNS AND BEAMS TO THE OUTSIDE OF TIES OR LIGATURES.
9. COVER TO BE MAINTAINED DURING POURING BY THE USE OF PLASTIC CHAIRS OR PLASTIC TIPPED METAL CHAIRS.
10. WHERE NO REINFORCEMENT IS SHOWN ON THE DRAWING AT RIGHT ANGLES TO THE MAIN REINFORCEMENT DISTRIBUTION REINFORCEMENT IS TO BE PROVIDED.
11. MINIMUM LAP FOR FABRICS SHALL BE TWO TRANSVERSE WIRES PLUS 25mm. MINIMUM LAP LENGTHS FOR DEFORMED BARS INCLUDING DISTRIBUTION REINFORCEMENT SHALL BE AS FOLLOWS U.N.O.:

BAR TYPE AND SIZE	VERTICAL BARS	HORIZONTAL BARS		90° COG LENGTH	135° or 180° COG LENGTH
		MORE THAN 300mm OF CONCRETE BELOW BAR	OTHER LOCATIONS		
12	450	550	450	170	70
16	700	800	700	200	70
20	1000	1250	1000	250	80
24	1200	1500	1200	300	95
28	1400	1750	1400	350	115
32	1550	1900	1550	400	130
36	1700	2150	1700	450	145

* COG LENGTH MEASURED FROM END OF BEND.

12. TYPICAL REINFORCEMENT NOTATION: 2N24-200.2
- 23 DENOTES NUMBER OF BARS REQUIRED
- N DENOTES GRADE OF REINFORCEMENT
- 24 DENOTES BAR DIAMETER IN MILLIMETRES
- 200 DENOTES BAR SPACING IN MILLIMETRES
- .2 DENOTES SECOND LAYER OF REINFORCEMENT LAID

MASONRY NOTES

1. ALL MASONRY SHALL COMPLY WITH AS3700 AND THE PROJECT SPECIFICATION.
2. CONCRETE MASONRY UNITS TO HAVE A MINIMUM CHARACTERISTIC UNCONFINED STRENGTH OF 20 MPa IN ACCORDANCE WITH AS2733.
3. MASONRY TO BE BEDDED IN FRESHLY PREPARED MORTAR.
- (a) CONCRETE BLOCKS :-
MORTAR MIX TO BE UNIFORMLY MIXED IN A RATIO OF ONE PART CEMENT, ONE PART LIME AND SIX PARTS SAND CONFORMING TO AS2701. 'BRICKIES LOAM' SHALL NOT BE USED.
- (b) CLAY BRICKS :-
MORTAR MIX TO BE UNIFORMLY MIXED IN THE RATIO OF ONE PART CEMENT, THREE PARTS SAND AND ONE FOURTH PART LIME CONFORMING TO AS2701. 'BRICKIES LOAM' SHALL NOT BE USED.
4. GROUT SHALL HAVE A COMPRESSIVE STRENGTH (F_c) OF 20 MPa AT 28 DAYS, A SLUMP OF 125mm IN A 150mm SLUMP CONE, A MAXIMUM AGGREGATE SIZE OF 10mm AND BE IN ACCORDANCE WITH AS3700.
5. BEDDING OF MASONRY SHALL BE FULL FACE WITH CROSS JOINTS COMPLETELY FILLED. JOINT THICKNESS SHALL NOT EXCEED 12mm.
6. PROVIDE WALL TIES AT 600mm MAXIMUM CENTRES VERTICALLY AND HORIZONTALLY. REFER TO MASONRY DETAILS FOR WALL TIE SETOUT AT OPENINGS.
7. THE CAVITY SHALL NOT EXCEED 100mm AND SHALL NOT BE SMALLER THAN 40mm UNLESS NOTED OTHERWISE. KEEP CAVITY CLEAN AND CLEAR OF OBSTRUCTIONS.
8. RAKING OF JOINTS IS NOT PERMITTED WITHOUT PRIOR APPROVAL FROM 'CAPITAL ENGINEERING CONSULTANTS.
9. ALL WALLS TO BE KEPT STABLE AT ALL STAGES OF CONSTRUCTION AND NOT BE OVER STRESSED AT ANY TIME.
10. NO HOLES OR CHASES SHALL BE CUT IN LOAD BEARING BRICKWORK OR BLOCKWORK WITHOUT THE PRIOR APPROVAL OF THE ENGINEER.
11. VERTICAL CONTROL JOINTS IN BLOCK RETAINING WALLS AND BLOCK WALLS TO BE SPACED AS SHOWN OR AT 6000mm MAX. APART. VERTICAL CONTROL JOINTS IN BRICKWORK TO BE SPACED AT 5000mm MAX. APART.
12. BRICK TIES TO COMPLY WITH AS3700 AND BE OF STAINLESS STEEL DUE TO REQUIRED EXPOSURE.

STRUCTURAL STEELWORK NOTES

1. ALL MATERIALS, WORKMANSHIP, FABRICATION AND ERECTION SHALL COMPLY WITH THE REQUIREMENTS OF AS4100, AS1538, AS1554 AND THE SPECIFICATION.
2. UNLESS SHOWN OTHERWISE, ALL STEEL SHALL BE IN ACCORDANCE WITH AS1204 GRADE 300. ALL STEEL HOLLOW SECTIONS SHALL BE GRADE 350 U.N.O. AND SHALL BE IN ACCORDANCE WITH AS1163. ALL PRESSED METAL PURLINS AND GIRTS SHALL BE GRADE 450 STEEL IN ACCORDANCE WITH AS1538
3. UNLESS SHOWN OTHERWISE ON THE DRAWINGS, ALL CONNECTIONS SHALL BE IN ACCORDANCE WITH THE FOLLOWING MINIMUM REQUIREMENTS:
- (i) ALL WELDS SHALL BE 6MM CONTINUOUS FILLET WELDS ALL ROUND.
- (ii) ALL BOLTS SHALL BE M20 - 8.8/S, WITH A MINIMUM OF 2 BOLTS PER CONNECTION. PURLIN BOLTS TO BE M12 - 4.6/S WITH A MINIMUM OF 2 BOLTS PER PURLIN END.
- (iii) ALL GUSSET AND CLEAT PLATES SHALL BE 10mm THICK. (U.N.O.)
- (iv) ALL CAP PLATES SHALL BE 10 mm THICK. (U.N.O.)
- (v) ALL BASE PLATES SHALL BE 10 mm THICK. (U.N.O.)
4. BOLT DESIGNATION:
- (i) 4.6/S REFERS TO COMMERCIAL BOLTS OF STRENGTH GRADE 4.6 TO AS1111 TIGHTENED TO A SNUG TIGHT CONDITION.
- (ii) 8.8/S REFERS TO HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 TIGHTENED TO A SNUG TIGHT CONDITION.
- (iii) 8.8/TB REFERS TO HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 FULLY TENSIONED TO AS4100 AS A BEARING JOINT.
- (iv) 8.8/TF REFERS TO HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 FULLY TENSIONED TO AS4100 AS A FRICTION JOINT.
5. HIGH STRENGTH BOLTED JOINTS SHALL BE IN ACCORDANCE WITH AS1511. THE SPECIFIED BOLT TENSION SHALL BE OBTAINED BY USE OF THE "PART TURN" METHOD OF TIGHTENING.
6. ALL WELDS SHALL BE SP (SPECIAL PURPOSE) IN ACCORDANCE WITH AS1554. ALL ELECTRODES SHALL BE CLASS E48. ALL BUTT WELDS SHALL BE FULL STRENGTH COMPLETE PENETRATION WELDS.
7. SUBSTITUTIONS FOR STEEL SECTIONS SHOWN ON DRAWINGS SHALL NOT BE MADE WITHOUT THE APPROVAL OF THE ENGINEER.
8. ALL BUILT INTO MASONRY OR EXPOSED TO THE WEATHER STEEL WORK SHALL BE HOT-DIPPED GALVANISED.
9. ALL STEEL WORK SHALL BE FIRE RATED IN ACCORDANCE WITH BCA REQUIREMENTS.
10. UNLESS SPECIFIED OTHERWISE, STEELWORK SHALL BE PREPARED TO CLASS 2.5 FINISH IN ACCORDANCE WITH AS1627.4 AND GIVEN TWO (2) COATS OF ZINC PHOSPHATE TO A TOTAL DRY FILM THICKNESS OF 70 MICRONS.

TIMBER NOTES

1. AS 1684 IS RELEVANT TO DOMESTIC CONSTRUCTION IN SHELTERED LOCATIONS.
2. SOFTWOOD MINIMUM GRADE F7 U.N.O. HARDWOOD MINIMUM GRADE F11 U.N.O.
3. EXTERNAL TIMBER TO BE EITHER HARDWOOD DURABILITY CLASS I OR II OR IMPREGNATED GRADE F7. PRESSURE TREATED TO AS1684 AND RE-DRILLED PRIOR TO USE. SUPPLEMENTARY TREATMENT SHALL BE APPLIED TO ALL CUT SURFACES. PROVIDE DOCUMENTATION.
4. ALL BOLTS IN TIMBER CONSTRUCTION TO BE MIN. M16 U.N.O. BOLT HOLES TO BE DRILLED EXACT SIZE. WASHERS UNDER HEADS AND NUTS TO BE AT LEAST 2.5 TIMES BOLT DIAMETER.
5. FINISHED TIMBER SIZES.
- | | |
|---------------------|-----------|
| SEASONED SOFTWOOD | +5,-0mm |
| UNSEASONED SOFTWOOD | F7+3,-3mm |
| | F7+2,-4mm |
| SEASONED HARDWOOD | +2,-0mm |
| UNSEASONED HARDWOOD | -3,-3mm |
- (SEE ALSO CLAUSE 1.6.2 IN AS 2082)
6. ALL TIMBER JOINTS AND NOTCHES TO BE 100mm MINIMUM FROM LOOSE KNOTS. SEVERE SLOPING GRAIN, GUM VEINS OR OTHER MINOR DEFECTS.
- FOR JOISTS SPANNING GREATER THAN 3m AND LESS THAN 4.2m PROVIDE ONE ROW OF BLOCKING MID-SPAN.
- FOR JOISTS SPANNING GREATER THAN 4.2m AND UP TO 6.0m PROVIDE TWO ROWS OF BLOCKING AT 1/3 POINTS.
- FOR DEEP JOISTED FLOORS WHERE A CONTINUOUS TRIMMING JOIST IS NOT PROVIDED AT END OF JOISTS. BLOCKING IS REQUIRED AT 1800 MAXIMUM CENTERS. (REFER TO AS 1684)
7. BLOCKING IS NOT REQUIRED FOR JOISTS SPANNING LESS THAN 3m.

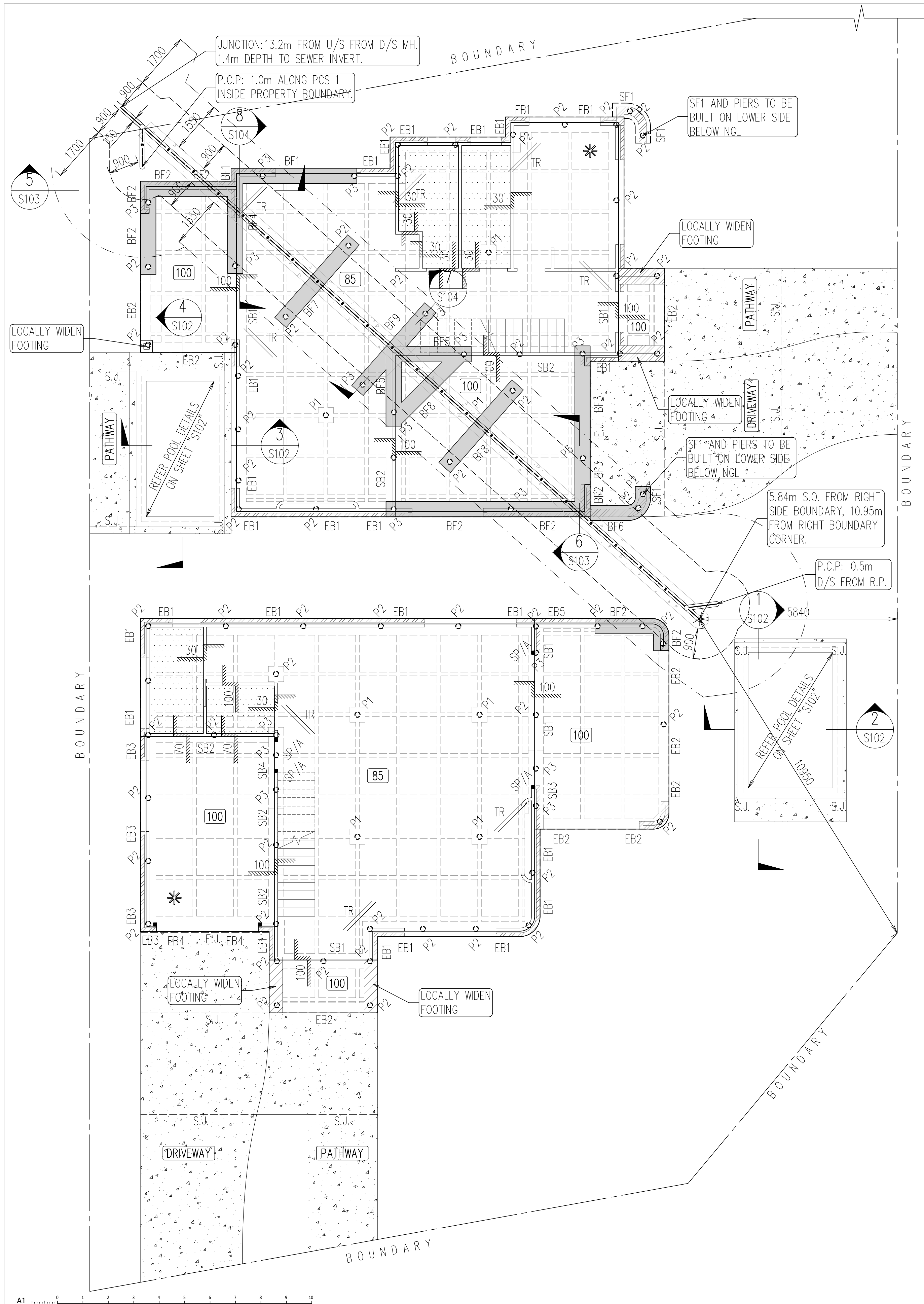
BRICK LINTEL SCHEDULE			
OPENING SIZE (mm)	INTERNAL SKIN	EXTERNAL SKIN	END BEARING
UP TO 900mm	100 x 8mm FLAT BAR	100 x 6mm FLAT BAR	100 mm
1200	100 x 10mm FLAT BAR	100 x 8mm FLAT BAR	100 mm
1500	100 x 100 x 8mm ANGLE	100 x 100 x 6mm ANGLE	150 mm
2100	150 x 100 x 8mm ANGLE	150 x 100 x 6mm ANGLE	150 mm
2400	150 x 100 x 8mm ANGLE	150 x 100 x 8mm ANGLE	150 mm
2700	150 x 100 x 10mm ANGLE	150 x 100 x 10mm ANGLE	150 mm
3000	150 x 100 x 12mm ANGLE	150 x 100 x 12mm ANGLE	150 mm

NOTES

- ALL STEEL LINTELS TO BE HOT DIPPED GALVANIZED.
- ABOVE ALL LINTELS, PROVIDE M.E.T MASONARY REINFORCEMENT FOR 2 COURSES AND CONTINUE 1000mm PAST OPENING.

A1 0 1 2 3 4 5 6 7 8 9 10

				E: ahmad@nceeng.com M: 0452489806 COPYRIGHT THIS DRAWING AND THE INFORMATION SHOWN HEREIN IS THE PROPERTY OF NANO CONSULTING ENGINEERS AND MAY NOT BE USED FOR ANY PURPOSES THAN FOR WHICH SUPPLIED.				DRAWN : A.N. DESIGNED : A.N. REVIEWED : A.N. DATE : 31.03.2026 CHECKED : A.N. APPROVED DATE : 31.03.2026				PROJECT : PROPOSED DETACHED DUAL OCCUPANCY ADDRESS : 42 DARTMOOR CIRCUIT, EMU HEIGHTS APPLICATION : C.D.C.				SHEET SUBJECT : GENERAL NOTES N/A				ALL DIMENSIONS IN mm UNLESS OTHERWISE STATED DO NOT SCALE DRAWING. USE FIGURED DIMENSIONS ONLY			
				APPROVAL TYPE: 1. PRELIMINARY 2. DEVELOPMENT APPLICATION 3. COMPLETED DEVELOPMENT 4. CONSTRUCTION CERTIFICATE 5. FOR TENDER 6. OTHERS				ARCHITECT: SLEIMAN PROPERTY GROUP				CLIENT: DEBRA FRANCIS				SCALE @ A1 N.T.S				JOB No ST2673			
A ISSUED FOR C.D.C.				A.N. A.N. 31.03.2026												AUTHORISED				DWG No S001			
REV DESCRIPTION				BY CHECK DATE												A.N.				REV A			



FOR STRUCTURAL NOTES REFER TO DRAWING S001

NOTES:

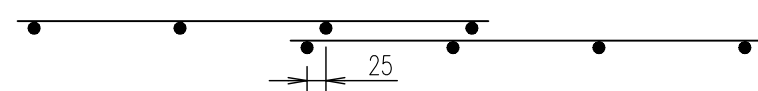
1. DRAWINGS TO BE READ IN CONJUNCTION WITH ARCHITECTURALS.
2. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS.
3. DO NOT SCALE ANY DIMENSIONS FROM STRUCTURAL DRAWINGS FOR SETTING OUT PURPOSES.

GROUND FLOOR – WAFFLE POD PLAN –
85/100 mm SLAB U.N.O

SITE CLASSIFICATION : S

NOTES :

1. SHALLOW FOOTING/SLAB SAFE BEARING CAPACITY TO BE A MIN OF 100 kPa.
2. SCREW PILES TO BE DESIGNED TO SPECIFIED LOADS BY SCREW PILING CONTRACTOR.
3. PROVIDE SL82 MESH TOP THROUGHOUT FOR ALL 85/100 mm THICK SLABS.
4. ALL INTERNAL PODS 225mm, GARAGE & ALFRESCO PODS 150mm.
5. ALL BRICKS WALLS TO BE ARTICULATED AND SPACED IN ACCORDANCE TO AS3700 AND ARTICULATED WALLING TECHNICAL GUIDELINE (TN61)
6. ARTICULATION JOINT SPACING TO BE 6000mm
7. ALL CORNERS TO HAVE L-BAR WITH 600mm LEGS TOP AND BOTTOM
8. PODS LESS THAN 400mm LONG CAN BE TAPED
9. PIERS TO BE POURED SEPARATE TO WAFFLE POD SLAB
10. ALL EXPOSED SLABS TO BE WATERPROOFED WITH APPROVED SYSTEMS
11. DRAWING TO BE READ IN CONJUNCTION WITH ARCHITECTURAL DRAWINGS. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC'S. ALL SLAB ON GROUND TO BE LAID ON CONTINUOUS VAPOUR BARRIER
12. SP/A DENOTES STEEL POST OVER 89 X 89 X 5mm SHS, TP/A DENOTES TIMBER POST OVER 90 X 90 POST
13. REFER TO STORM WATER DRAINAGE DRAWINGS FOR LOCATION OF SUMP PITS.
14. REFER TO ARCHITECTURAL DRAWINGS FOR LEVELS, STEPS AND FOLDS.
15. MESH SHALL BE LAPPED SUCH THAT THE TWO OUTERMOST WIRES OF ONE SHEET OVERLAP THE TWO OUTERMOST WIRES OF THE OTHER SHEET BY 25mm MINIMUM.

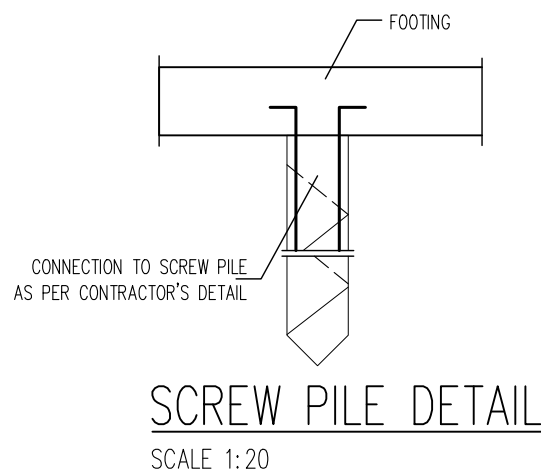


REINFORCEMENT COVER SCHEDULE				
MEMBER	COVER (mm)			EXPOSURE CLASSIFICATION
	TOP	BOTTOM	SIDES	
FOOTINGS	50mm	50mm	50mm	A2
INTERNAL SLAB	30mm	30mm	30mm	A1
EXTERNAL SLAB	40mm	40mm	40mm	B1

CONCRETE QUALITY				
ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f'c
WAFFLE SLAB	80mm	20mm	G.P.	32 MPa
FOOTINGS	80mm	20mm	G.P.	32 MPa

LEGEND	
	DENOTES BRICK WALLS
	DENOTES CONCRETE BLOCK WALLS
	DENOTES TIMBER STUD WALLS
	DENOTES STEP ON SLAB
	DENOTES SCREW PILES – REFER TO SCHEDULE
	DENOTES SAWN CONTROL JOINT
	DENOTES EXPANSION JOINT
	30mm SETDOWN IN WET AREAS
	STEEL POST ABOVE (89 x 89 x 5mm SHS)
	2N12 1200 LONG TRIMMER BARS
	DENOTES POD STARTING POINT

SCREW PILES SCHEDULE		
MARK	SIZE (Ø x THICKNESS)	SWL
S101	CONTRACTOR TO CONFIRM SPECS	50kN
S102	CONTRACTOR TO CONFIRM SPECS	90kN
S103	CONTRACTOR TO CONFIRM SPECS	130kN



WAFFLE RAFT SLAB SPECIFICATIONS	
SLAB THICKNESS	85/100 mm
REINFORCEMENT	SL82 TOP
REINFORCEMENT COVER	30

DRIVEWAY– 150mm SLAB WITH SL82 MESH TOP & BOTTOM U.N.O
NOTE : 1. MAX 6m SAWN CONTROL JOINT
2. MAX 15m KEYED CONSTRUCTION JOINT
3. MAX 30m EXPANSION JOINT

PATHWAY– 100mm SLAB WITH SL82 MESH TOP & BOTTOM U.N.O
NOTE : 1. MAX 4m SAWN CONTROL JOINT
2. MAX 10m KEYED CONSTRUCTION JOINT
3. MAX 20m EXPANSION JOINT

NOTE:

BUILDER TO PROVIDE TREATMENT FOR THE PROTECTION OF THE BUILDING FROM SUBTERRANEAN TERMITES IN ACCORDANCE WITH AS3660.

NOTE:

ALL FILLS AND SOILS UNDER FOOTING TO BE COMPACTED IN LAYERS AS PER AS2870 AND AS378.

NOTE:

BUILDER TO CONFIRM EXTERNAL FINISH LEVELS, DROP EDGE BEAMS MAY BE REQUIRED, REFER TO DETAILS DEB1 AND DEB2 ON S101 SHEET.

NOTE :

1. IF GROUND CONDITIONS CHANGE DURING EXCAVATION, PLEASE NOTIFY ENGINEER, AND SEEK FURTHER INSTRUCTIONS.
2. ENGINEER TO BE ADVISED IF ACID SULPHATE SOIL ARE PRESENT. IN THE EVENT THAT ACID SULPHATE SOILS ARE PRESENT, THE BUILDERS IS STRONGLY ADVISED TO ENSURE PROTECTIVE MEASURES ARE TAKEN TO MINIMISE THE EFFECT OF AN ACID SULPHATE ATTACK ON THE FOOTING STRUCTURAL ELEMENTS.

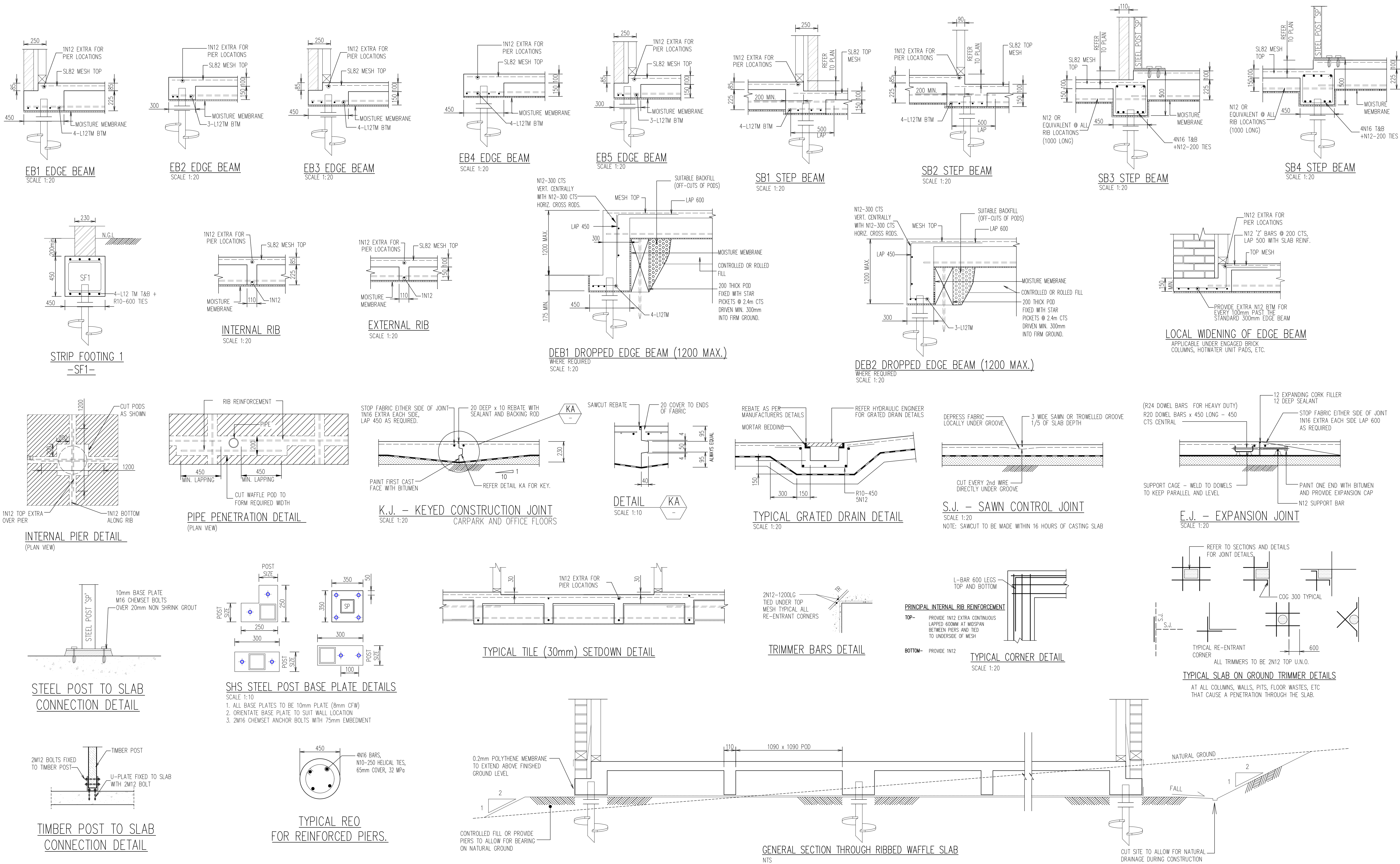
ALL RETAINING WALLS MAY NOT BE SHOWN ON PLANS & MAY BE REQUIRED. TO BE CONFIRMED DURING CONSTRUCTION

BEARING CAPACITIES TO BE CHECKED AND CONFIRMED DURING CONSTRUCTION. FOOTING DIMENSIONS ARE SUBJECTED TO CHANGES. ENGINEER TO CONFIRM.

IF ROCK IS ENCOUNTERED WITH THE DEEPER PIERS, THEN ALL PIERS NEED TO BE BEARING ON THE SAME ROCK MATERIAL. PLEASE ADVISE NANO DURING CONSTRUCTION.

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A	ISSUED FOR C.D.C.	A.N.	A.N.	31.03.2026	APPROVAL TYPE: 1. PRELIMINARY 2. DEVELOPMENT APPLICATION 3. COMPLETION DEVELOPMENT 4. CONSTRUCTION CERTIFICATE 5. FOR TENDER 6. OTHERS	ARCHITECT: SLEIMAN PROPERTY GROUP		CLIENT: DEBRA FRANCIS			



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A				ISSUED FOR C.D.C.				A.N. A.N. 31.03.2026				APPROVAL TYPE: 1. PRELIMINARY 2. DEVELOPMENT APPLICATION 3. COMPLETION DEVELOPMENT 4. CONSTRUCTION CERTIFICATE 5. FOR TENDER 6. OTHERS				PH: 0452489806 EMAIL: ahmad@nceeng.com				ARCHITECT: SLEIMAN PROPERTY GROUP				CLIENT: DEBRA FRANCIS			

POOL SPECIFICATION

1. PLUMBING IS TO BE IN ACCORDANCE WITH WRITTEN RECOMMENDATIONS OF FILTER MANUFACTURER
2. WALKWAYS ARE NOT DESIGNED TO SUPPORT MASONARY WALLS UNLESS OTHERWISE NOTED
3. DESIGN LIVE LOAD FOR WALKWAYS AND CONCOURSES 3 kPa

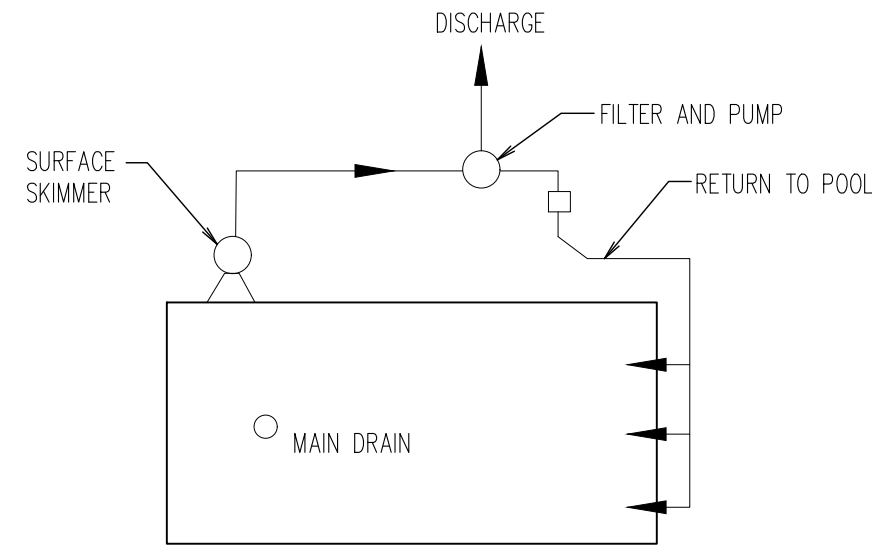
CONSTRUCTION NOTES

1. ALL DIMENSIONS LOCATING POOL ARE BE TAKEN FROM ARCHITECTURAL DRAWINGS
2. WRITTEN DIMENSIONS TO BE TAKEN IN REFERENCE TO SCALE
3. ENGINEER TO BE ADVISED IF EXCAVATION IS IN FILL OR IF EXCESSIVE GROUND WATER IS ENCOUNTER
4. SUPPORTING FOUNDATION MATERIAL TO BE SILTY SAND OF UNIFORM MOISTURE CONTENT WITH SAFE BEARING CAPACITY OF 200 kPa
5. WHERE IT IS CONSIDERED THAT GROUND WATER CAN BUILD UP TO A LEVEL 500mm ABOVE THE FLOOR OF THE EXCAVATION ADEQUATE DRAINAGE SHALL BE PROVIDED UNDER THE POOL FLOOR
6. CONCRETE TO HAVE A MINIMUM DESIGN STRENGTH OF $f_c = 32 \text{ MPa}$ AT 28 DAYS CONCRETE TO BE PNEUMATICALLY APPLIED
7. UPON COMPLETION OF CONCRETING THE HYDROSTATIC VALVE IS TO BE CHECKED TO ENSURE EFFECTIVE AND SUFFICIENT OPERATION
8. ALL REINFORCEMENT TO BE SUPPORTED BY BAR CHAIRS
9. REINFORCEMENT IS TO BE STRUCTURAL GRADE DEFORMED BAR GRADE TO 230 TO AUSTRALIAN STANDARD AS1302
10. WATER FACE REINFORCEMENT TO HAVE 65mm CONCRETE COVER REAR FACE REINFORCEMENT HAVE 50mm COVER FROM REAR REAR FACE IF FORMED AND 65mm COVER ISF SPRAYED AGAINST GROUND
11. ALL BARS SHALL BE SPLICED 40 BAR DIAMETERS MIN.
12. SPLICES IN BOND BEAM BARS SHALL BE STAGGERED
13. ALTERNATIVE REINFORCEMENT TO BE TEMCORE BARS IN ACCORDANCE WITH AS 1302 410 Y

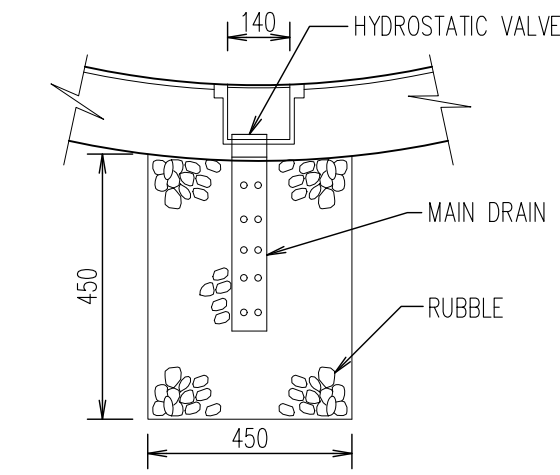
CONCRETE QUALITY				
ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f_c
POOL BASE / WALLS	80mm	20mm	SR	32 MPa

IF WATER IS ENCOUNTERED
DURING EXCAVATION
ENGINEER NEEDS TO BE
IMMEDIATELY NOTIFIED

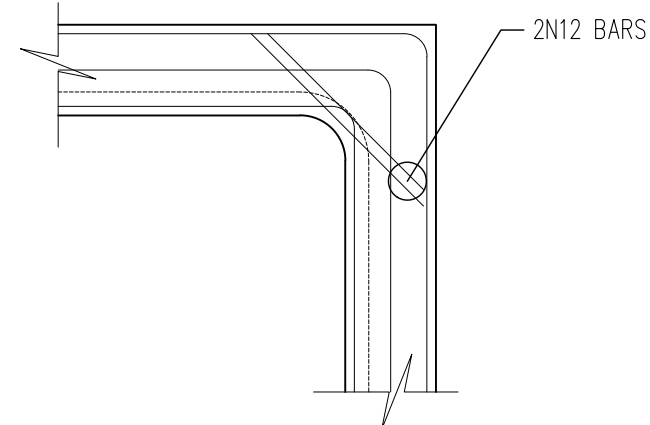
ALL MEASUREMENTS HAVE
BEEN ASSUMED & ARE ONLY
APPROXIMATE INDICATIONS –
TO BE CONFIRMED BY POOL
BUILDER OR ARCHITECT



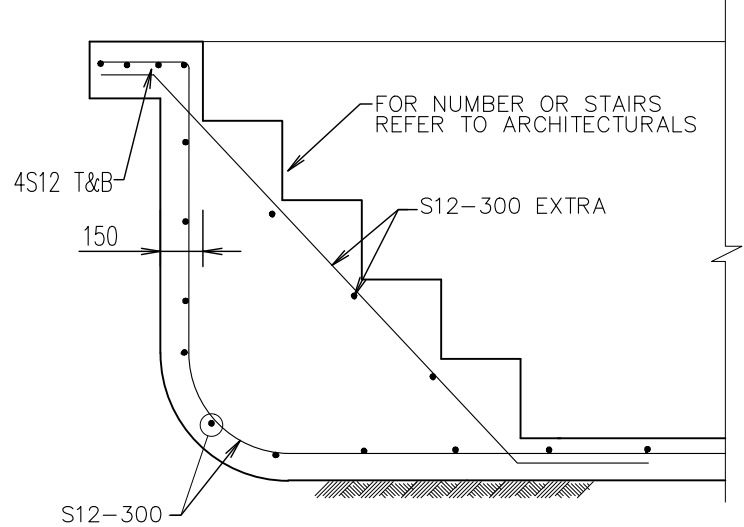
PLUMBING LAYOUT
DIAGRAMATIC ONLY



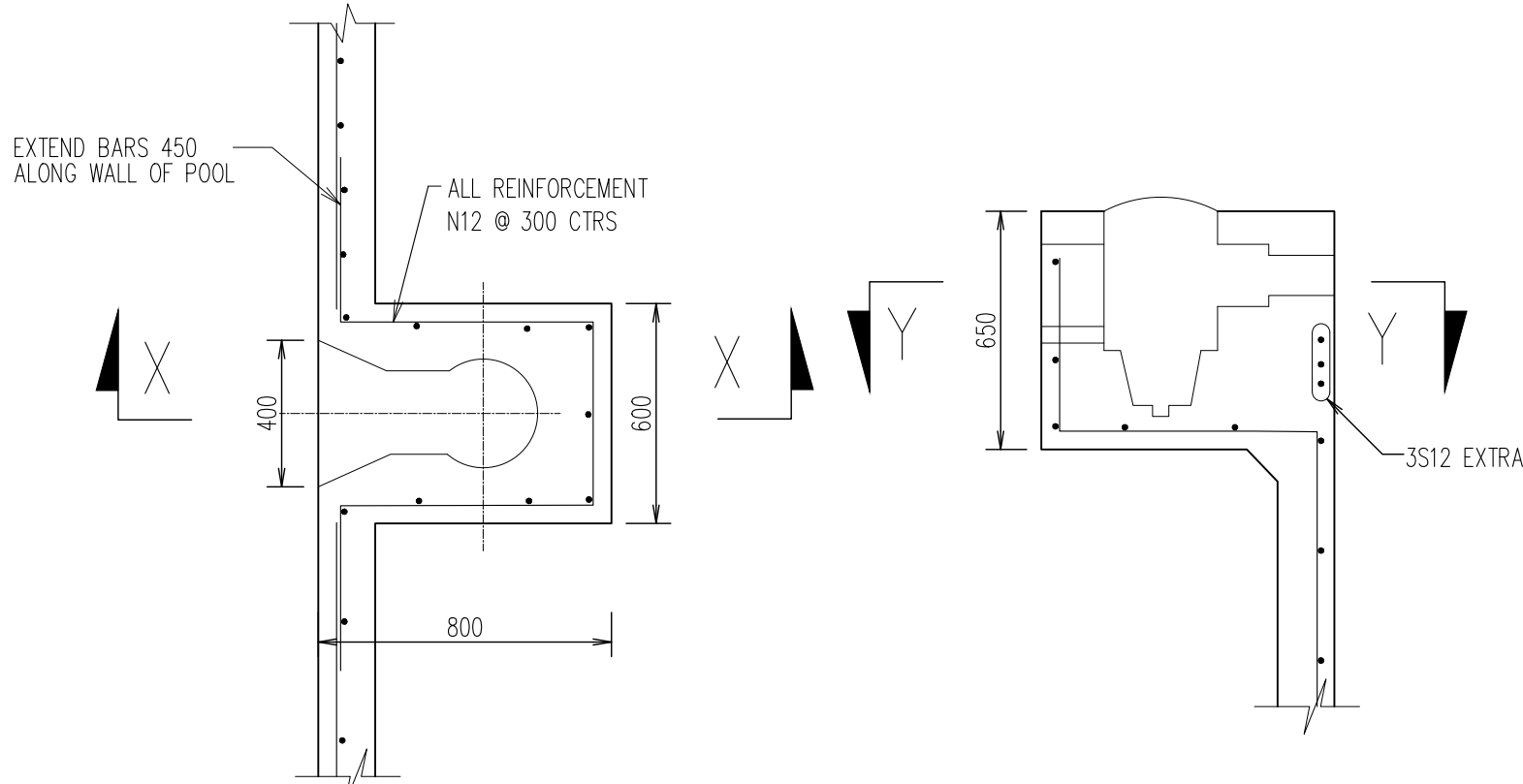
MAIN DRAIN DETAIL
SCALE 1:20



BOND BEAM CORNER DETAIL
SCALE 1:50



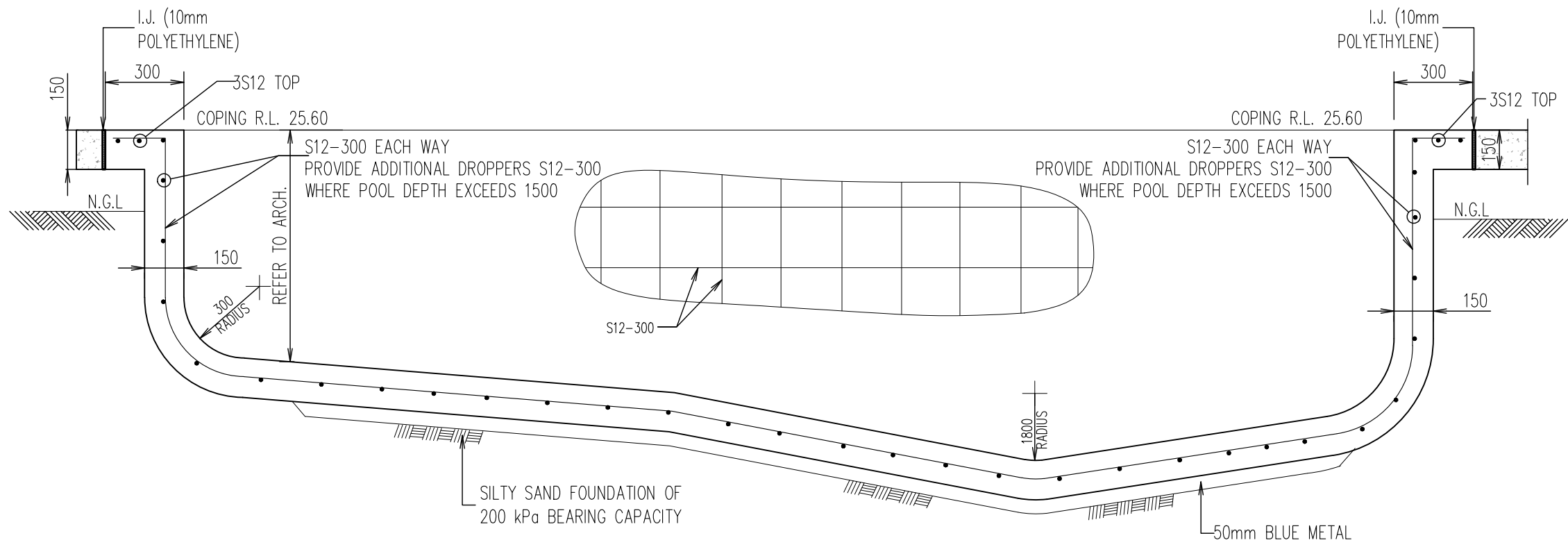
TYPICAL STAIR SECTION



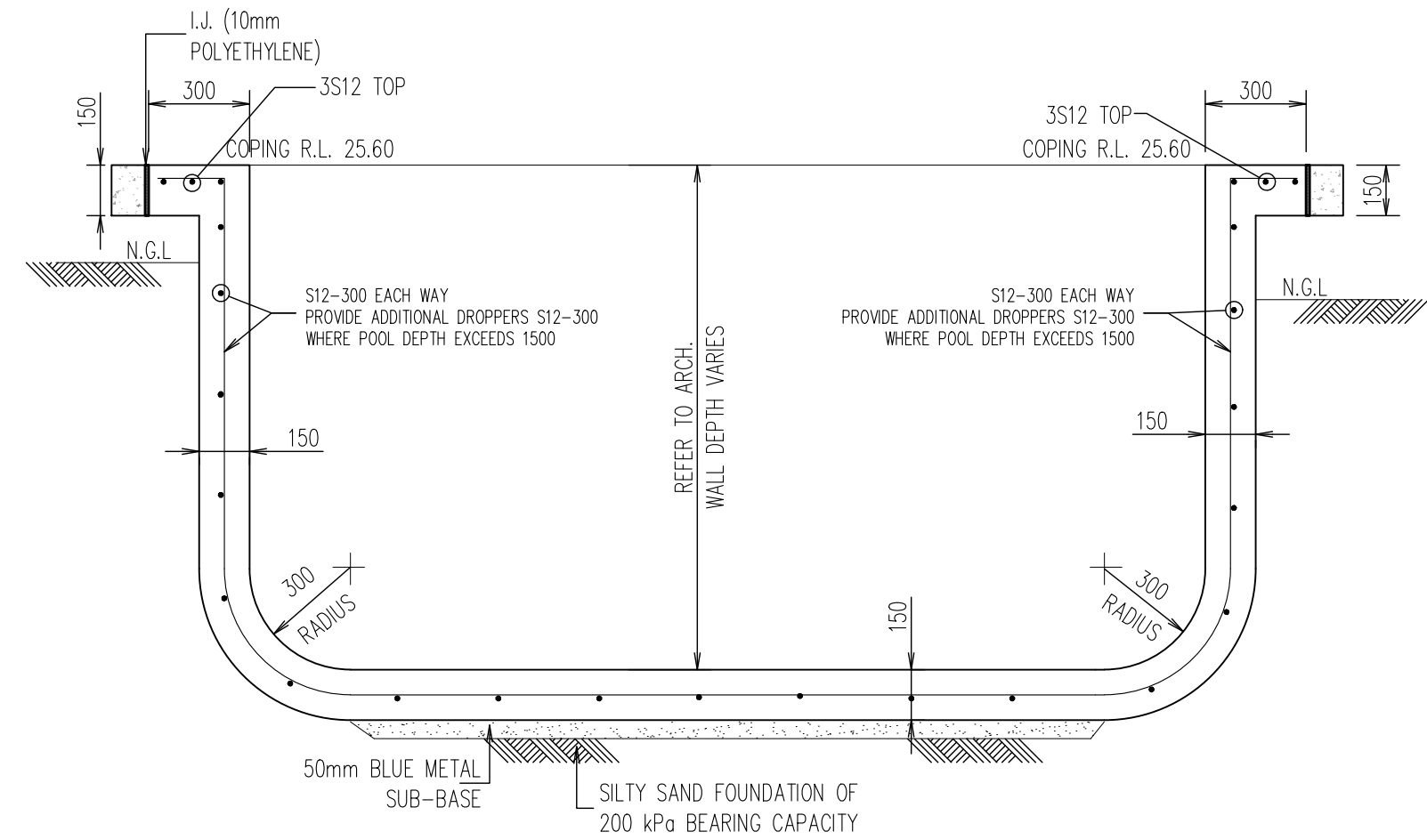
PLAN - SECTION Y-Y

ELEVATION - SECTION X-X

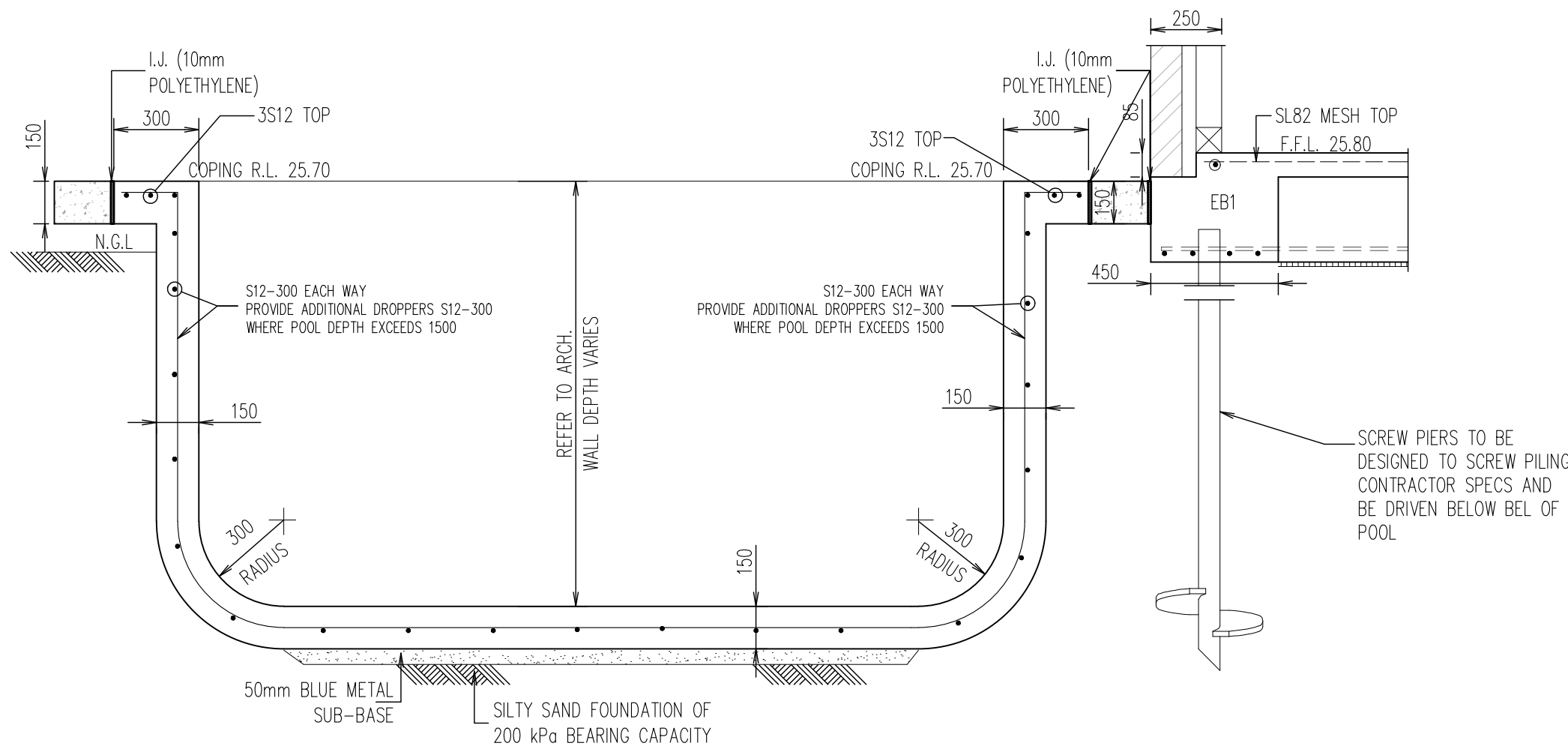
SKIMMER BOX DETAIL
SCALE 1:20



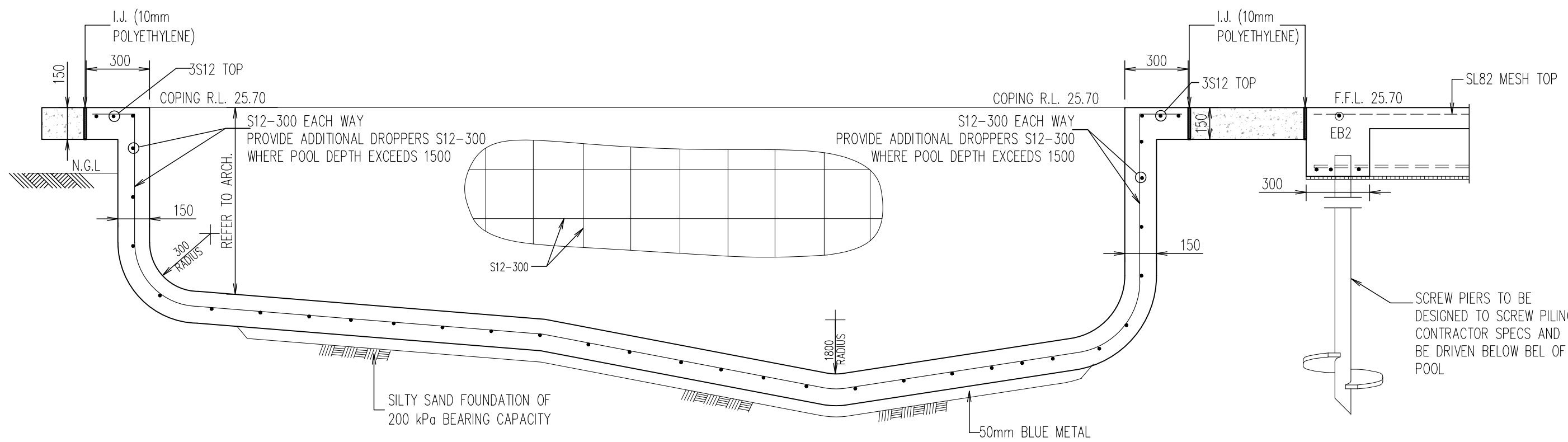
SECTION 1
SCALE 1:20



SECTION 2
SCALE 1:20



SECTION 3
SCALE 1:20

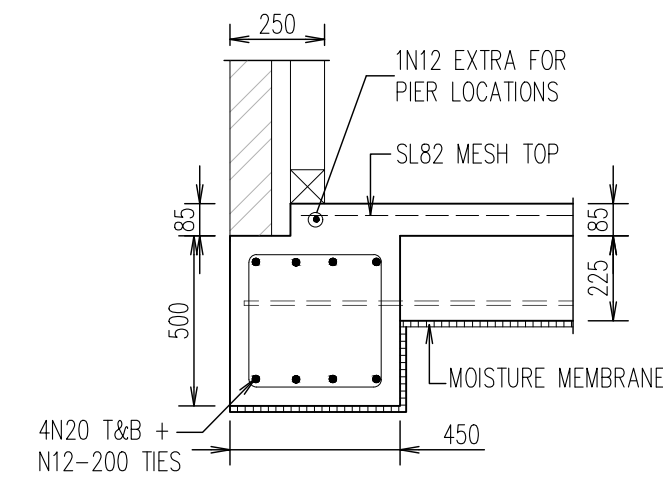


SECTION 4
SCALE 1:20

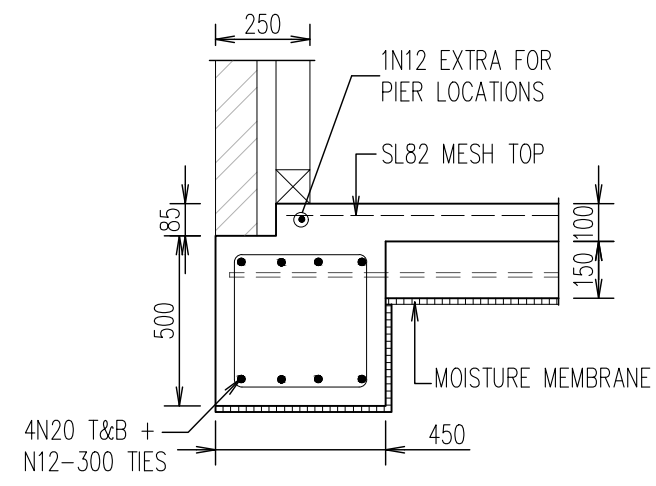
A1 0 1 2 3 4 5 6 7 8 9 10

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REV	DESCRIPTION	BY	CHECK	DATE	APPROVAL TYPE: 1. PRELIMINARY 2. DEVELOPMENT APPLICATION 3. COMPLETED DEVELOPMENT 4. CONSTRUCTION CERTIFICATE 5. FOR TENDER 6. OTHERS	PH: 0452489806 EMAIL: ahmad@nceeng.com		ARCHITECT: SLEIMAN PROPERTY GROUP		CLIENT: DEBRA FRANCIS					

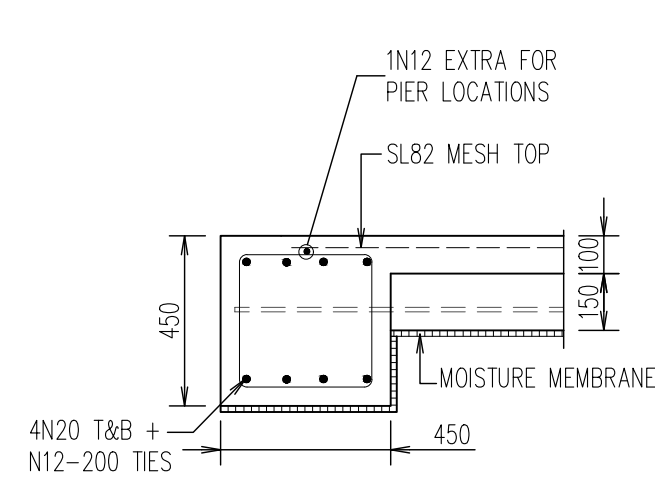
ISSUED FOR CDC.



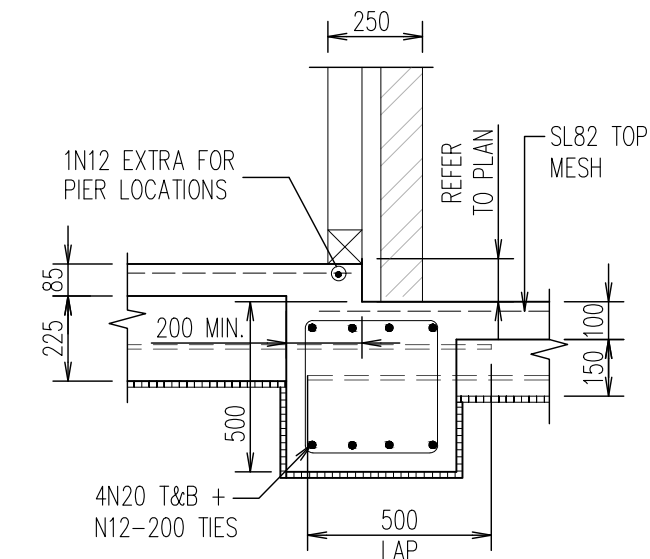
BF1 BRIDGE FOOTING
SCALE 1:20



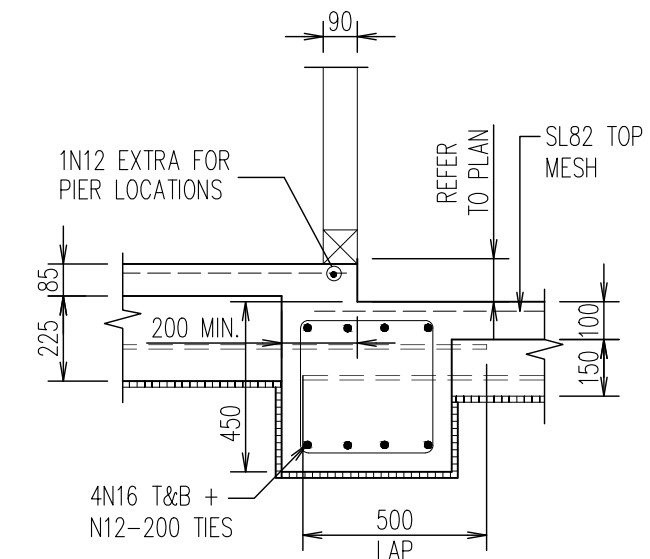
BF2 BRIDGE FOOTING
SCALE 1:20



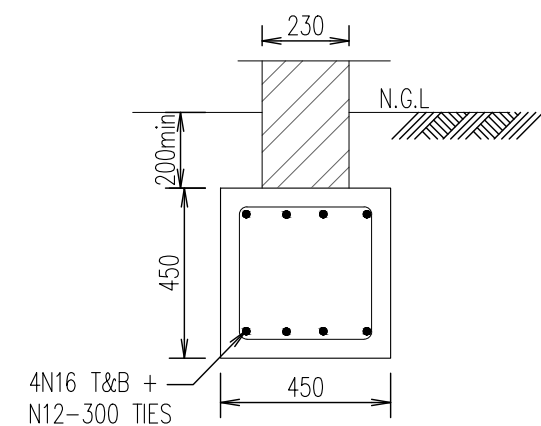
BF3 BRIDGE FOOTING
SCALE 1:20



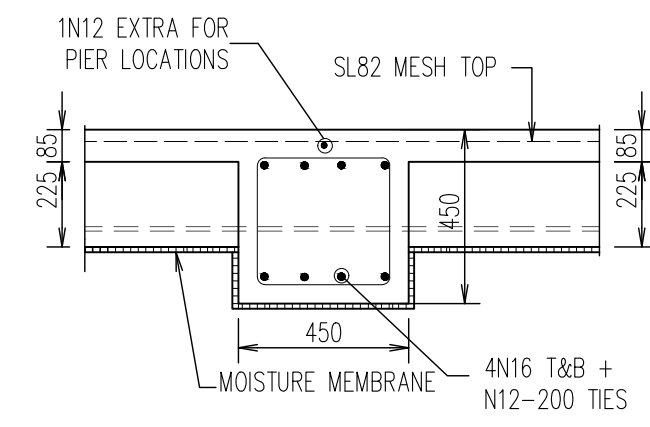
BF4 BRIDGE FOOTING
SCALE 1:20



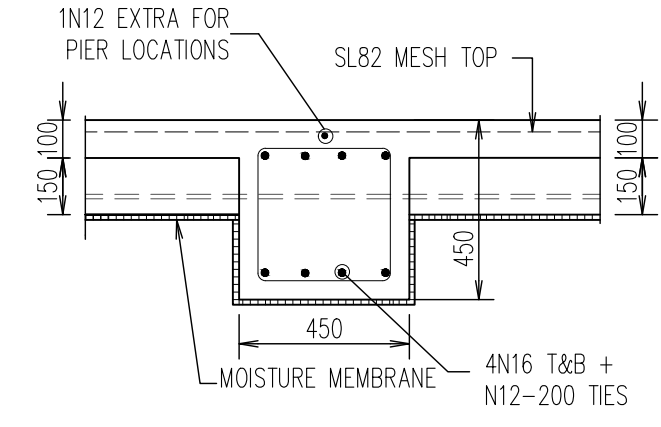
BF5 BRIDGE FOOTING
SCALE 1:20



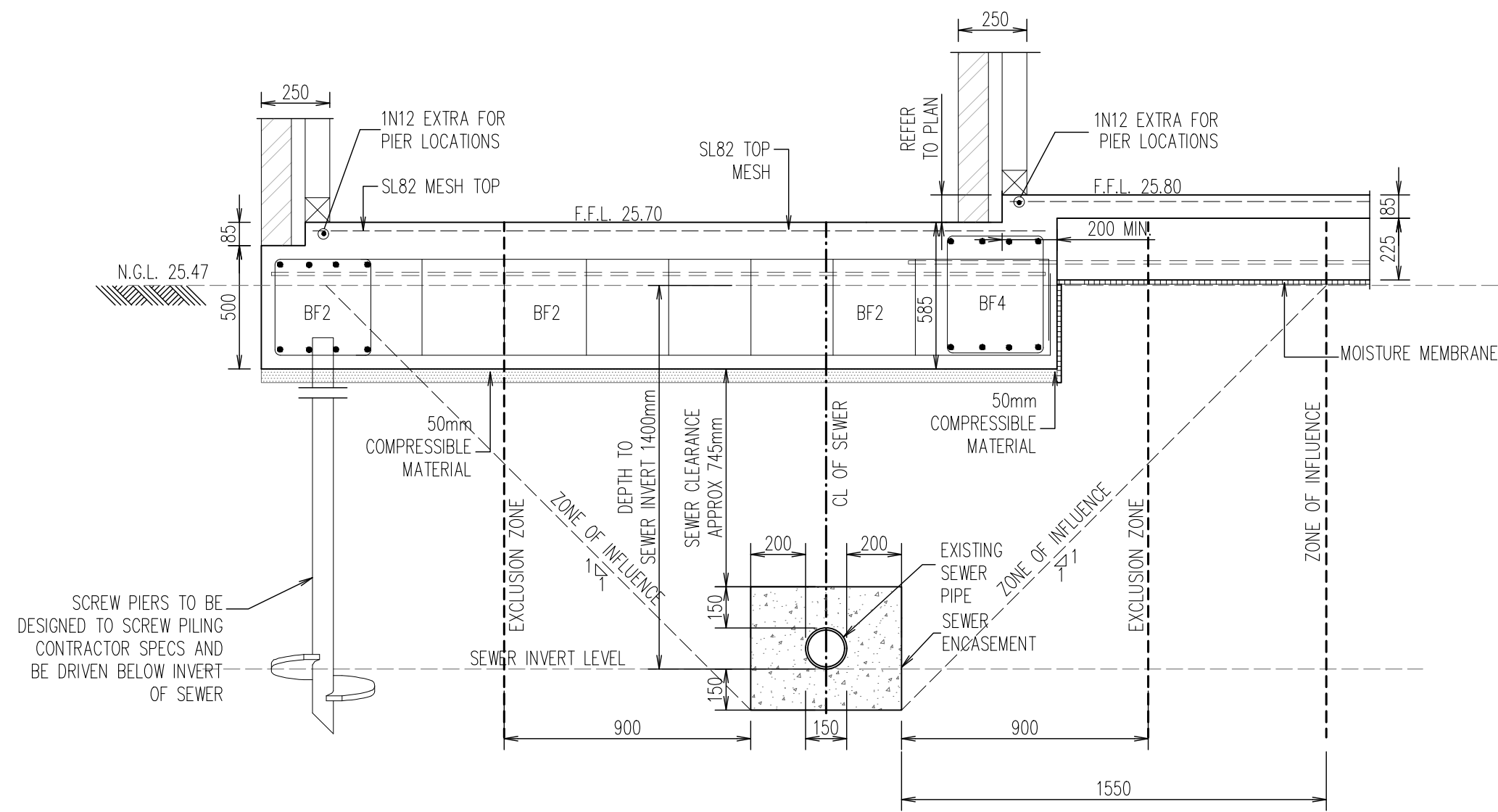
BF6 BRIDGE FOOTING
SCALE 1:20



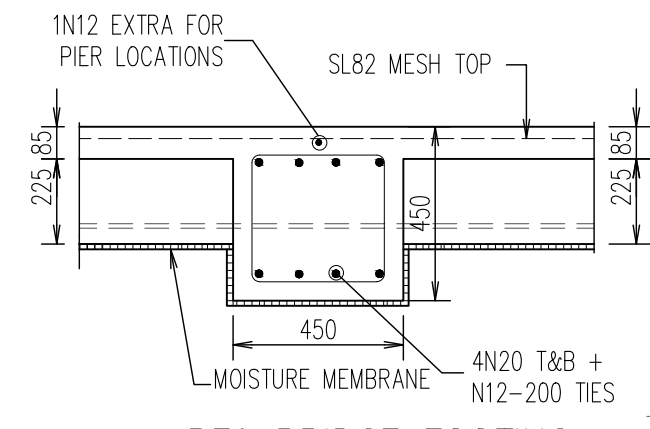
BF7 BRIDGE FOOTING
SCALE 1:20



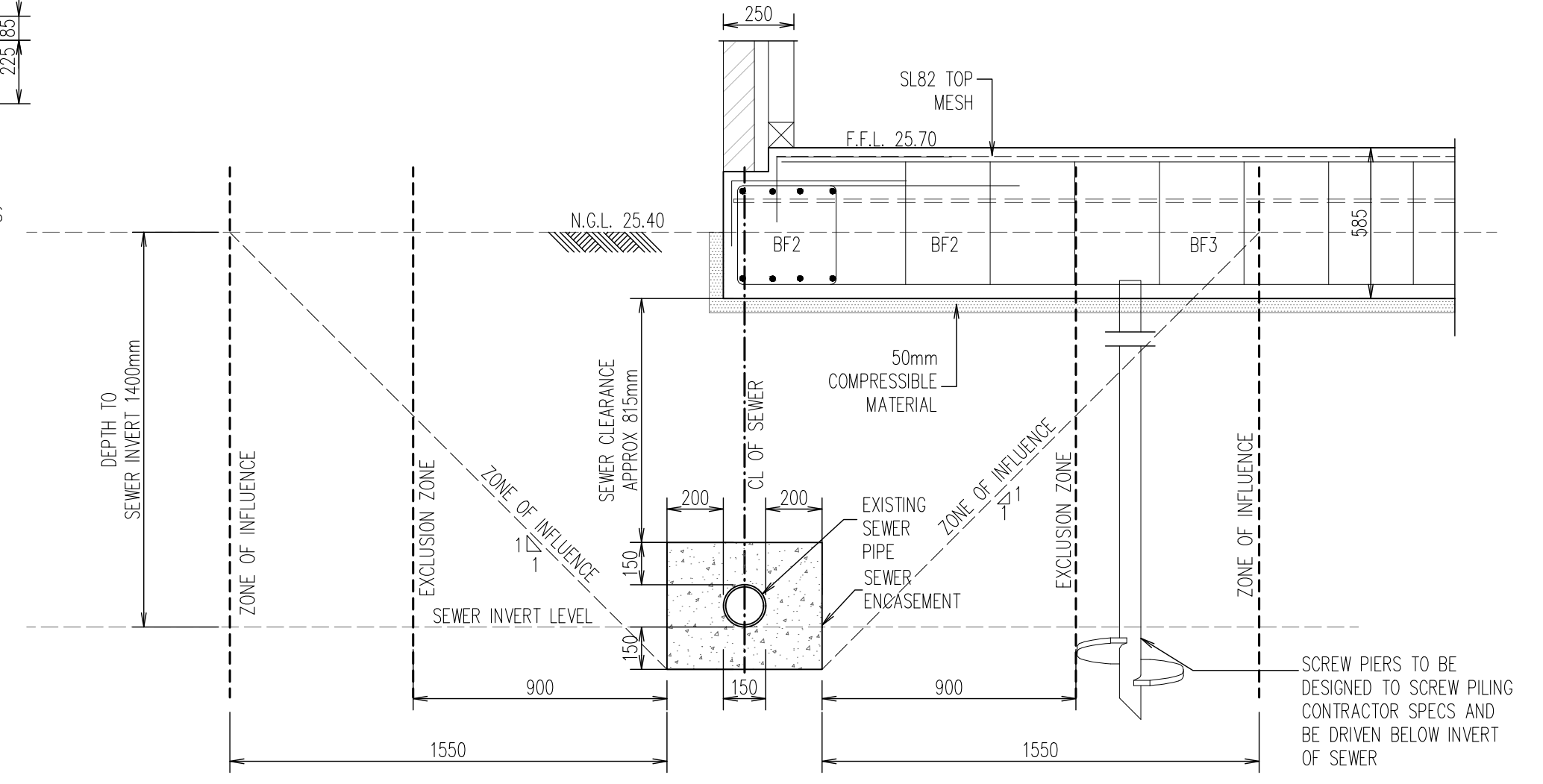
BF8 BRIDGE FOOTING
SCALE 1:20



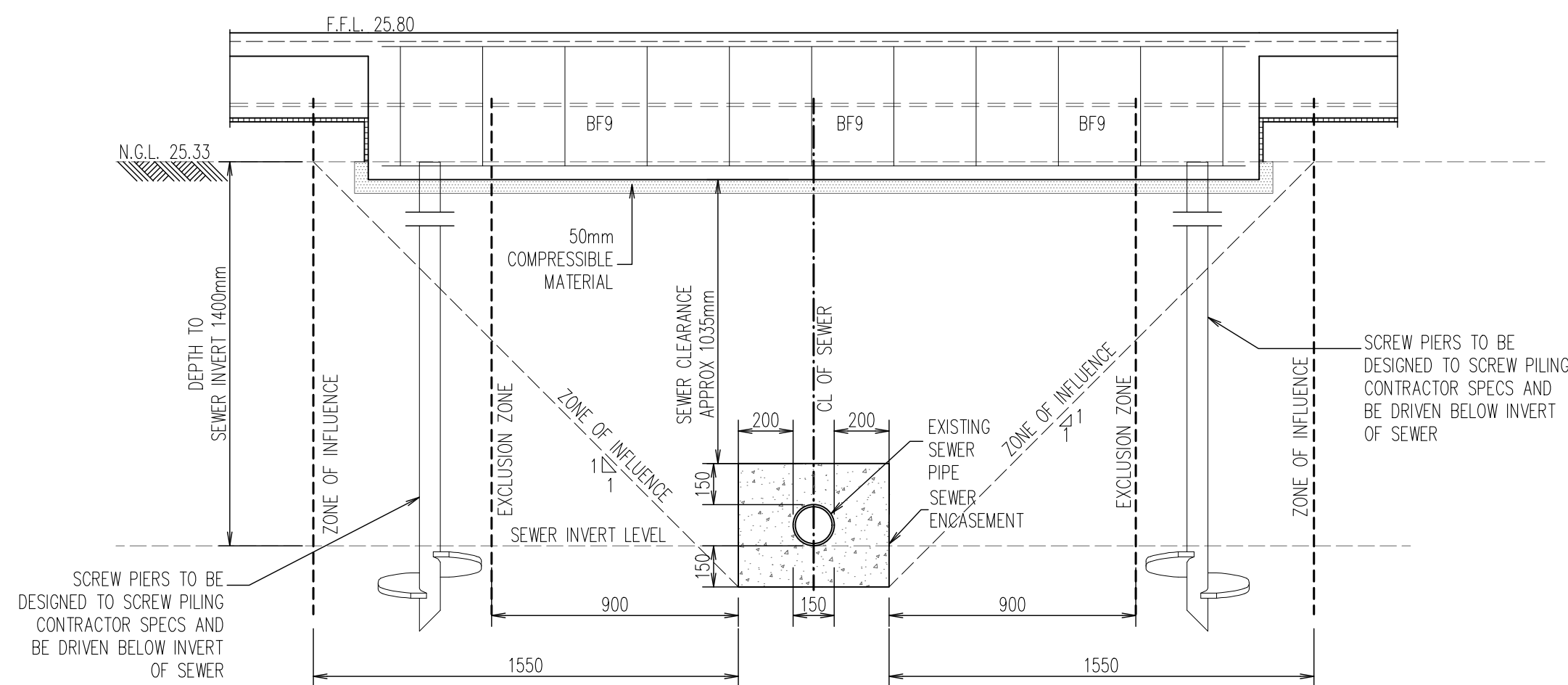
SECTION 5
SCALE 1:20
S100



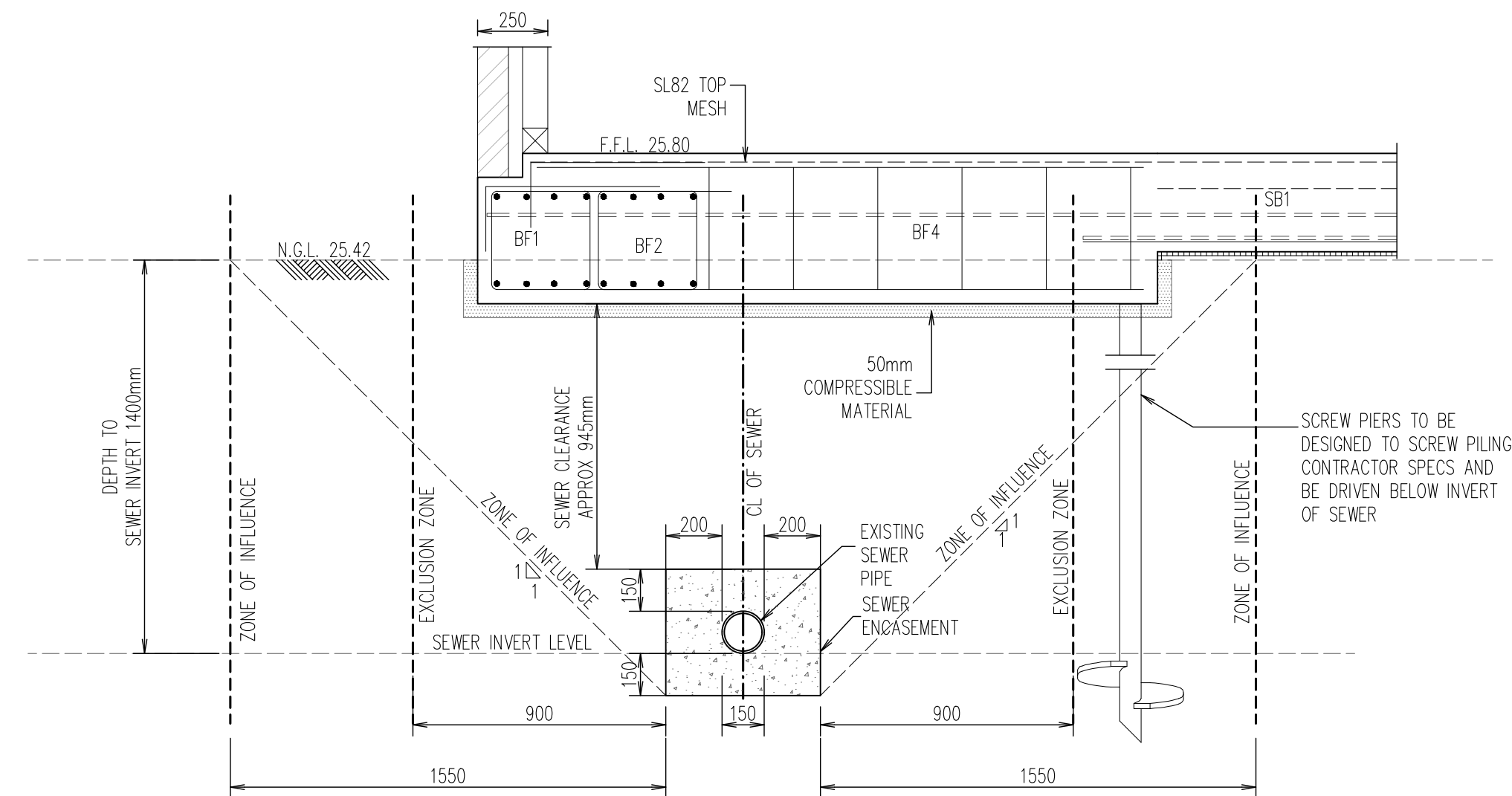
BF9 BRIDGE FOOTING
SCALE 1:20



SECTION 6
SCALE 1:20
S100



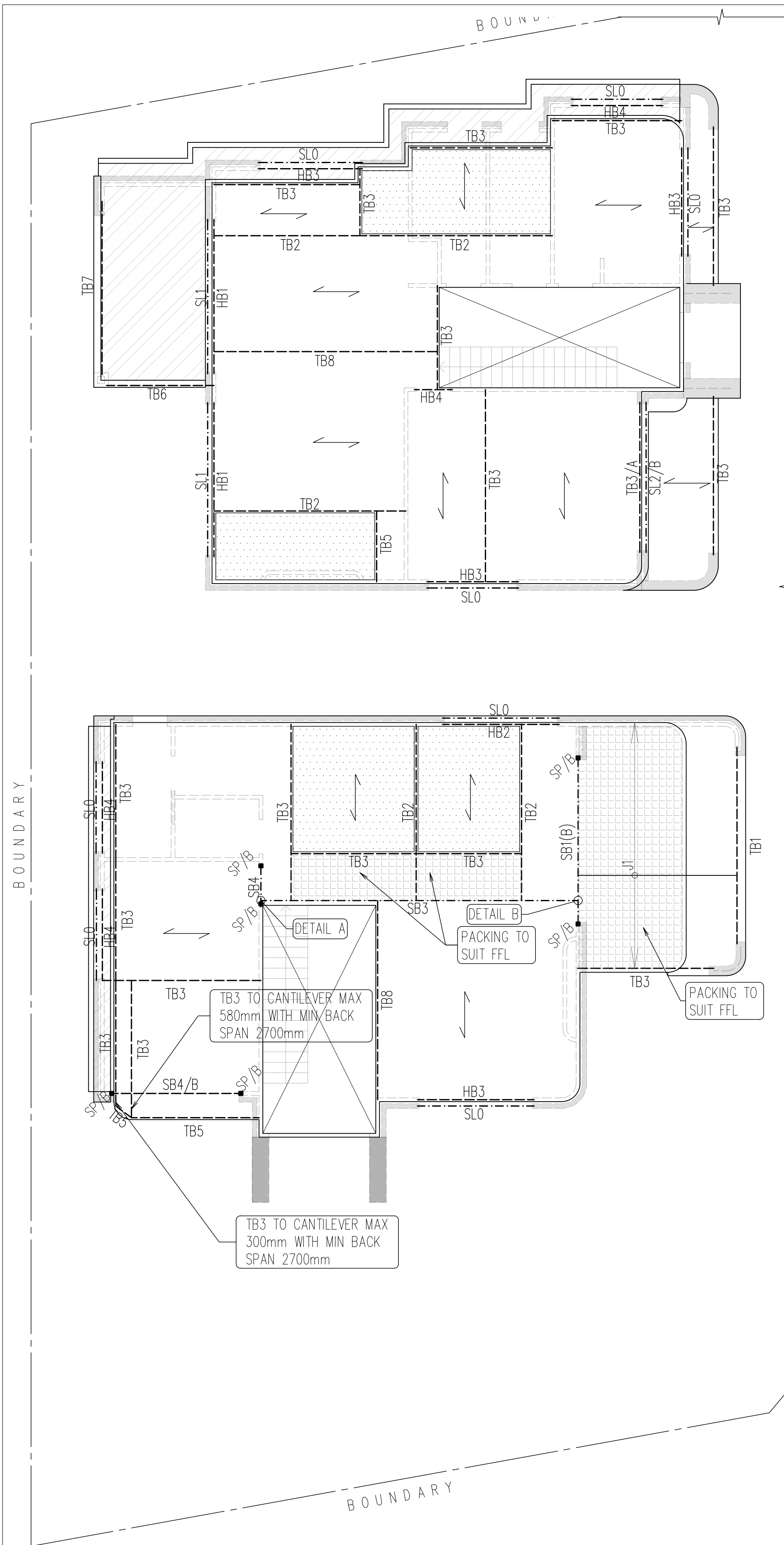
SECTION 7
SCALE 1:20
S100



SECTION 8
SCALE 1:20
S100

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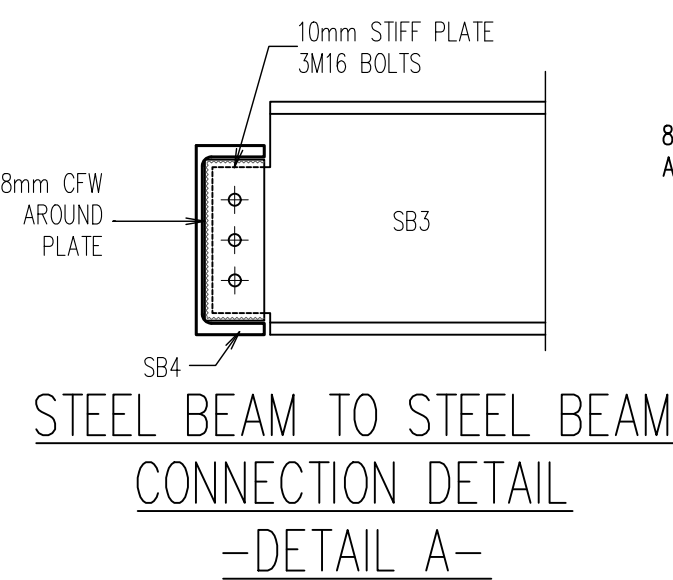
FIRST FLOOR FRAMING PLAN

SCALE: 1:75
NOTES:

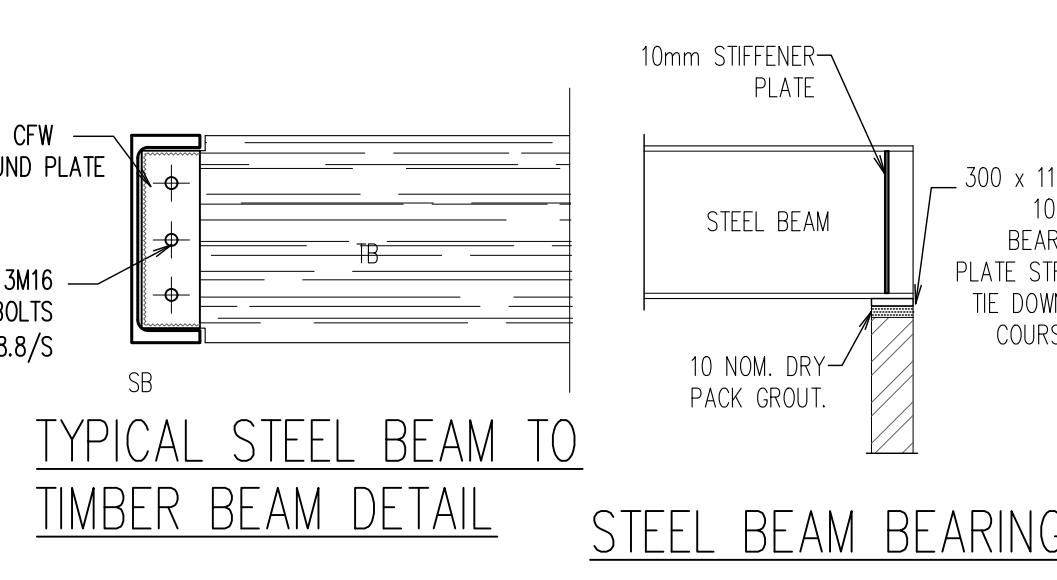
1. ALL EXPOSED STEEL TO BE PRESSURE GALVANIZED.
2. ALL STEEL FIXINGS TO BE IN ACCORDANCE TO AS4100.
3. ALL EXPOSED TIMBER TO BE PRESSURE TREATED TO AS1684.
4. ALL DETAILS TO BE CONFIRMED DURING CONSTRUCTION.
5. ALL TIMBER TO BE IN ACCORDANCE TO AS1684.
6. ROOF TO BE BUILT IN ACCORDANCE TO AS1684.

FLOOR LOADING ASSUMPTION	
GROUND FLOOR	TILED FLOOR
FIRST FLOOR	CARPET / FLOOR BOARDS
ALL WET AREAS	TILED FLOOR

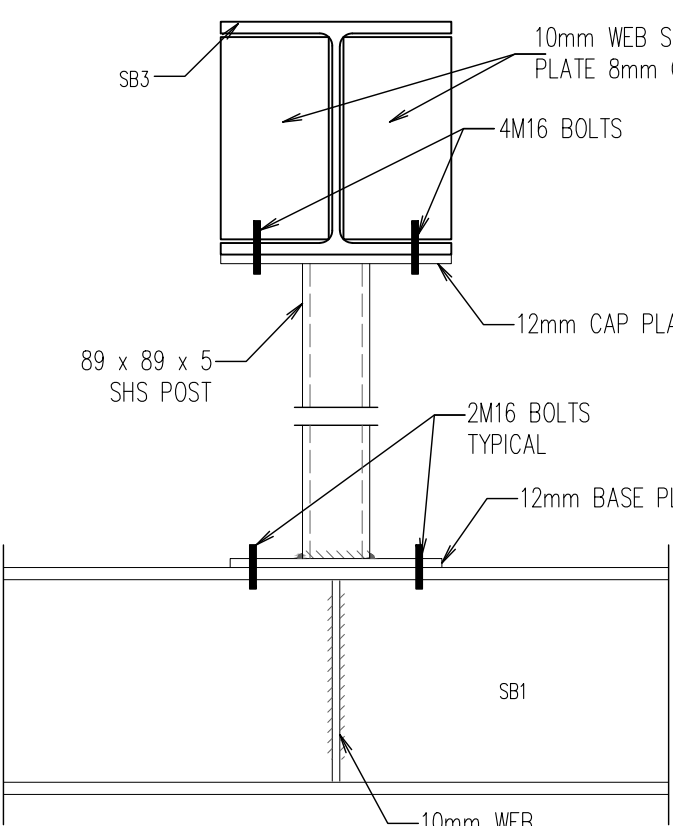
ROOF LOADING ASSUMPTION	
OF LOWER ROOF	SHEET ROOF (40kg/m ²)
FIRST FLOOR MAIN ROOF	SHEET ROOF (40kg/m ²)



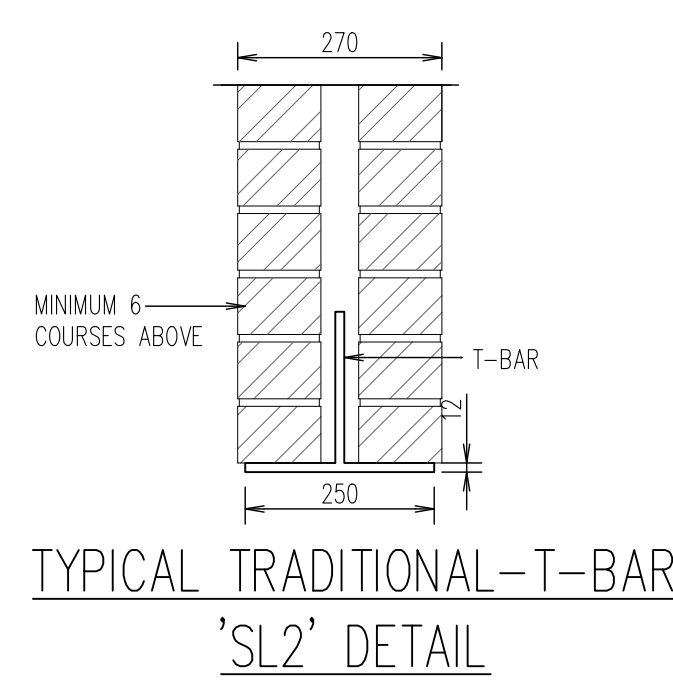
STEEL BEAM TO STEEL BEAM
CONNECTION DETAIL
-DETAIL A-



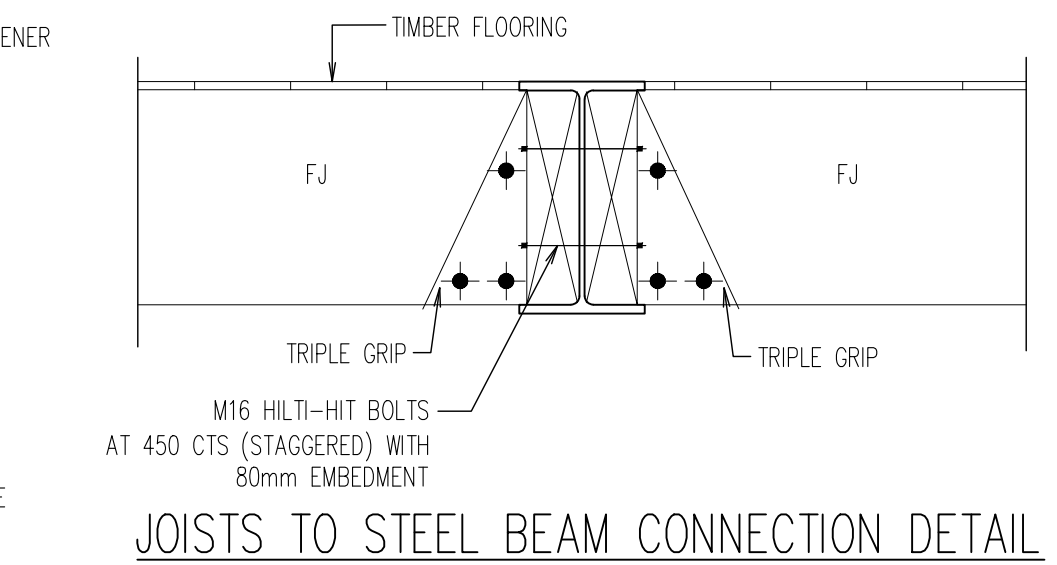
TYPICAL STEEL BEAM TO
TIMBER BEAM DETAIL
STEEL BEAM BEARING



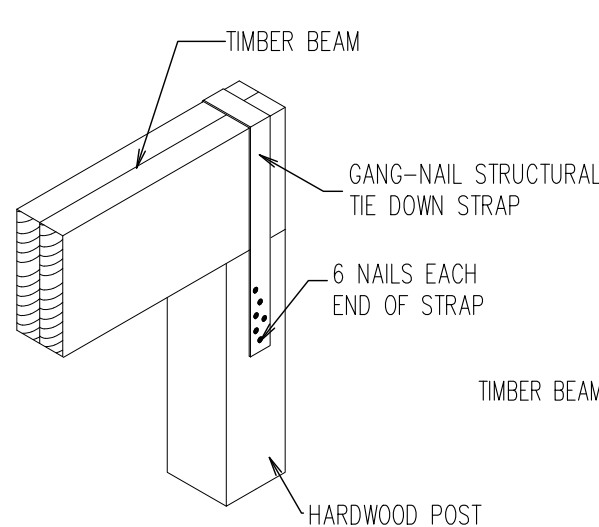
STEEL BEAM TO STEEL BEAM
CONNECTION DETAIL
-DETAIL B-



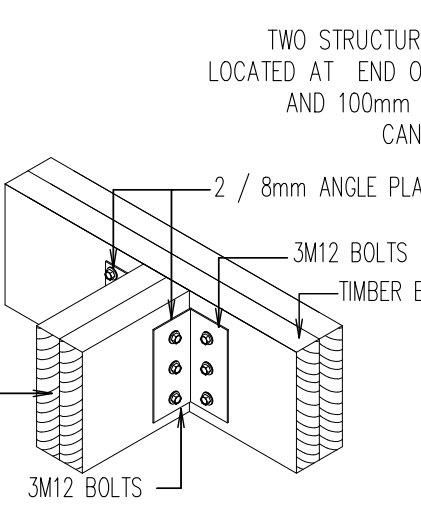
TYPICAL TRADITIONAL-T-BAR
'SL2' DETAIL



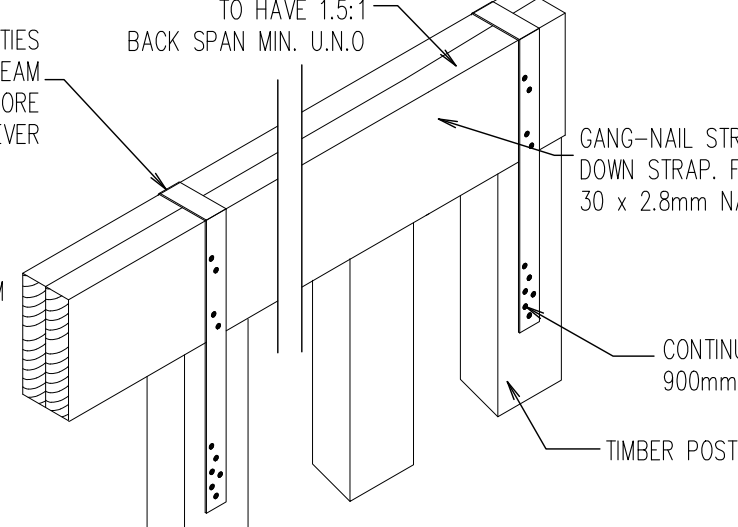
JOISTS TO STEEL BEAM CONNECTION DETAIL



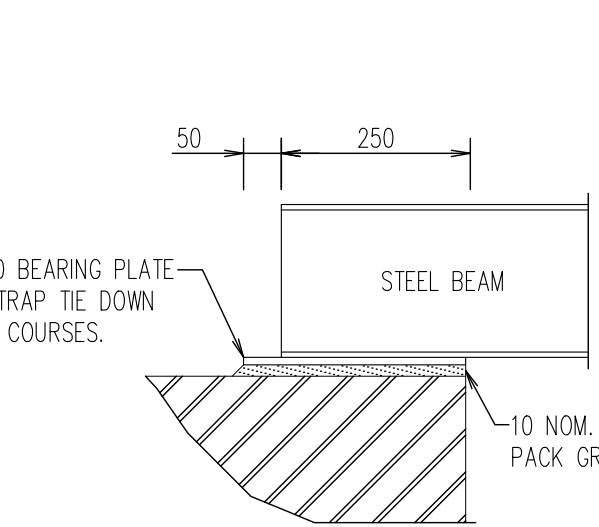
TIMBER BEAM TO
POST CONNECTION



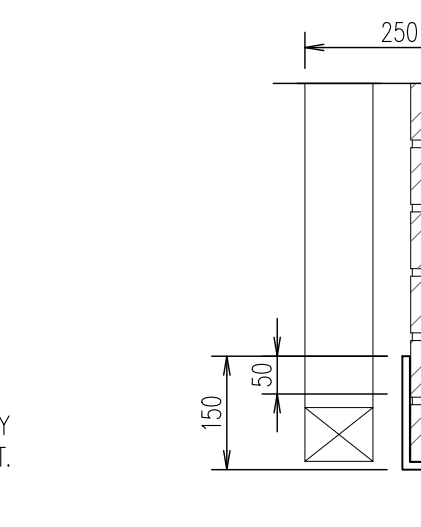
TIMBER BEAM
CONNECTION DETAIL



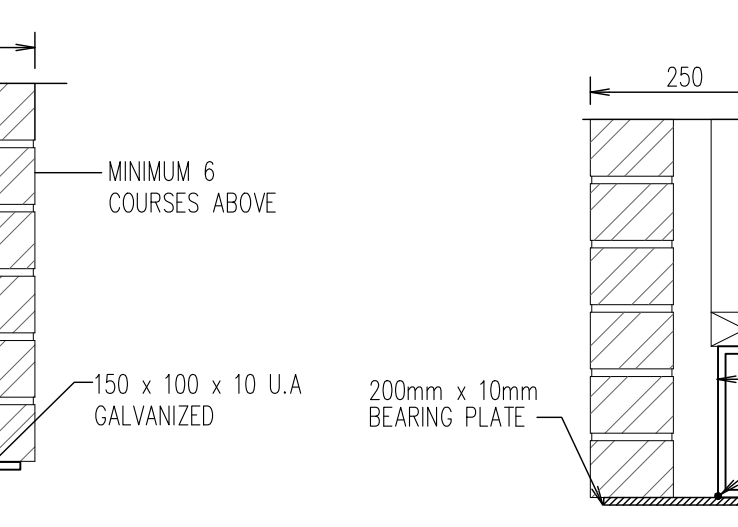
TIMBER BEAM CANTILEVER DETAIL



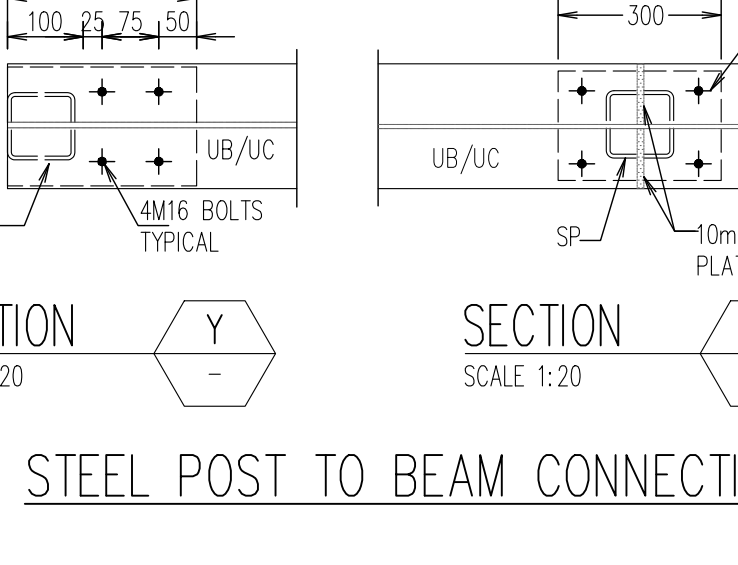
TYPICAL BEAM BEARING DETAIL



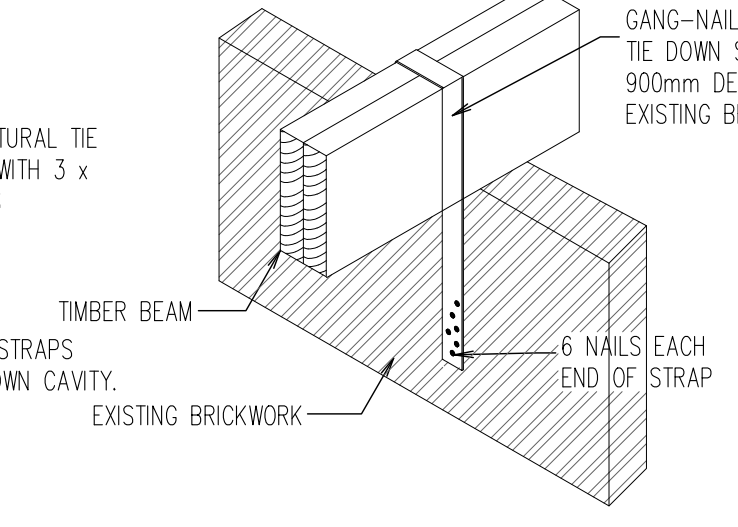
STEEL LINTEL 'SL1' DETAIL



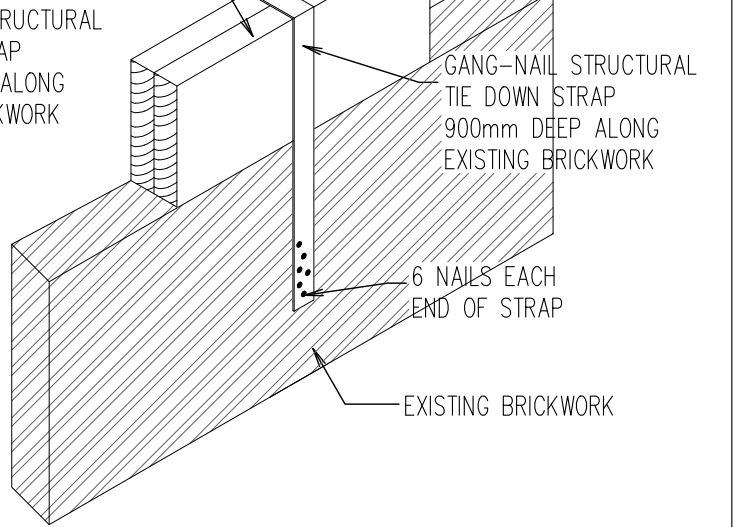
TYPICAL PFC+PLATE DETAIL



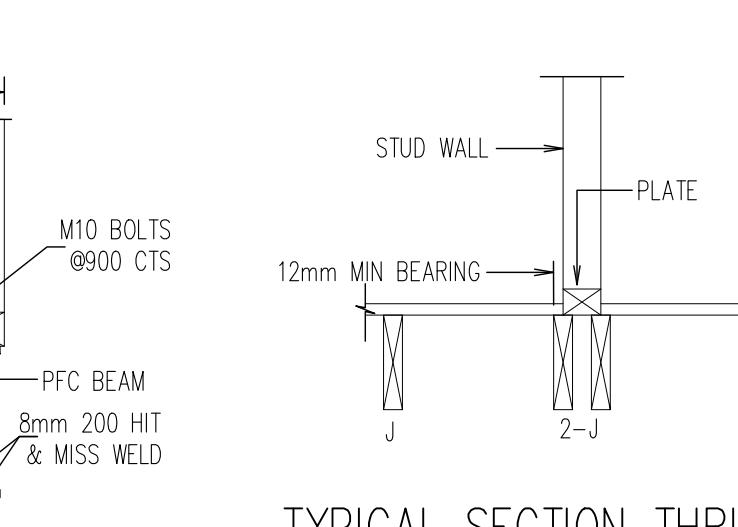
STEEL POST TO BEAM CONNECTION DETAIL



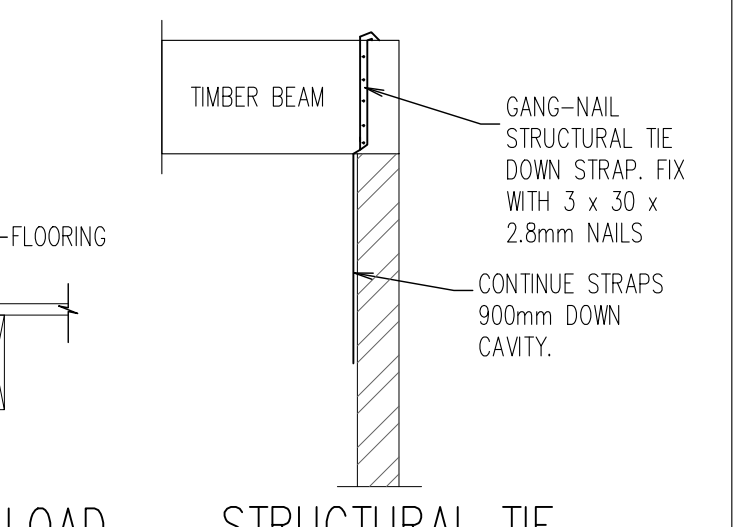
TIMBER BEAM TO BRICKWORK
CONNECTION



TIMBER BEAM TO BRICKWORK
CONNECTION



TYPICAL SECTION THRU LOAD
BEARING WALL



STRUCTURAL TIE
DOWN DETAILS

TIMBER MEMBER SCHEDULE		
MARK	TYPE & GRADE	SIZE
HB1	HEADER BEAM	2/300 x 45 LVL
HB2	HEADER BEAM	1/300 x 63 LVL
HB3	HEADER BEAM	1/240 x 63 LVL
HB4	HEADER BEAM	1/200 x 63 LVL
TB1	TIMBER BEAM	2/360 x 45 LVL
TB2	TIMBER BEAM	2/300 x 63 LVL
TB3	TIMBER BEAM	2/300 x 45 LVL
TB4	TIMBER BEAM	2/240 x 63 LVL
TB5	TIMBER BEAM	300 x 63 LVL
TB6	TIMBER BEAM	200 x 63 LVL
TB7	TIMBER BEAM	240 x 63 LVL
TB8	TIMBER BEAM	3/300 x 63 LVL
J1	TIMBER JOIST	200 x 45 LVL JOISTS @ 450 CTS

STEEL MEMBER SCHEDULE		
SB1	STEEL BEAM	300PFC + 10mm PLATE
SB2	STEEL BEAM	300PFC
SB3	STEEL BEAM	310UC 96.8
SB4	STEEL BEAM	250 PFC + 10mm PLATE
SL1	STEEL LINTEL	150 x 100 x 10mm L-ANGLE
SL2	STEEL LINTEL	250 x 12 TRADITIONAL T BAR (GALV.)
SLO	STEEL LINTEL	REFER TO BRICK LINTEL SCHEDULE ON SHEET "S001"

NOTES

- TRIPLE BEAMS TO BE BOLTED TOGETHER @ 300 C.C.
- DOUBLE BEAMS TO BE BUGLED TOGETHER AT @450 C.C. STAGGERED

NOTE: ALL EXTERNAL/ EXPOSED STEEL BEAMS, POSTS, PLATES, BOLTS, WELDS TO BE FULLY GALVANIZED

DOUBLE JOISTS TO BE FOUND UNDER LOAD BEARING WALLS

FIRST FLOOR WALLS TO BE LIGHTWEIGHT WITH FOAM/CLADDING

TIMBER JOIST SCHEDULE		
MAX SPAN	SIZE	SPACING
4200	240 - 45 HYJOIST OR 240 x 45 F7	450 CTS
4400	240 - 63 HYJOIST OR 290 x 35 F7	450 CTS
4600	300 - 45 HYJOIST OR 290 x 45 F7	450 CTS
5000	300 - 63 HYJOIST	450 CTS
5400	300 - 90 HYJOIST	450 CTS

NOTES

- ALL LOAD BEARING WALLS SUPPORTING ROOF TO BE CARRIED BY BEAMS IN ACCORDANCE WITH TIMBER SPAN TABLES.
- ENGINEER TO BE NOTIFIED IF FURTHER DESIGN REQUIRED
- ALL STEEL BEAMS TO BE 300 PLUS GRADE

FLOOR JOISTS ARE NOT DESIGNED TO CARRY ROOF LOADS. ENGINEER TO BE NOTIFIED IF SIZES TO SUPPORT LOAD BEARING WALLS ARE REQUIRED

ALL BEAMS TO BE SUPPORTED BY TRIPLE STUDS + BUGLES IF NO HARDWOOD POST IS DENOTED ON PLAN

LEGEND	
	BRICK WALLS BELOW
	BLOCK WALLS BELOW
	TIMBER STUD WALLS BELOW
	VOID
	30mm SETDOWN IN WET AREAS
	SHEET ROOF TO AS1684
	DENOTES TIMBER JOISTS
	STEEL POST BELOW (89 x 89 x 5mm SHS)
	DENOTES PACKING TO SUIT FLOOR LEVEL
(B)	DENOTES BEAM BELOW

A1 0 1 2 3 4 5 6 7 8 9 10

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A	ISSUED FOR C.D.C.	A.N.	31.03.2026
REV	DESCRIPTION	BY	CHECK



DRAWN: A.N.	DESIGNED: A.N.
REVIEWED: A.N.	DATE: 31.03.2026
CHECKED: A.N.	
APPROVED DATE: 31.03.2026	

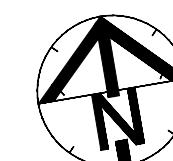
SLEIMAN
PROPERTY GROUP

ARCHITECT:
SLEIMAN PROPERTY GROUP

CLIENT:
DEBRA FRANCIS

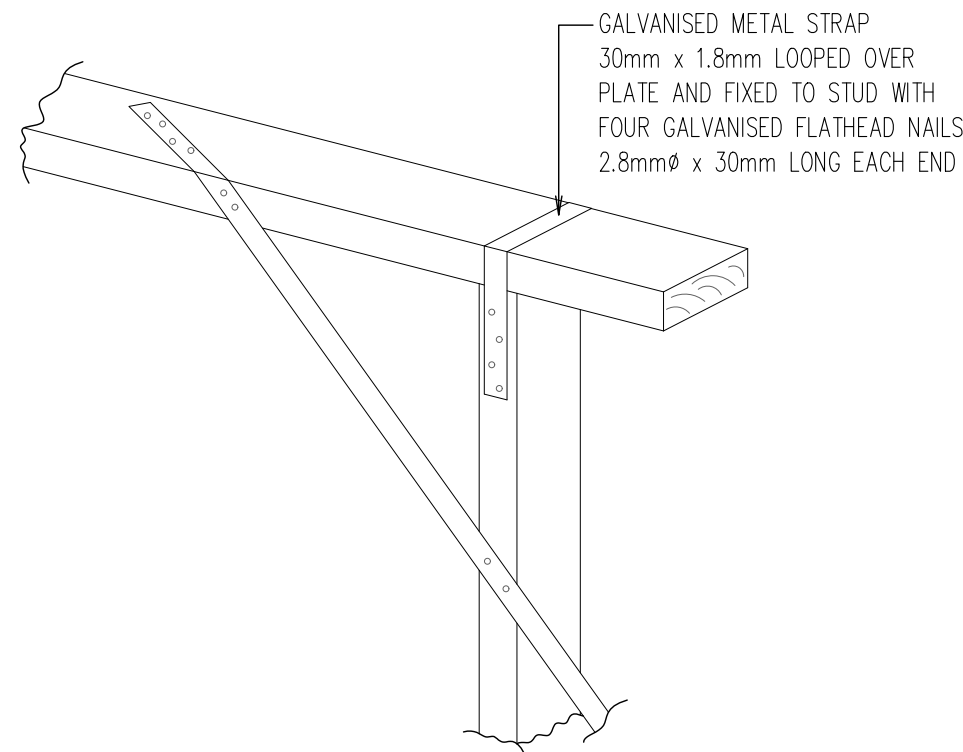
PROJECT:
PROPOSED DETACHED DUAL OCCUPANCY
ADDRESS:
42 DARTMOOR CIRCUIT, EMU HEIGHTS
APPLICATION:
C.D.C.

SHEET SUBJECT:
FIRST FLOOR MAIN
BEAM PLAN AND
DETAILS

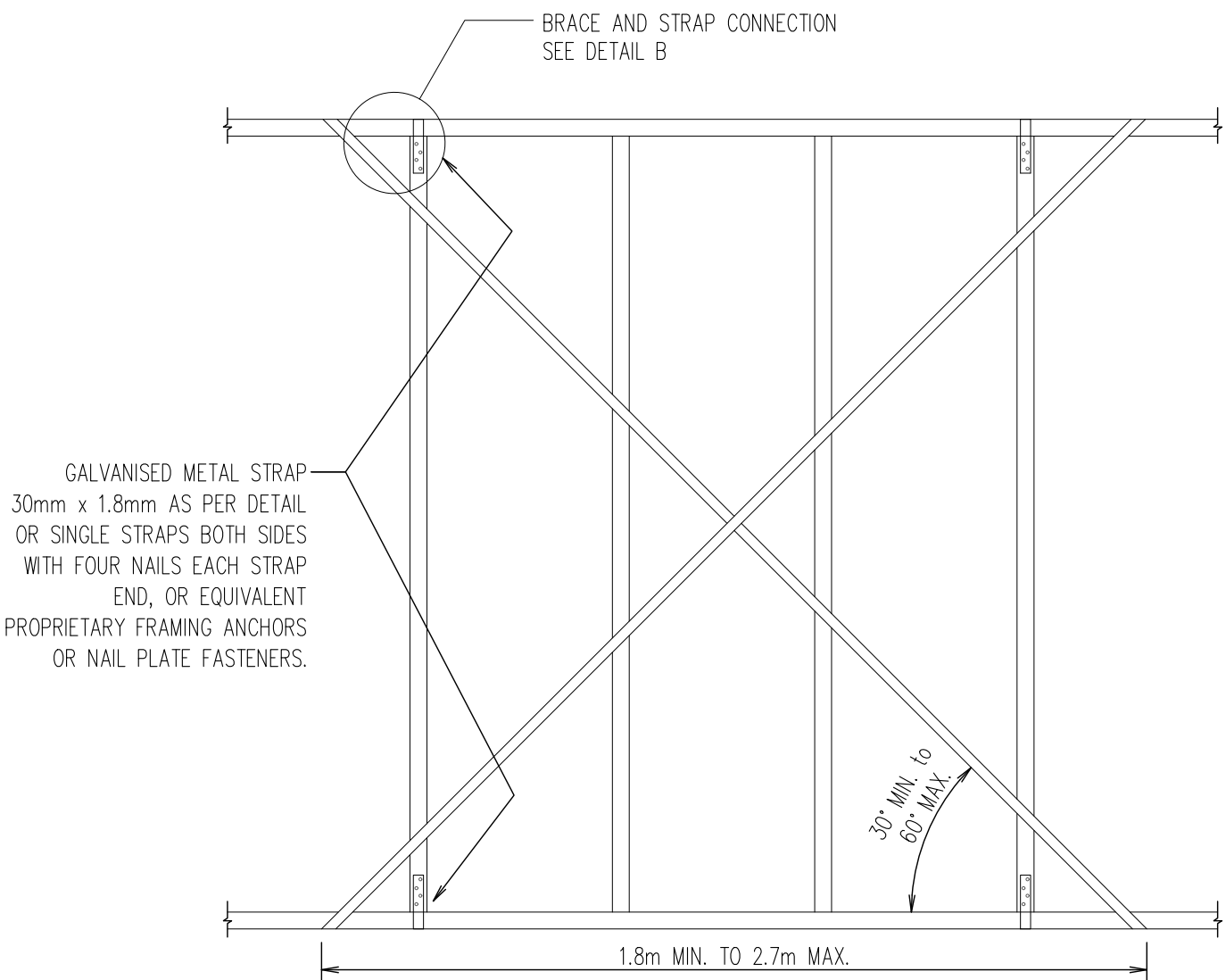


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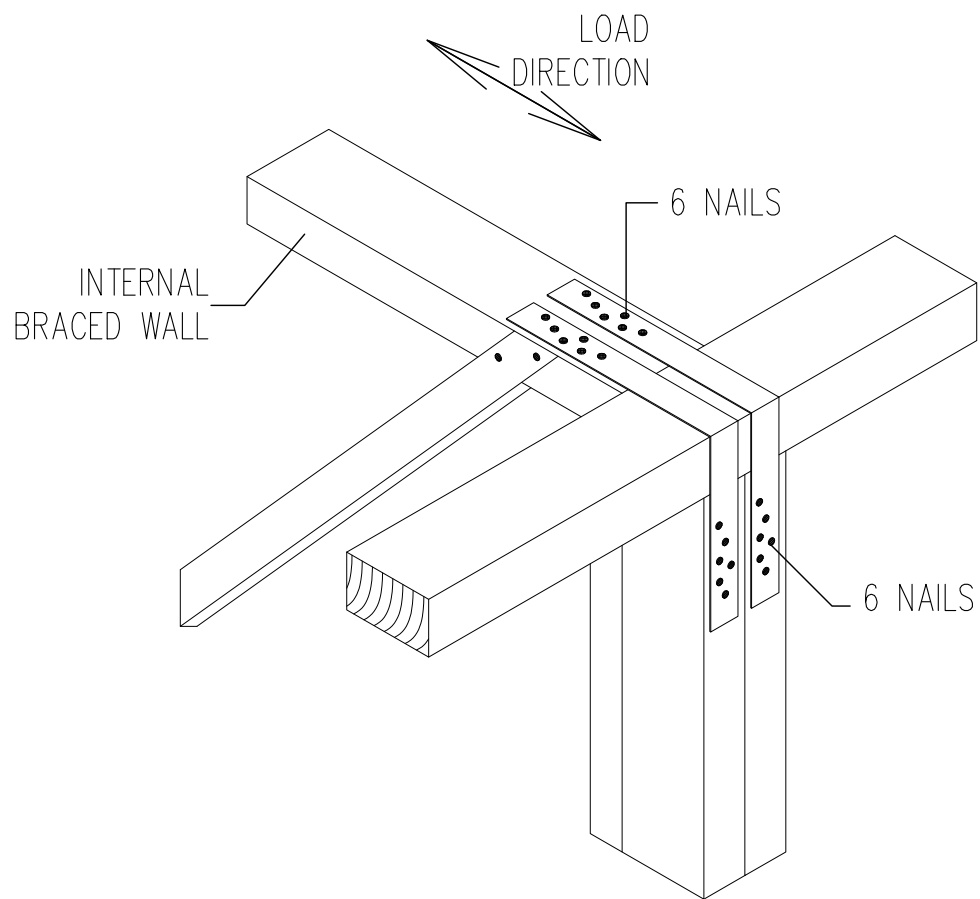
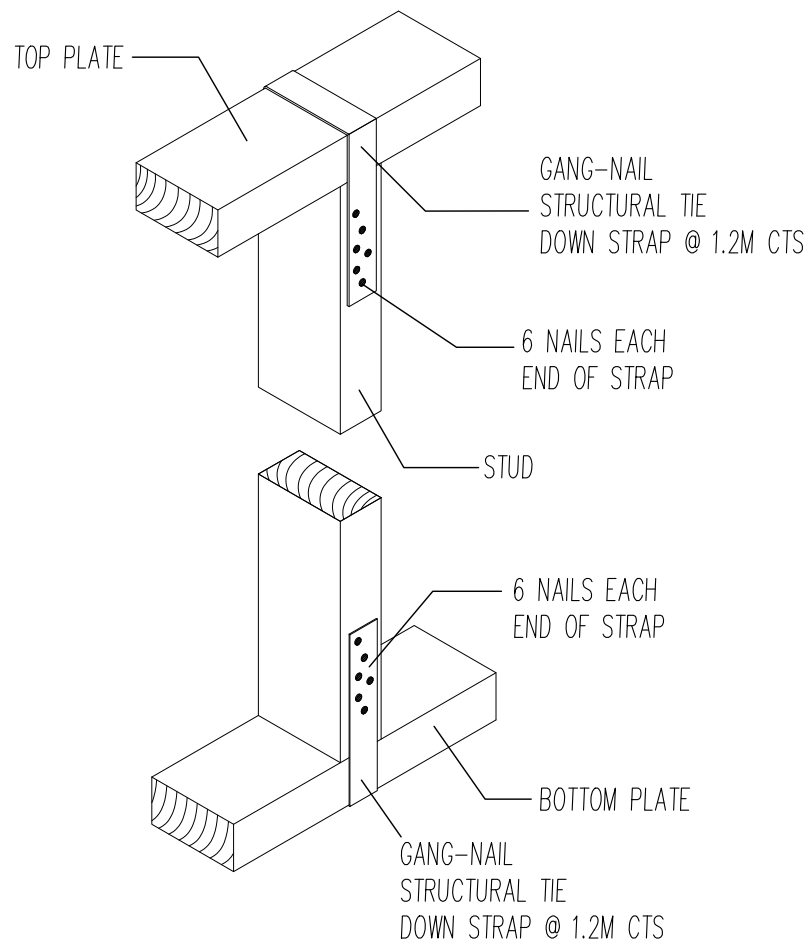
ALL DIMENSIONS IN mm UNLESS OTHERWISE STATED DO NOT SCALE DRAWING, USE FIGURED DIMENSIONS ONLY	
SCALE @ A1 1:75	JOB No. ST2673
AUTHORISED A.N.	DWG No. S200
	REV A



METAL TENSION STRAP BRACING
CORROSION PROTECTED FLAT METAL TENSION STRAPPING FIXED WITH TWO GALVANISED FLATHEAD NAILS 3.15mmØ x 30mm LONG TO EACH END STUD AND THE FACE OF THE TOP AND BOTTOM PLATE AND FOUR GALVANISED FLATHEAD NAILS 3.15mmØ x 30mm LONG TO THE STRAP RETURN OVER THE TOP PLATE AND UNDER THE BOTTOM PLATE.

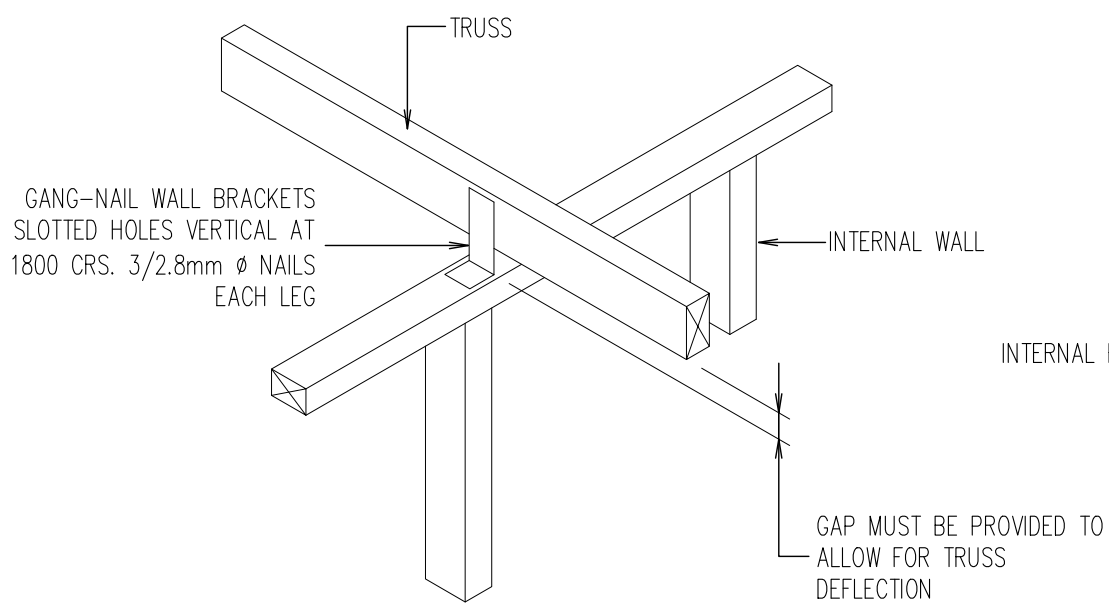


NOTE: NOGGINGS HAVE BEEN OMITTED FOR CLARITY
TYPICAL WALL BRACING DETAIL
SCALE 1:20
NOTE: IN ACCORDANCE WITH REQUIREMENTS OF AS1684 NATIONAL FRAMING CODE.

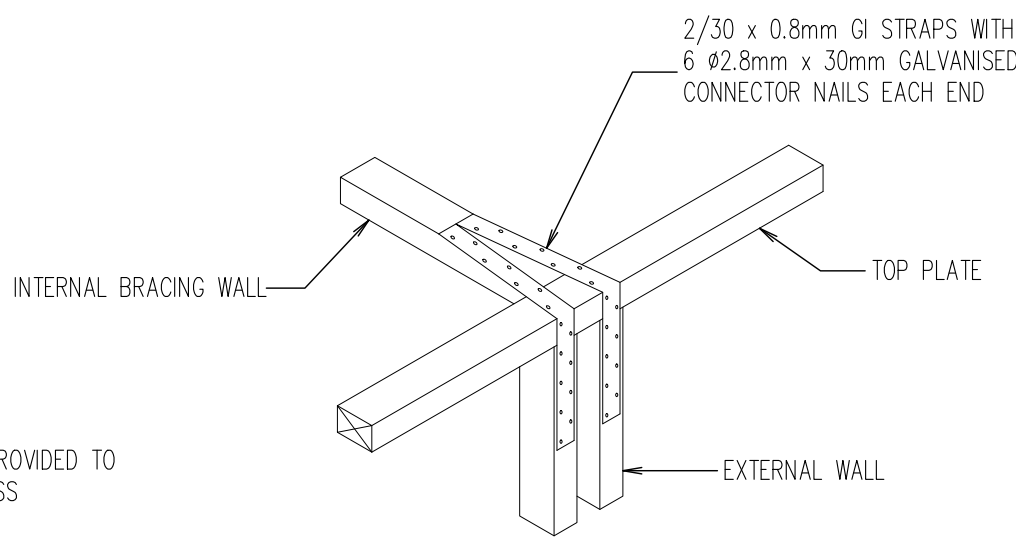


NOTE :
ALL BRACING TO COMPLY WITH RELEVANT SECTIONS OF AS1684 RESIDENTIAL TIMBER FRAMED CONSTRUCTION, FOR FURTHER INFORMATION REFER TO SECTION 8 OF PART 2 AS1684.2-2010

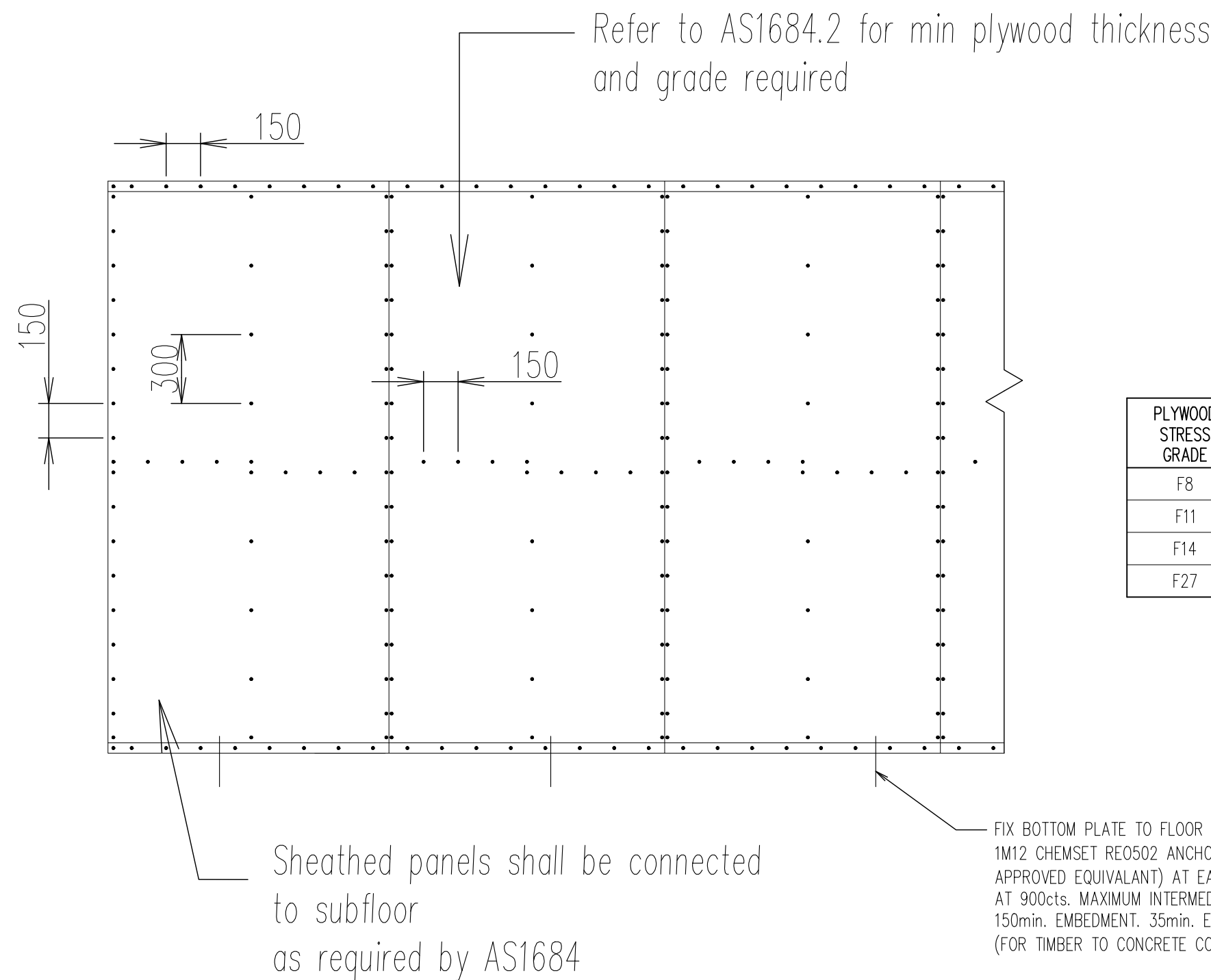
STRUCTURAL TIE DOWN STRAP



NON LOAD BEARING WALL PERPENDICULAR TO TRUSSES



INTERNAL BRACING WALL TO EXTERNAL WALL CONNECTION
SCALE 1:20



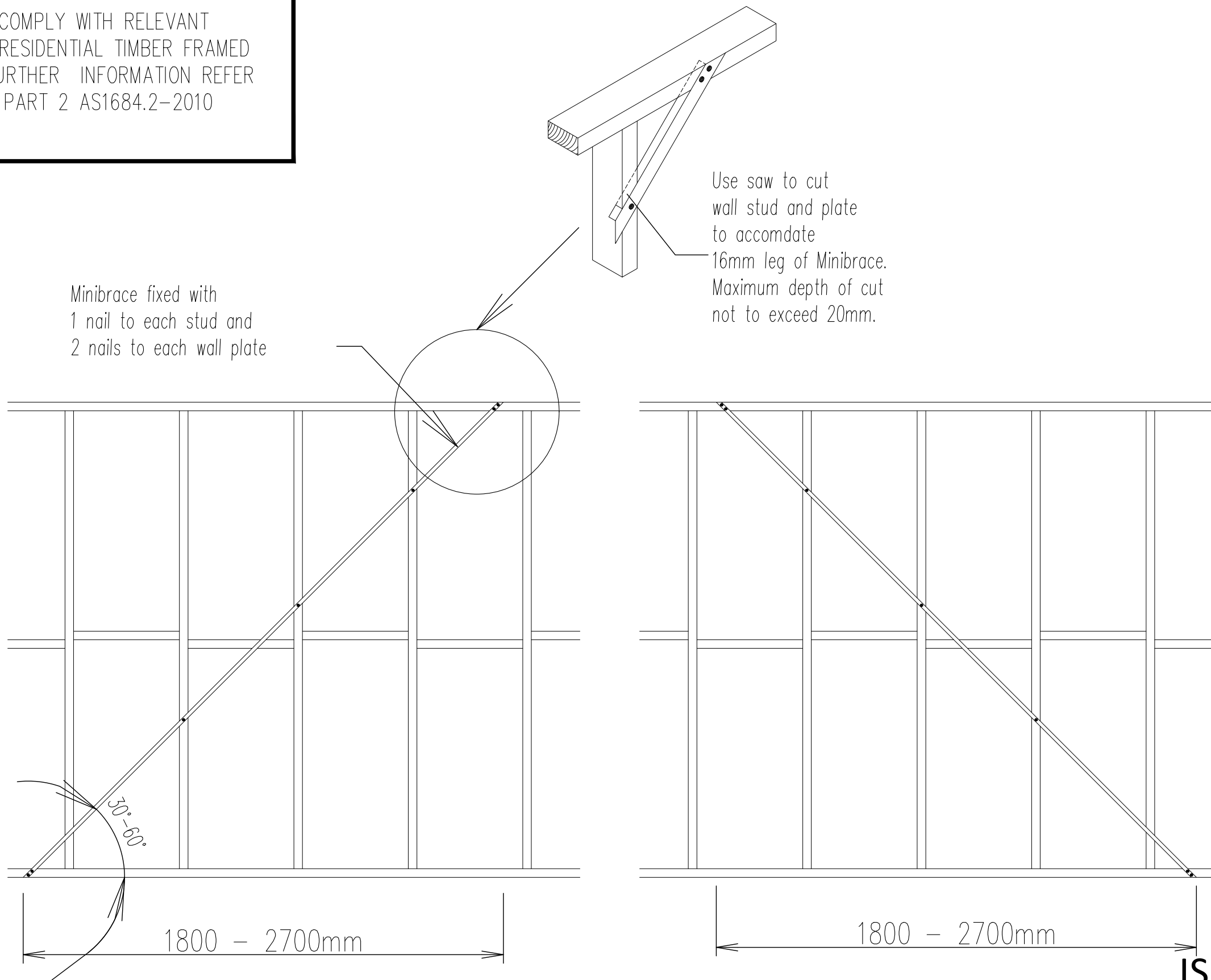
PLYWOOD STRESS GRADE	PLYWOOD THICKNESS WHEN STUDS AT 450
F8	7 mm
F11	4.5 mm
F14	4 mm
F27	3 mm

PLYWOOD BRACING DETAIL
SCALE 1:20

- PLYWOOD BRACING IN ACCORDANCE WITH TABLE 8.18(g) OF AS 1684.2
- ULTIMATE BRACING CAPACITY 3.4kN/m

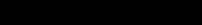
OR

FIX BOTTOM PLATE TO FLOOR WITH M10 4.6/S BOLTS AND AT 900cts. MAXIMUM 35min. EDGE DISTANCE (FOR TIMBER TO TIMBER CONNECTION)



ISSUED FOR CDC.

A1 0 1 2 3 4 5 6 7 8 9 10

				E: ahmad@nceeng.com M: 0452489806 COPYRIGHT THIS DRAWING AND THE INFORMATION SHOWN HEREIN IS THE PROPERTY OF NANO CONSULTING ENGINEERS AND MAY NOT BE USED FOR ANY PURPOSES THAN FOR WHICH SUPPLIED.						DRAWN : A.N. DESIGNED : A.N. REVIEWED : A.N. DATE : 31.03.2026 CHECKED : A.N. APPROVED DATE : 31.03.2026				ARCHITECT: SLEIMAN PROPERTY GROUP		CLIENT: DEBRA FRANCIS		PROJECT: PROPOSED DETACHED DUAL OCCUPANCY ADDRESS: 42 DARTMOOR CIRCUIT, EMU HEIGHTS APPLICATION: C.D.C.		SHEET SUBJECT: TIMBER BRACING DETAILS N/A				ALL DIMENSIONS IN mm UNLESS OTHERWISE STATED DO NOT SCALE DRAWING, USE FIGURED DIMENSIONS ONLY			
A				ISSUED FOR C.D.C.				A.N.		A.N.		31.03.2026		APPROVAL TYPE: 1. PRELIMINARY 2. DEVELOPMENT APPLICATION 3. COMPLETION DEVELOPMENT 4. CONSTRUCTION CERTIFICATE 5. FOR TENDER 6. OTHERS		PH: 0452489806 EMAIL: ahmad@nceeng.com						SCALE @ A1 1:20		JOB No ST2673			
REV				DESCRIPTION				BY		CHECK		DATE										AUTHORISED A.N.		DWG No S210		REV A	

TIE DOWN NOTES

N1 TIE-DOWNS FOR SHEET ROOF RAFTERS AND BATTENS

1. ALL BATTENS TO BE FIXED TO RAFTERS WITH 12 GAUGE BUGLE SCREW.
2. RAFTERS SITTING ON EXTERNAL (PERIMETER) WALLS: FIX ALL RAFTERS TO TOP PLATE WITH TRIPLGRIP OR 12 GAUGE BUGLE SCREW.
3. RAFTERS SITTING ON EXTERNAL (PERIMETER) ROOF BEAMS: FIX ALL RAFTERS TO BEAM WITH TRIPLGRIP OR 12 GAUGE BUGLE SCREW OR STRAP.
4. RAFTERS SITTING ON INTERNAL TOP PLATES OR INTERNAL ROOF BEAMS (SUPPORTS MORE THAN 2.0M AWAY FROM EXTERNAL WALLS): FIX EVERY 2ND RAFTER TO TOP PLATE/BEAM WITH TRIPLGRIP OR 12 GAUGE BUGLE SCREW OR STRAP.
5. CROSS STRAP BRACE ALONG ROOF PROFILE, FIX TO ALL RAFTERS/BATTENS AND TIE DOWN TO WALL FRAME AT ENDS.

PARAPET WALLS AND ROOF ('DWARF') BUILD-UP WALLS

1. STRAP TOP PLATE TO UPPER STUDS AT 1200 CTS. SIMILARLY STRAP BOTTOM PLATE TO STUDS AT 1200 CTS.
2. STRAP, TRIPLGRIP OR SCREW UPPER STUDS TO MAIN TOP PLATE OR ROOF BEAM UNDER AT 900 CTS.
3. FIX EXTERNAL PARAPET WALLS TO RAFTER BEHIND AT 600 CTS WITH 2/12 GAUGE BUGLE SCREWS.
4. WHEN EXTERNAL PARAPET WALLS DO NOT HAVE A RAFTER TO FIX INTO AND BRACE THE WALL FROM WIND, INSTALL 150X150X10 EA BRACKET X 100MM LONG. LOCATE BRACKETS AT EVERY STUD. FIX BRACKET TO PARAPET STUD WALL WITH 4/M10 COACH SCREWS OR 14 GAUGE BUGLE SCREWS AND FIX BRACKET TO TOP PLATE/ROOF BEAM WITH 4/M10 COACH SCREWS OR 14 GAUGE BUGLE SCREWS. IF FIXING PARAPET WALL TO A ROOF BEAM UNDER, ENSURE ROOF BEAM IS BRACED TO CEILING FRAMING WITH BLOCKING AT 1200 CTS.

ROOF/STRUTTING BEAMS

1. ALL ROOF AND STRUTTING BEAMS TO BE STRAPPED DOWN OR TRIPLEGRIPPED DIRECTLY TO DOUBLE STUD BELOW. COMBINE FIXINGS MAY BE USED TO CONNECTED ROOF BEAM INTO TOP PLATE AND TOP PLATE INTO THE DOUBLE STUD UNDER.
 - 1.1. IF UNDERPURLINS ARE USED: STRAP (LOOP OVER) OR BUGLE EVERY 2ND RAFTER TO UNDERPURLIN.
 - 1.2. IF STRUTS ARE USED: STRAP UNDERPURLIN TO STRUT AT ALL STRUTTING LOCATIONS. STRAP STRUT TO TOP PLATE (LOOP UNDER TOP PLATE).

LINTELS

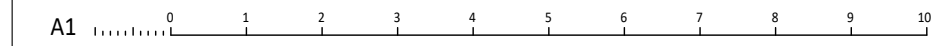
1. FIX TOP PLATE TO WINDOW/DOOR LINTEL WITH STRAPS OR 12 GAUGE SCREWS AT 1200 CTS.
2. FIX ENDS OF LINTEL (LINTEL SPAN LESS THAN 3.6M) TO FULL HEIGHT DOUBLE STUD WITH 3/12 GAUGE BUGLE SCREWS DRIVEN FROM SIDE OF DOUBLE STUD AND INTO END OF LINTEL. ALTERNATIVELY, USE STRAPS TOP AND BOTTOM TO TIE DOWN LINTEL TO THE BOTTOM PLATE.
3. ALL ROOF BEAMS/LINTELS TO HAVE A DOUBLE STUD AT THE ENDS, FOR SUPPORT AND TIE-DOWN PURPOSES.

TOP PLATE, STUDS AND BOTTOM PLATES

1. ALL INTERNAL AND EXTERNAL WALLS: STRAP OR BUGLE TOP PLATE TO STUDS AND BOTTOM PLATE TO STUDS AT 1350 CTS. PROVIDE ADDITIONAL STRAPS TOP AND BOTTOM AT ALL BEAM/LINTEL, WALL BRACING AND ROOF BRACING LOCATIONS.
2. ALL INTERNAL AND EXTERNAL WALLS: FIX ALL BOTTOM PLATES TO FLOOR BEAM/JOISTS/SOLID BLOCKING UNDER WITH 14 GAUGE BUGLE SCREWS AT 900 CTS. EXTRA FIXINGS REQUIRED AT ROOF BEAM/LINTEL, WALL BRACING AND ROOF BRACING LOCATIONS.

NOTES:

- STRAPS TO HAVE AT LEAST 4-2.8 DIAMETER NAILS EACH END OF THE STRAP. STRAP TO LOOP OVER/UNDER MEMBER TO BE TIED DOWN AND EXTEND AT LEAST 300MM OVER THE SUPPORTING MEMBER. IF IT IS NOT FEASIBLE TO HAVE A LOOPED STRAPPED 2/FLAT STRAPS COULD BE USED WITH 8 NAILS TOTAL EACH MEMBER.
- BUGLE SCREWS TO EMBED AT LEAST 45MM INTO THE SUPPORTING TIMBER MEMBER.



ALL DIMENSIONS IN mm UNLESS OTHERWISE STATED DO NOT SCALE DRAWING, USE FIGURED DIMENSIONS ONLY		
SCALE @ A1 1: 75	JOB No ST2673	
AUTHORISED A.N.	DWG No S300	REV A

STORMWATER MANAGEMENT PLAN

42 DARTMOOR CIRCUIT, EMU HEIGHTS

GENERAL NOTES:

ALL WORK SHALL BE CARRIED OUT IN ACCORDANCE WITH COUNCIL'S REQUIREMENTS, BUILDING CODE OF AUSTRALIA, NSW CODE OF PRACTICE AND THE TO THE RELEVANT SERVICE CODES.

THESE DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL ARCHITECTURAL AND OTHER CONSULTANTS' DRAWINGS AND SPECIFICATIONS AND WITH SUCH OTHER WRITTEN INSTRUCTIONS AS MAY BE ISSUED DURING THE COURSE OF THE CONTRACT. ALL DISCREPANCIES SHALL BE REFERRED TO THE SUPERINTENDENT FOR DECISION BEFORE PROCEEDING WITH THE WORK.

ALL DIMENSIONS SHOWN ON THE DRAWINGS ARE IN MILLIMETERS (U.N.O.). DIMENSIONS SHALL NOT BE OBTAINED BY SCALING OF THESE DRAWINGS. USE FIGURED DIMENSIONS ONLY.

BENCHMARKS HAVE BEEN ESTABLISHED WHERE INDICATED ON THE DRAWINGS. ALL LEVELS ARE TO AUSTRALIAN HEIGHT DATUM (A.H.D.). THE CONTRACTOR SHALL UNDERTAKE ALL NECESSARY SURVEY WORK TO ENSURE THAT THE WORKS ARE CONSTRUCTED TO DESIGN LINE AND LEVEL.

SETTING OUT DIMENSIONS AND LEVELS SHOWN ON THE DRAWINGS SHALL BE VERIFIED BY THE CONTRACTOR.

ALL MATERIALS SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF THE RELEVANT SAA CODES AND THE BY-LAWS AND ORDINANCES OF THE RELEVANT BUILDING AUTHORITIES.

IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE ALL SAFETY FENCES, WARNING SIGNS, TRAFFIC DIVERSIONS AND THE LIKE DURING CONSTRUCTION. ALL WORKS TO COMPLY WITH WORK HEALTH AND SAFETY REQUIREMENTS AND OTHER RELEVANT AUTHORITY SAFETY REQUIREMENTS.

NO TREES SHALL BE REMOVED, CUTBACK OR RELOCATED WITHOUT THE WRITTEN INSTRUCTION FROM THE SUPERINTENDENT.

WHERE NEW WORKS ABOUT EXISTING THE CONTRACTOR SHALL ENSURE THAT A SMOOTH EVEN PROFILE, FREE FROM ABRUPT CHANGES IS OBTAINED.

ALL WORKS SHALL BE CARRIED OUT IN ACCORDANCE WITH THE DETAILS SHOWN ON THE DRAWINGS AND THESE SPECIFICATIONS.

DESIGN LEVELS GIVEN ARE TO FINISHED SURFACE LEVEL AND INCLUSIVE OF TOPSOIL. (TOPSOIL DEPTH VARIES)

THE CONTRACTOR SHALL ARRANGE ALL SURVEY SETOUT TO BE CARRIED OUT BY A N.A.T.A. REGISTERED SURVEYOR.

CARE IS TO BE TAKEN WHEN EXCAVATING NEAR EXISTING SERVICES. NO MECHANICAL EXCAVATIONS ARE TO BE UNDERTAKEN OVER TELECOMMUNICATIONS OR ELECTRICAL SERVICES. HAND EXCAVATE IN THESE AREAS.

THE LOCATIONS OF UNDERGROUND SERVICES SHOWN ON THE DRAWING HAVE BEEN PLOTTED FROM DIAGRAMS PROVIDED BY SERVICE AUTHORITIES. THIS INFORMATION HAS BEEN PREPARED SOLELY FOR THE AUTHORITIES OWN USE AND MAY NOT NECESSARILY BE UPDATED OR ACCURATE.

THE POSITION OF SERVICES AS RECORDED BY THE AUTHORITY AT THE TIME OF INSTALLATION MAY NOT REFLECT CHANGES IN THE PHYSICAL ENVIRONMENT SUBSEQUENT TO INSTALLATION.

NANO CONSULTING ENGINEERS DOES NOT GUARANTEE THAT THE SERVICES INFORMATION SHOWN ON THE DRAWING SHOWS MORE THAN THE PRESENCE OR ABSENCE OF SERVICES, AND WILL ACCEPT NO LIABILITY FOR INACCURACIES IN THE SERVICES INFORMATION SHOWN FROM ANY CAUSE WHATSOEVER.

IT IS THE CONTRACTOR'S RESPONSIBILITY TO OBTAIN FROM THE UTILITY SERVICES AUTHORITIES A CURRENT COPY OF UNDERGROUND SERVICES SEARCH FOR THE LOCATION OF ALL EXISTING SERVICES PRIOR TO COMMENCEMENT OF ANY WORK AND NOTIFY ANY CONFLICT WITH THE DRAWINGS IMMEDIATELY. CLEARANCE SHALL BE OBTAINED FROM THE RELEVANT REGULATORY AUTHORITY. CONTRACTOR TO KEEP COPY OF UNDERGROUND SERVICES SEARCH ON SITE AT ALL TIMES. ANY DAMAGES TO SERVICES OR SERVICES ADJUSTMENTS SHALL BE CARRIED OUT BY THE CONTRACTOR OR RELEVANT AUTHORITY AT THE CONTRACTOR'S EXPENSE.

VISIT THE SITE BEFORE SUBMITTING THE FINAL TENDER PRICE TO ASSESS 'ON SITE' CONDITIONS. FAILURE TO DO SO WILL FORFEIT ANY CLAIM FOR NOT BEING AWARE OF CONDITIONS AFFECTING THE TENDER.

THE CONTRACTOR SHALL PREPARE ACCURATE WORK-AS-EXECUTED DRAWINGS FOLLOWING THE COMPLETION OF ALL WORKS.

IT IS THE CONTRACTOR'S RESPONSIBILITY TO HAVE IN PLACE & MAINTAIN TRAFFIC FACILITIES AT ALL TIMES DURING CONSTRUCTION.

STORMWATER NOTES:

ALL WORK SHALL BE CARRIED OUT IN ACCORDANCE AS3500.3-2018: 'STORMWATER DRAINAGE'.

FOR STORMWATER DRAINAGE PIPES THAT EXCEED 1:5 GRADE, REINFORCED CONCRETE ANCHOR BLOCKS SHALL BE INSTALLED. ANCHOR BLOCKS TO BE CONSTRUCTED TO SPECIFICATIONS SET OUT IN AS3500.3-2018.

COORDINATE THE INSTALLATION OF NEW SERVICES WITH ALL NEW & EXISTING SERVICES & STRUCTURAL PROVISIONS AS DETERMINED ON SITE.

ALL PIPEWORK TO BE SUPPORTED IN ACCORDANCE WITH AS3500.3-2018.

ALL PIPEWORK IS TO BE TESTED IN ACCORDANCE WITH THE REQUIREMENTS AS SET DOWN IN AS3500.3-2018. ALL IN-GROUND PIPEWORK TO BE INSPECTED BY THE SUPERINTENDENT UNDER TEST CONDITIONS PRIOR TO BACKFILLING.

PIPES SHALL BE TRUE TO GRADES SHOWN AND ALIGNED SO THAT THE CENTRE OF THE INLET PIPE INTERSECTS WITH THE CENTRE OF THE OUTLET PIPE AT THE DOWNSTREAM FACE OF THE PIT.

BED ALL PIPES FIRMLY AND EVENLY WITH IMPORTED FILL ONLY. THICKNESS OF BEDDING LAYER SHALL BE 75mm IN SOIL AND 200mm IN ROCK.

LAY AND JOINT ALL PIPES IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS AND AS3725-2007: 'DESIGN FOR INSTALLATION OF BURIED CONCRETE PIPES'.

ALLOW TO TEST ALL PIPES AND PITS TO LOCAL AUTHORITY'S REQUIREMENTS.

EXCAVATE TRENCHES AND STOCKPILE ALL MATERIAL FOR INSPECTION WITH REGARD TO REUSE FOR TRENCH BACKFILL. REMAINING MATERIAL TO BE REMOVED FROM SITE.

BACKFILL PIPES WITH IMPORTED FILL. PROVIDE 200mm SIDE SUPPORT AND 150mm OVERLAY ABOVE PIPE CROWN. TRENCH FILL ABOVE THE EMBEDMENT ZONE TO THE UNDERSIDE OF THE ROAD PAVEMENT OR THE FOOTWAY SHALL BE AS FOLLOW:-

UNDER ROADWAY
TRENCH FILL MATERIAL SHALL CONSIST OF IMPORTED FILL AS SPECIFIED HEREIN OF EITHER HIGH GRADE COMPACTION SAND OR APPROVED CRUSHED ROAD GRAVEL CONFORMING TO RMS QA SPECIFICATION 3051 OR SIMILAR.

OTHER THAN ROADWAY
TRENCH MATERIAL EXCAVATED SHALL CONSIST OF SELECT FILL AS SPECIFIED HEREIN AND SHALL NOT CONTAIN MORE THAN 20% OF STONES OF SIZE BETWEEN 25mm AND 75mm AND NONE LARGER THAN 75mm. PRIOR TO USE OF THE EXCAVATED MATERIAL IT SHALL BE INSPECTED AND APPROVED BY THE ENGINEER.

COMPACT BEDDING. EMBEDMENT AND TRENCH FILL MATERIALS AS FOLLOW:-

EMBEDMENT:-
FOR GRANULAR FILL MATERIAL (NON-COHESIVE SOIL) e.g. COARSE AGGREGATE FILL, THE DENSITY INDEX (ID) SHALL BE NOT LESS THAN 70%.

TRENCH FILL:-
FOR GRANULAR MATERIAL (NON COHESIVE SOILS), THE DENSITY INDEX (ID) SHALL BE NOT LESS THAN 70%. FOR NON-GRANULAR FILL MATERIAL (COHESIVE SOILS), THE DRY DENSITY RATIO (RD) SHALL BE NOT LESS THAN 95%.

UTILITY INFORMATION SHOWN ON THE PLANS IS NOT INTENDED TO DEPICT MORE THAN THE PRESENCE OF ANY SERVICES. ACTUAL LOCATIONS SHOULD BE VERIFIED BY HAND EXCAVATION PRIOR TO CONSTRUCTION.

THE CONTRACTOR SHALL ALLOW FOR THE CAPPING OFF, EXCAVATION AND REMOVAL (IF REQUIRED) OF ALL EXISTING SERVICES IN AREAS AFFECTED BY THE WORKS.

GEOTEXTILE FABRIC MATERIAL TO BE BIDIM A24 OR APPROVED EQUIVALENT AND SHALL COMPLY WITH AS3705-2012: 'GEOTEXTILES - IDENTIFICATION, MARKING AND GENERAL DATA'

THE CONTRACTOR SHALL ENSURE THAT SERVICES TO ALL BUILDINGS NOT AFFECTED BY THE WORKS ARE NOT DISRUPTED AT ALL TIMES. THE CONTRACTOR SHALL CONSTRUCT TEMPORARY SERVICES TO MAINTAIN EXISTING SUPPLY TO BUILDINGS REMAINING WHERE REQUIRED. ONCE THE WORKS ARE COMPLETE AND COMMISSIONED THE CONTRACTOR SHALL REMOVE ALL SUCH TEMPORARY SERVICES AND MAKE GOOD ALL DISTURBED AREAS.

STORMWATER NOTES (CONT):

EXISTING PIPES WHICH FORM NO PART OF THE DRAINAGE SYSTEM SHALL BE REMOVED OR SEALED AS INDICATED ON THE PLANS. PIPES UP TO 300mm DIAMETER SHALL BE SEWER GRADE uPVC WITH SOLVENT WELDED JOINTS (U.N.O.). ALL PIPE JUNCTIONS AND TAPERS SHALL BE VIA PURPOSE MADE FITTINGS.

WHERE DOWNPIPES PASS UNDER FLOOR SLABS, SEWER GRADE uPVC WITH RUBBER RING JOINTS ARE TO BE USED.

MINIMUM GRADE TO DRAINAGE PIPES TO BE 1% (U.N.O.), MIN. SIZE 100mm DIAMETER (U.N.O.).

PIPES LARGER THAN OR EQUAL TO 300mm DIAMETER TO BE REINFORCED CONCRETE RUBBER RING JOINTED TYPE (CLASS 2) MANUFACTURED TO AS4058 (U.N.O.).

PIPE INSTALLATION UNDER TRAFFICABLE AREAS SHALL BE IN ACCORDANCE WITH CONCRETE PIPE ASSOCIATION OF AUSTRALIA PUBLICATION "CONCRETE PIPE SELECTION & INSTALLATION" TYPE HS3 SUPPORT.

EQUIVALENT STRENGTH FRC PIPES MAY BE USED SUBJECT TO AUTHORITY APPROVAL.

MINIMUM PIPE COVER TO BE 600mm UNDER TRAFFICABLE AREAS AND 300mm ELSEWHERE (U.N.O.).

CONTRACTOR TO SUPPLY AND INSTALL ALL FITTINGS AND SPECIALS INCLUDING VARIOUS PIPE ADAPTORS TO ENSURE PROPER CONNECTION BETWEEN DISSIMILAR PIPEWORK.

PROVIDE CLEANING EYES TO ALL DOWNPIPES NOT DIRECTLY CONNECTED TO PITS.

STORMWATER DRAINAGE CONNECTIONS TO COUNCIL'S SYSTEM SHALL BE TO THE REQUIREMENTS AND THE SATISFACTION OF LOCAL COUNCIL.

PITS DEEPER THAN 1200mm TO BE FITTED WITH STEP IRONS AT 300 CENTRES TO AS1657-2013: 'FIXED PLATFORMS, WALKWAYS, STAIRWAYS AND LADDERS - DESIGN, CONSTRUCTION AND INSTALLATION'.

ALL EXPOSED EDGES TO BE ROUNDED WITH 20mm RADIUS, OR CHAMFERED 20mm x 20mm.

PIT REINFORCEMENT - MESH SL82 LAP TO BE 400mm MIN. CLEAR COVER 40 MIN. CAST AGAINST BLINDING OR FORMWORK. CORNER RETURNS MAY BE FABRIC OR EQUIVALENT BARS.

BENCHING TO BE HALF OUTGOING PIPE DEPTH. CONCRETE FOR BENCHING TO BE 20MPa MASS CONCRETE.

BRICKWORK, BLOCKWORK, CONCRETE OR APPROVED PRECAST PITS ARE TO BE USED IN TRAFFICABLE AREAS SUBJECT TO APPROVAL.

FIBREGLASS, HARD-PLASTIC OR APPROVED PRECAST PITS ARE TO BE USED IN NON-TRAFFICABLE AREAS SUBJECT TO APPROVAL.

100mm DIAMETER HOLE FOR SUBSOIL DRAINAGE OUTLET TO BE LOCATED 100mm ABOVE INVERT OF ALL INLET PIPES. SUBSOIL DRAINAGE TO EXTEND FOR A DISTANCE OF 3m UPSTREAM OF PIT (AT EACH INLET TRENCH) WITH THE UPSTREAM END SEALED.

ALL CONNECTIONS TO EXISTING DRAINAGE PITS SHALL BE MADE IN TRADESMAN-LIKE MANNER AND THE INTERNAL WALL OF THE PIT AT THE POINT OF ENTRY SHALL BE CEMENT RENDERED TO ENSURE A SMOOTH FINISH.

PIT GRATE, FRAMES AND SOLID COVERS SHALL BE CLASS B IN NON TRAFFIC AREAS AND CLASS C IN TRAFFICABLE AREAS IN ACCORDANCE WITH AS3996 U.N.O.

ALL GRATES SHALL BE PROVIDED WITH A 'J-LOCK' TYPE LOCKING CLIPS.

GRATES TO PITS IN FOOTPATH AREAS SHALL BE HEEL SAFE COMPLYING WITH THE DISABLED ACCESS CODE

PIT GRATING TO BE GALVANISED STEEL TYPE 'WELDLOK' OR APPROVED EQUIVALENT.

SUBSOIL PIPES SHALL BE LAID AT A MIN GRADE OF 1% (U.N.O.).

ADDITIONAL SUBSOIL DRAINAGE SHALL BE LAID TO SUIT SITE CONDITIONS AND GROUNDWATER PRESENCE AS DIRECTED. SUBSOIL PIPES SHALL BE LAID BEHIND KERBS IN CUT AREAS OF THE SITE.

PROVIDE A MINIMUM OF 150mm GRAVEL AROUND SUBSOIL PIPE. TRENCH TO BE LINED WITH GEOTEXTILE FABRIC TYPE BIDIM A24

SURVEY

THE EXISTING SITE CONDITIONS SHOWN ON THE FOLLOWING DRAWINGS HAVE BEEN INVESTIGATED BY REGISTERED SURVEYORS. THE INFORMATION IS SHOWN TO PROVIDE A BASIS FOR DESIGN.

NANO CONSULTING ENGINEERS DOES NOT GUARANTEE THE ACCURACY OR COMPLETENESS OF THE SURVEY BASE OR ITS SUITABILITY AS A BASIS FOR CONSTRUCTION OR DESIGN.

SHOULD DISCREPANCIES BE ENCOUNTERED DURING CONSTRUCTION BETWEEN THE SURVEY DATA AND ACTUAL FIELD DATA, CONTACT NANO CONSULTING ENGINEERS.

ABBREVIATIONS:

ø or DIA	DIAMETER
CBR	CALIFORNIA BEARING RATIO
CH	CHAINAGE
CL	CENTER LINE
CO	CLEAR OUT
DD	DISH DRAIN
DDO	DISH DRAIN OUTLET
DEJ	DOWELLED EXPANSION JOINT
DGB	DENSE GRADED BASECOURSE
DGS	DENSE GRADED SUB-BASE
DP	DOWNPIPE
e	EXISTING
FFL	FINISHED FLOOR LEVEL
GTD	GRADED TRENCH DRAIN
GSIP	GRADED SURFACE INLET PIT
HYD	HYDRANT
IJ	ISOLATING JOINT
IK	INTEGRAL KERB
IL	INVERT LEVEL
IP	INTERSECTION POINT
KIP	KERB INLET PIT
KO	KERB ONLY
K&G	KERB & GUTTER
KR	KERB RETURN
NGL	NATURAL GROUND LEVEL
OPF	OVERLAND FLOW PATH
OSD	ON-SITE DETENTION
R	RADIUS
RCP	REINFORCED CONCRETE PIPE
RK	ROLL KERB & GUTTER
RL	REDUCED LEVEL
RW	RETAINING WALL
RWT	RAINWATER TANK
SJ	SAWN CONTROL JOINT
SMH	SEWER MAN HOLE
SW	STORMWATER
SWP	STORMWATER PIT
SWRM	STORMWATER RISING MAIN
SWS	STORMWATER SUMP
SV	STOP VALVE
TOK	TOP OF KERB
TOW	TOP OF WALL
TWL	TOP WATER LEVEL
TP	TANGENT POINT
UPVC	UNPLASTICISED POLYVINYL CHLORIDE
UNO	UNLESS NOTED OTHERWISE
WPJ	WEAKENED PLANE JOINT
FF	FIRST FLUSH DEVICE
TYP	TYPICAL
BM	BENCH MARK



DIAL BEFORE YOU DIG SHOULD BE CONTACTED PRIOR TO ANY EXCAVATION ON SITE

TM: TRADE MARK OF THE ASSOCIATION OF DIAL BEFORE YOU DIG SERVICES LTD. USED UNDER LICENSE.

DRAWING REGISTER		
NUMBER	NAME	REVISION
SW001	COVER SHEET	01
SW010	GROUND FLOOR PLAN, NOTES & DETAILS	01
SW011	SITE CATCHMENT PLAN	01
SW020	FIRST FLOOR & ROOF PLAN, NOTES & DETAILS	01
ER001	EROSION AND SEDIMENT CONTROL PLAN	01

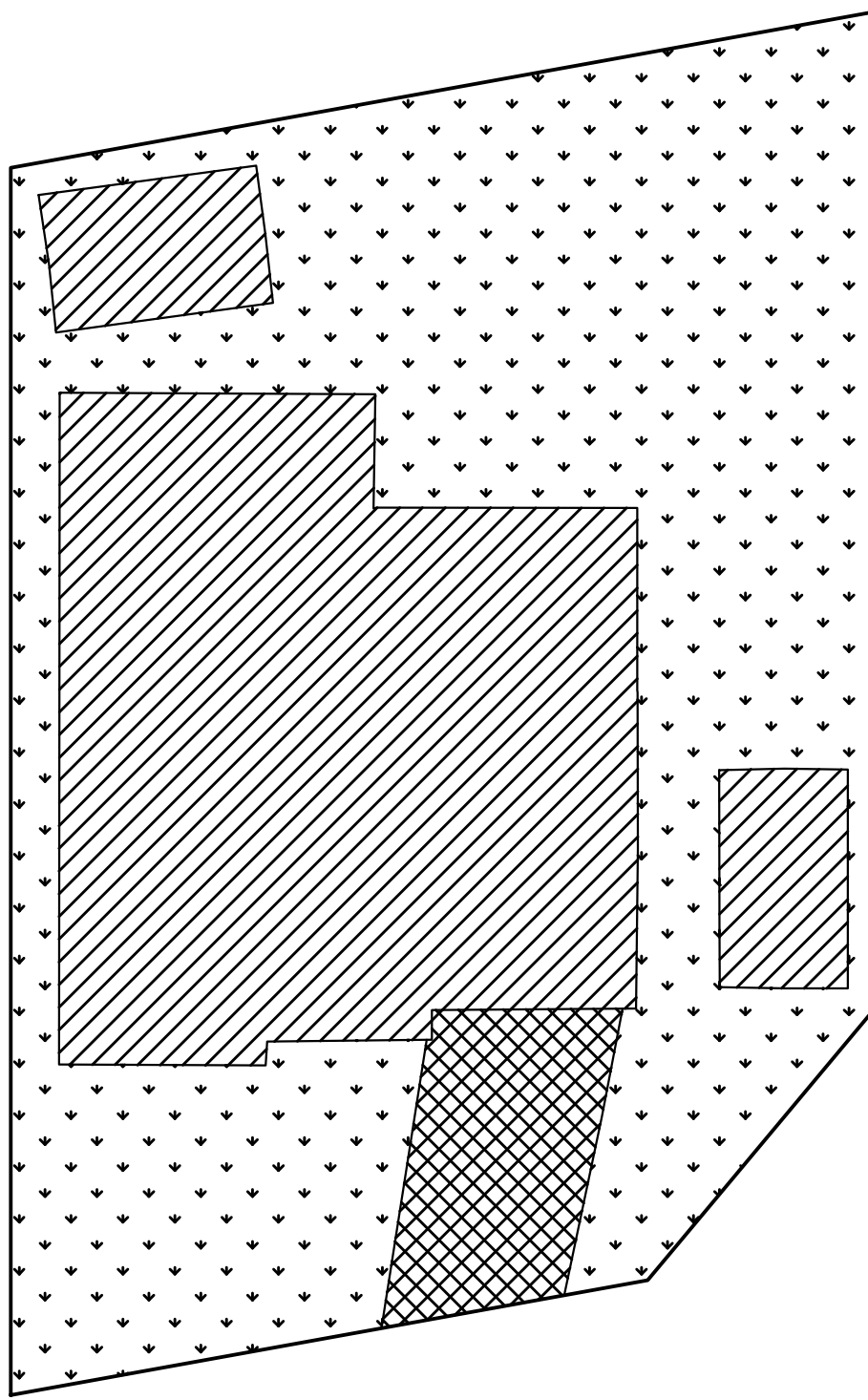
LEGEND:

	DP	DOWNPIPE		eSMH	EXISTING SEWER MANHOLE
	>>>	STORMWATER LINE			EXISTING JUNCTION PIT
	>>>	STORMWATER LINE DRAINING TO RWT			EXISTING KERB INLET PIT
	OF	OVER FLOW PIPE		eTEL	EXISTING TELSTRA PIT
	SSD	SUBSOIL LINE		eHYD	EXISTING HYDRANT
	SWRM	STORMWATER RISING MAIN		eSV	EXISTING STOP VALVE
	e	EXISTING STORMWATER LINE		eGAS	EXISTING GAS VALVE
	S	AUTHORITY SEWER LINE		ePP	EXISTING POWER POLE
	W	AUTHORITY WATER LINE			EXISTING GRATED SURFACE INLET PIT
	G	AUTHORITY GAS LINE		FF	FIRST FLUSH
	E	AUTHORITY ELECTRICITY LINE		RWO	RAINWATER OUTLET
	FO	AUTHORITY FIBRE OPTIC LINE		CO	CLEAR OUT POINT
	TEL	AUTHORITY COMMS LINE		DDO	DISH DRAIN OUTLET
	/	SEDIMENT FENCE		PD	PLANTER DRAIN
		GRATED SURFACE INLET PIT			CAPPING
		GRATED SURFACE INLET PIT WITH OCEANGUARD INSERT		RH	RAINHEAD
		SEALED JUNCTION PIT		SP	DOWNPIPE SPREADER
		PROPOSED KERB INLET PIT			WARNING LIGHT
		GRATED TRENCH DRAIN		144.37	SPOT LEVELS
	R/W TANK	RAINWATER RE-USE TANK			BENCHMARK
		PROPOSED RETAINING WALL		OPF	OVERLAND FLOW PATH

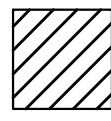
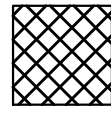
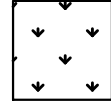
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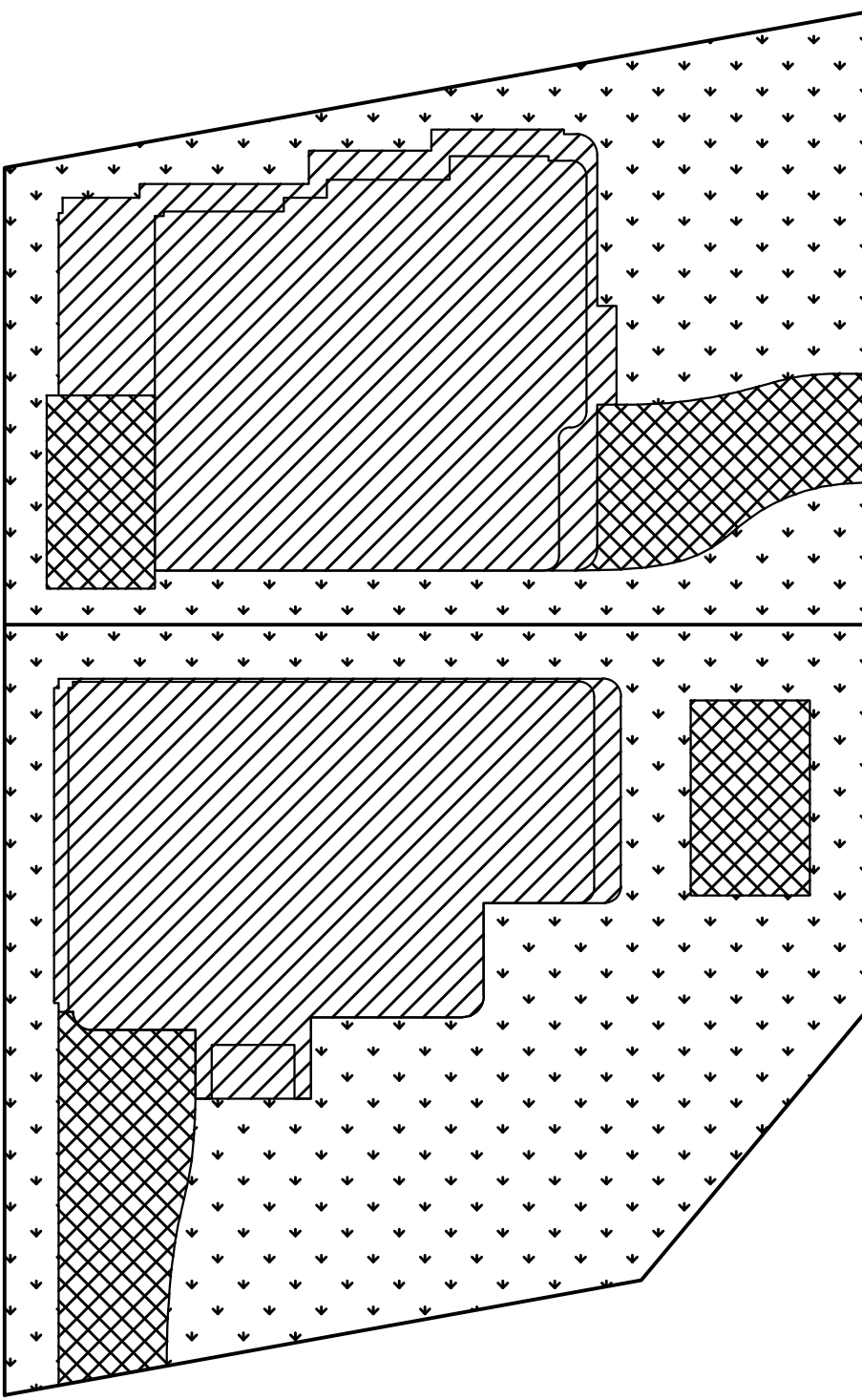
A1				E: ahmad@nceeng.com M: 0452489806 COPYRIGHT THIS DRAWING AND THE INFORMATION SHOWN HEREIN IS THE PROPERTY OF NANO CONSULTING ENGINEERS AND MAY NOT BE USED FOR ANY PURPOSES THAN FOR WHICH SUPPLIED.								DRAWN : A.N. DESIGNED : A.N. REVIEWED : A.N. DATE : 14/03/2026 CHECKED : A.N. APPROVED DATE : 14/03/2026								ARCHITECT : SLEIMAN PROPERTY GROUP				CLIENT : SLEIMAN PROPERTY GROUP				PROJECT : PROPOSED DETACHED DUAL OCCUPANCY ADDRESS : 42 DARTMOOR CIRCUIT, EMU HEIGHTS APPLICATION : CDC LGA : PENRITH CITY COUNCIL				SHEET SUBJECT : COVER SHEET								ALL DIMENSIONS IN mm UNLESS OTHERWISE STATED DO NOT SCALE DRAWING, USE FIGURED DIMENSIONS ONLY			
01 FOR CDC APPROVAL				A.N.	A.N.	14/03/2026		APPROVAL TYPE: 1. PRELIMINARY 2. DEVELOPMENT APPLICATION 3. CONCEPT DEVELOPMENT 4. CONTRACTOR/CONCEPT 5. DEVELOPMENT APPLICATION 6. OTHER				ADDRESS : 5 BURROWS AVENUE, CHESTER HILL PH: 0452489806 EMAIL: ahmad@nceeng.com																				SCALE @ A1 1:100				JOB No SW26076							
REV	DESCRIPTION			BY	CHECK	DATE																										AUTHORISED A.N.				DWG No SW001				REV 01			



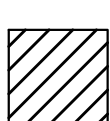
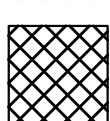
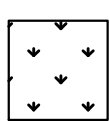
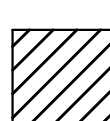
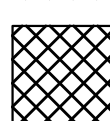
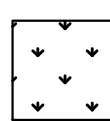


PRE-DEVELOPED CATCHMENT PLAN
SCALE 1: 200

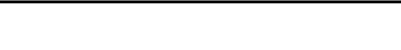
-  ROOF AREA = 307.73 Sq.m
-  HARDSTAND AREA = 42.77 Sq.m
-  PERVIOUS AREA = 439.82 Sq.m



POST-DEVELOPED CATCHMENT PLAN
SCALE 1: 200

-  UNIT 1 – ROOF AREA = 143.28 Sq.m
-  UNIT 1 – HARDSTAND AREA = 50.86 Sq.m
-  UNIT 1 – PERVIOUS AREA = 243.63 Sq.m
-  UNIT 2 – ROOF AREA = 157.21 Sq.m
-  UNIT 2 – HARDSTAND AREA = 44.53 Sq.m
-  UNIT 2 – PERVIOUS AREA = 150.81 Sq.m

A1 0 1 2 3 4 5 6 7 8 9 10

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01 FOR CDC APPROVAL					APPROVAL TYPE: 1. PRELIMINARY 4. CONTRACTOR/DEVELOPER 2. DEVELOPMENT APPLICATION 5. FOR TENDER 3. CONTRACT DEVELOPMENT 6. OTHER		ADDRESS : 5 BURROWS AVENUE, CHESTER HILL PH: 0452489806 EMAIL: ahmad@nceeng.com														SCALE @ A1 1:100		JOB No SW26076			
REV	DESCRIPTION				BY	CHECK	DATE													AUTHORISED A.N.		DWG No SW012		REV 01		

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FIRST FLOOR & ROOF NOTES:

INSTALL 50mm uPVC SPITTER PIPES 20mm ABOVE SURFACE LEVEL FOR BALCONY AND CONCRETE ROOF AREAS TO ALLOW FOR EMERGENCY OVERFLOW INCASE OF BLOCKAGES DURING HEAVY STORMS. PLUMBER TO CONFIRM LOCATION DURING CONSTRUCTION.

ALL BUILDING AND HYDRAULIC SERVICES TO BE PROPERLY CO-ORDINATED WITH STORMWATER PIPES AND ENSURE NO CLASHES ARE PRESENT DURING CONSTRUCTION (TYP).

STORMWATER PIPE ARRANGEMENT TO BE CO-ORDINTED WITH STRUCTURAL SLAB AND BEAMS WHERE REQUIRED (TYP).

BALCONY, TERRACE & CONCRETE ROOF AREAS TO SLOPE TOWARDS RAINWATER OUTLETS WHERE REQUIRED (TYP).

ARROW DENOTES THE SLOPE OF FINISHED SURFACE LEVEL (TYP).

DOWNPipes SHOWN ON PLAN ARE TO BE ø100mm uPVC U.N.O. (TYP).

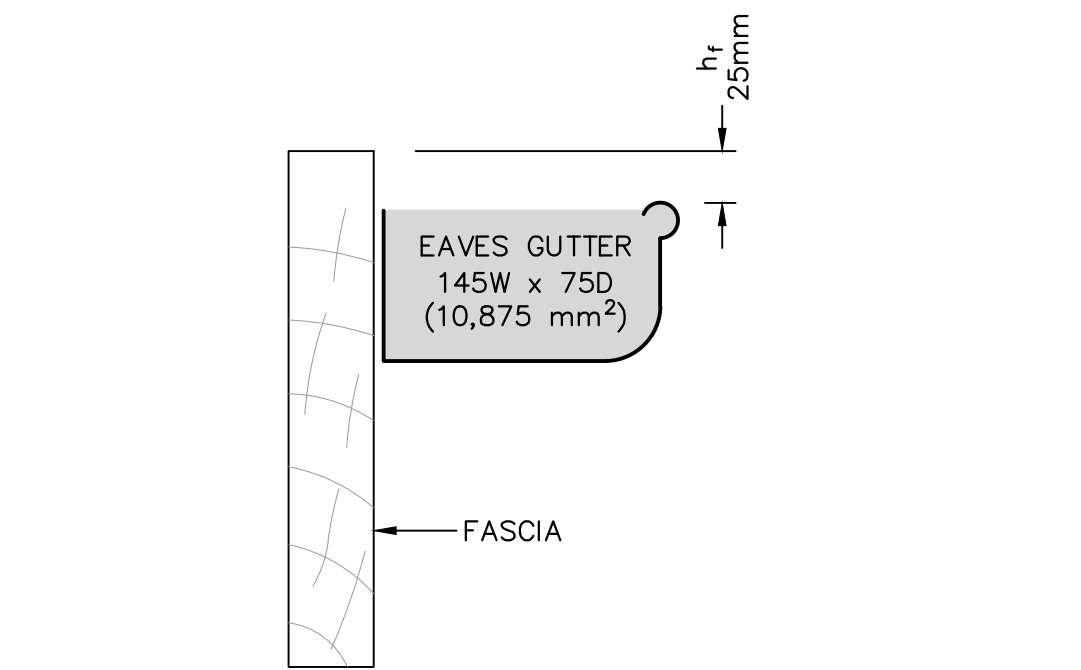
ALL EAVES GUTTERS SHALL BE 145mm WIDE x 75mm DEEP (OR EQUIVALENT) AND LAID AT MIN. 1:500 SLOPE.

ALL GUTTERS TO BE FITTED WITH ADEQUATE OVERFLOW MEASURES IN ACCORDANCE WITH AS3500.3:2018.

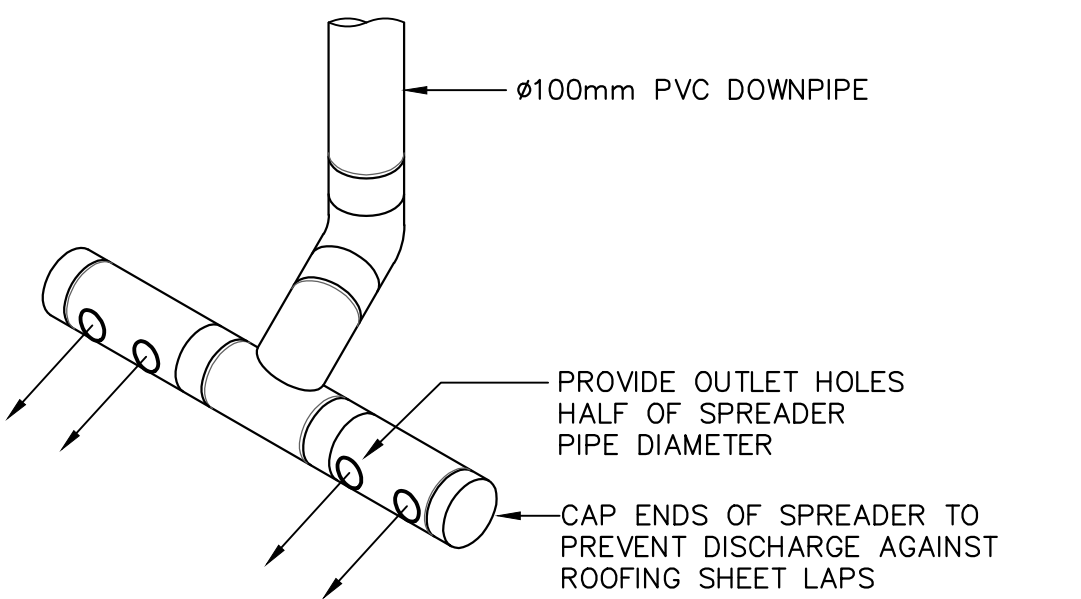
PROPOSED DOWNPIPE LOCATIONS ARE NOMINAL AND TO BE CONFIRMED DURING CONSTRUCTION (TYP).

INSTALL DOWNPIPE WITH SPREADER (IF REQUIRED) TO DISPERSE STORMWATER ONTO LOWER ROOF AREAS EFFECTIVELY.

PROVIDE SURFACE DRAINAGE FOR ALL CONCRETE AND BALCONY ROOF AREAS WHERE REQUIRED.

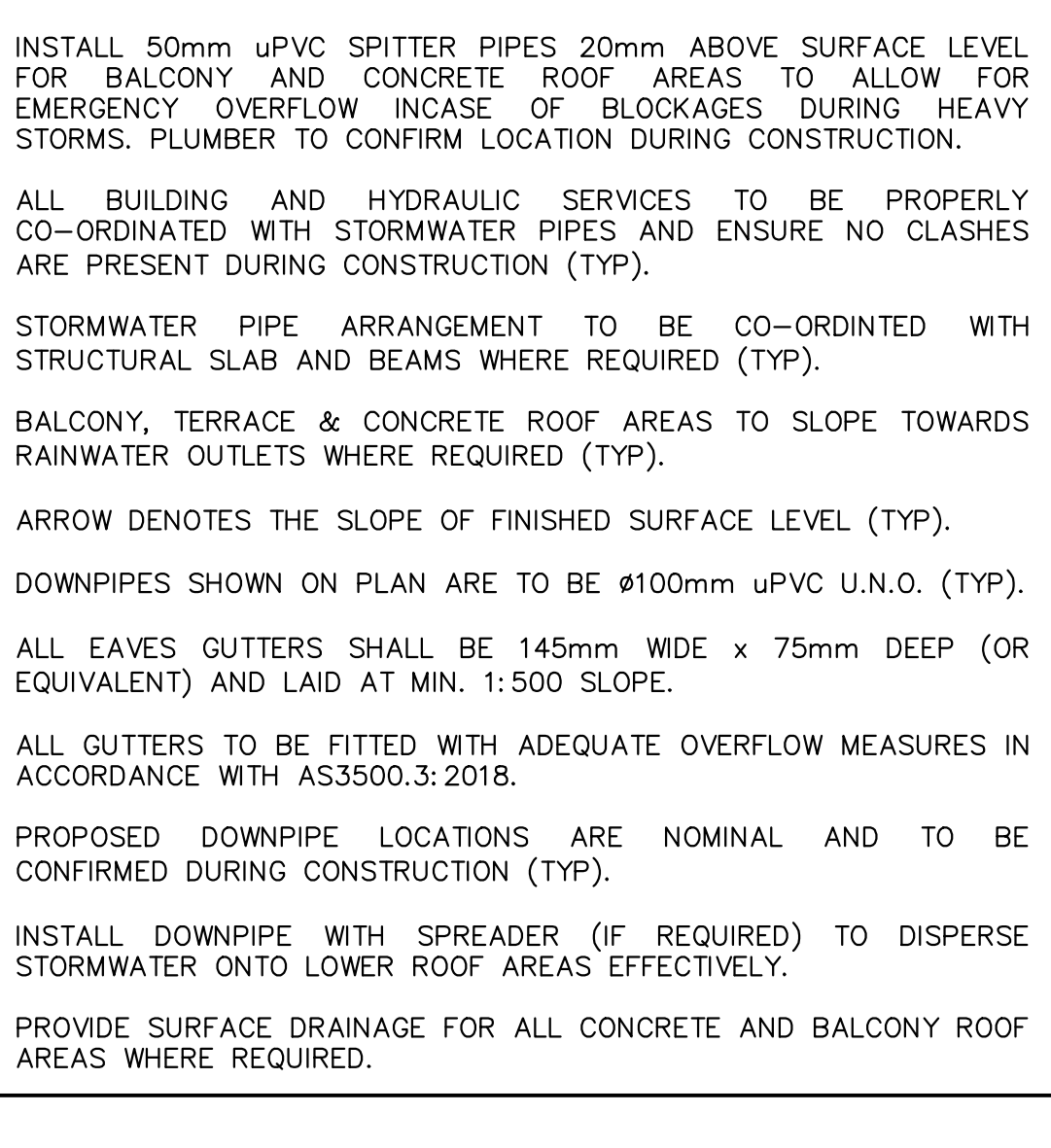


TYPICAL EAVES GUTTER
DETAIL (LOW FRONT)
NOT TO SCALE

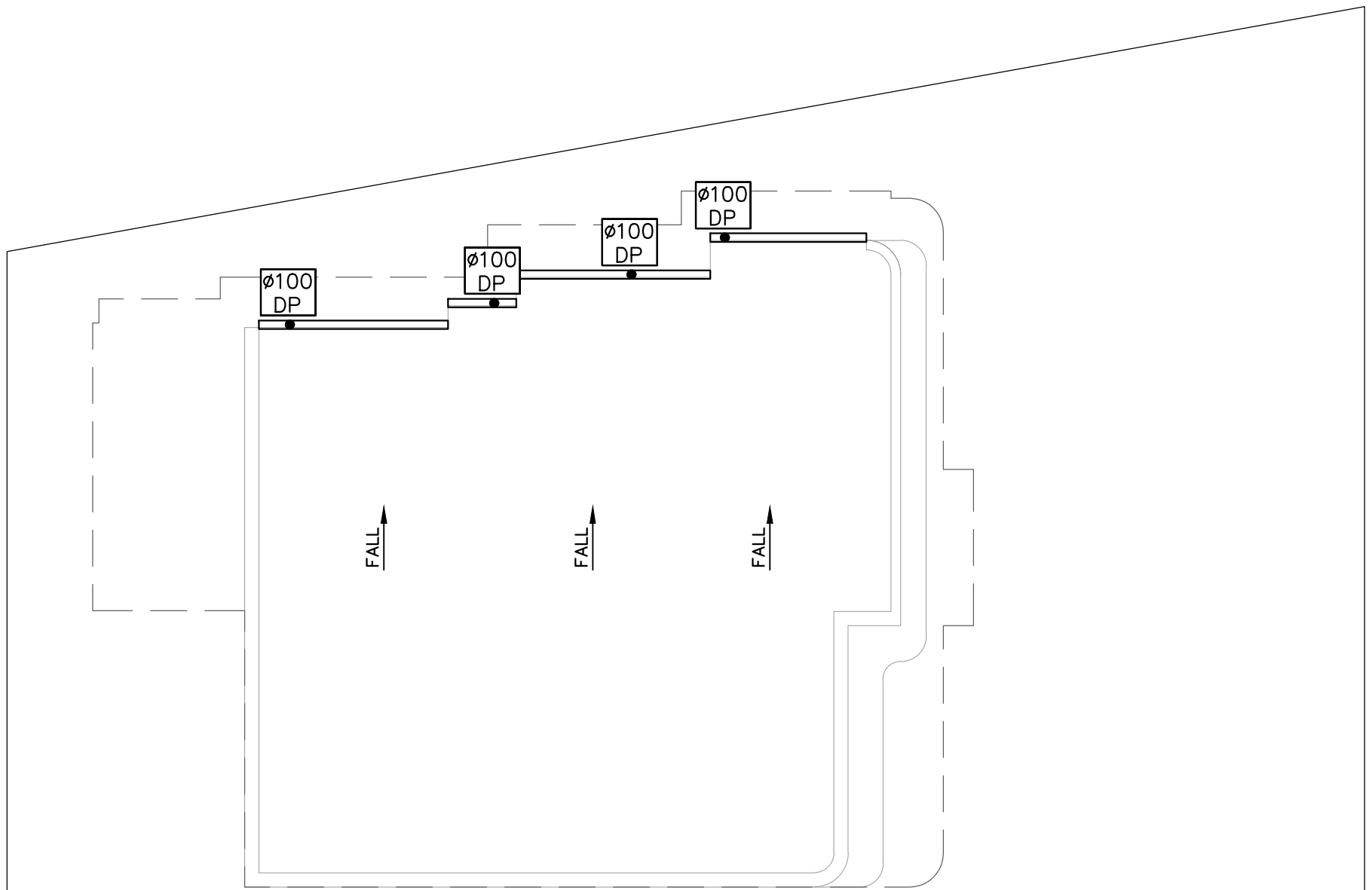


SPREADER PIPE DETAIL
SCALE: 1:10

- NOTES: –
- OUTLET HOLES POSITIONS SHALL BE POSITIONED TO AVOID JOINTS IN ROOFING
 - WHEN DOWNPIPE IS LOCATED IN CORNER, SPREADER PIPE TO BE L-SHAPED OR SIMILAR



STORMWATER LAYOUT PLAN
FIRST FLOOR LEVEL
SCALE 1:100



STORMWATER LAYOUT PLAN
ROOF LEVEL
SCALE 1:100

A1 1:100

REV	DESCRIPTION	BY	CHECK	DATE
01	FOR CDC APPROVAL	A.N.	A.N.	14/03/2026

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APPROVAL TYPE:
1. PRELIMINARY
2. DEVELOPMENT APPLICATION
3. CONSTRUCTION PERMIT
4. CONSTRUCTION PERMIT
5. FOR TENDER
6. OTHER

NANO
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ADDRESS : 5 BURROWS AVENUE, CHESTER HILL
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EMAIL: ahmad@nceeng.com

DRAWN : A.N.	DESIGNED : A.N.
REVIEWED : A.N.	DATE : 14/03/2026
CHECKED : A.N.	
APPROVED DATE : 14/03/2026	

SLEIMAN
PROPERTY GROUP

ARCHITECT:
SLEIMAN PROPERTY GROUP

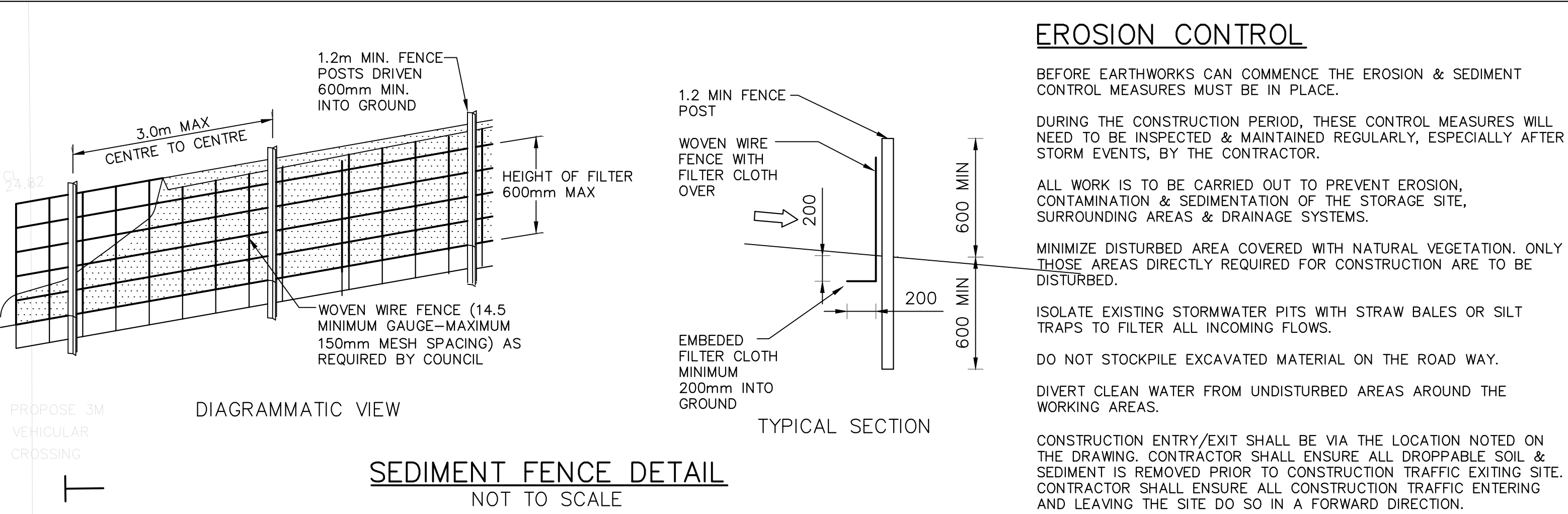
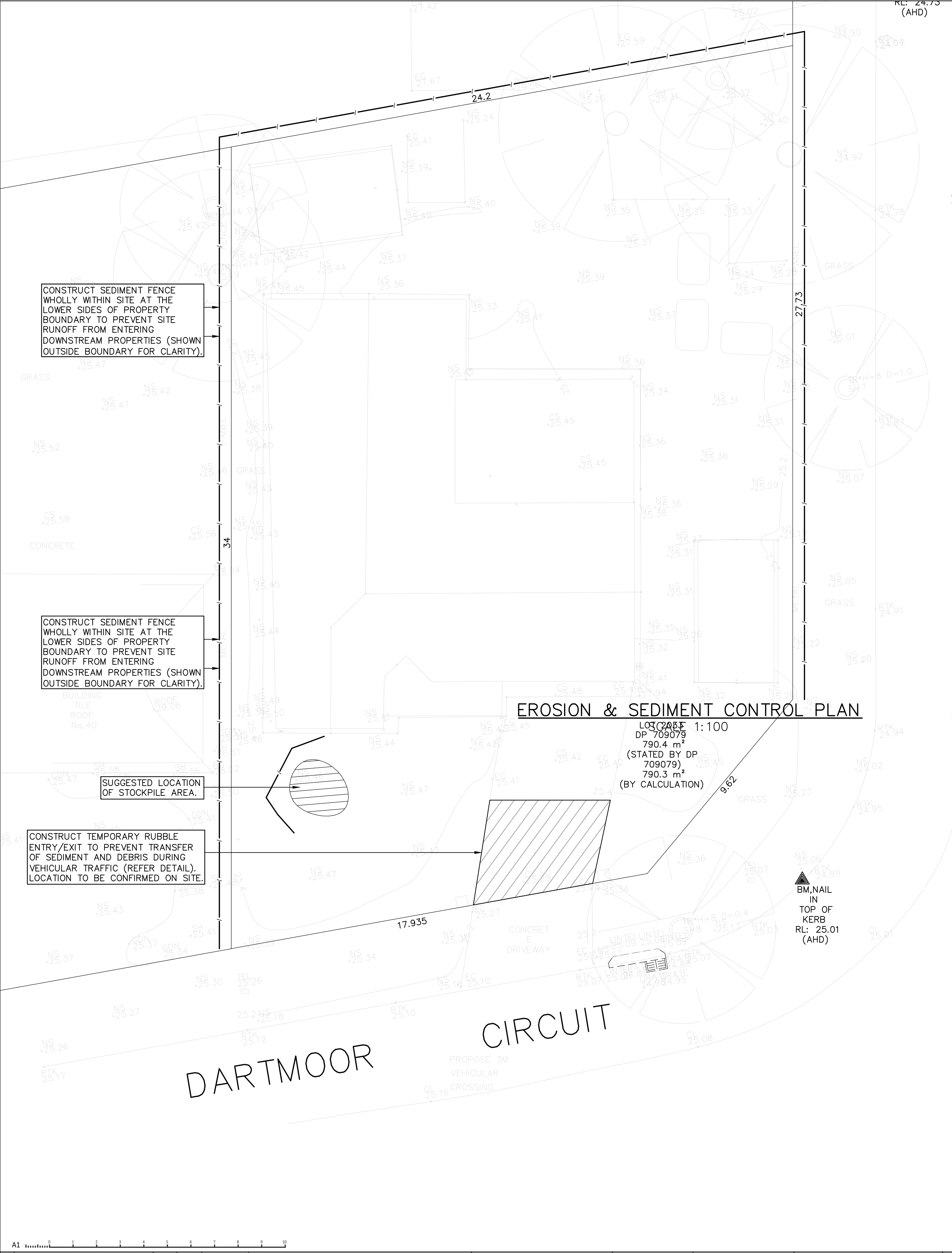
CLIENT:
SLEIMAN PROPERTY GROUP

PROJECT:
PROPOSED DETACHED DUAL OCCUPANCY
ADDRESS:
42 DARTMOOR CIRCUIT, EMU HEIGHTS
APPLICATION:
CDC
LGA:
PENRITH CITY COUNCIL

SHEET SUBJECT:
FIRST FLOOR & ROOF
PLAN, NOTES & DETAILS

ALL DIMENSIONS IN mm UNLESS OTHERWISE STATED DO NOT SCALE DRAWING, USE FIGURED DIMENSIONS ONLY		
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EROSION CONTROL

BEFORE EARTHWORKS CAN COMMENCE THE EROSION & SEDIMENT CONTROL MEASURES MUST BE IN PLACE.

DURING THE CONSTRUCTION PERIOD, THESE CONTROL MEASURES WILL NEED TO BE INSPECTED & MAINTAINED REGULARLY, ESPECIALLY AFTER STORM EVENTS, BY THE CONTRACTOR.

ALL WORK IS TO BE CARRIED OUT TO PREVENT EROSION, CONTAMINATION & SEDIMENTATION OF THE STORAGE SITE, SURROUNDING AREAS & DRAINAGE SYSTEMS.

MINIMIZE DISTURBED AREA COVERED WITH NATURAL VEGETATION. ONLY THOSE AREAS DIRECTLY REQUIRED FOR CONSTRUCTION ARE TO BE DISTURBED.

ISOLATE EXISTING STORMWATER PITS WITH STRAW BALES OR SILT TRAPS TO FILTER ALL INCOMING FLOWS.

DO NOT STOCKPILE EXCAVATED MATERIAL ON THE ROAD WAY.

DIVERT CLEAN WATER FROM UNDISTURBED AREAS AROUND THE WORKING AREAS.

CONSTRUCTION ENTRY/EXIT SHALL BE VIA THE LOCATION NOTED ON THE DRAWING. CONTRACTOR SHALL ENSURE ALL DROPPABLE SOIL & SEDIMENT IS REMOVED PRIOR TO CONSTRUCTION TRAFFIC EXITING SITE. CONTRACTOR SHALL ENSURE ALL CONSTRUCTION TRAFFIC ENTERING AND LEAVING THE SITE DO SO IN A FORWARD DIRECTION.

ADOPT TEMPORARY MEASURES AS MAY BE NECESSARY FOR EROSION & SEDIMENT CONTROL, INCLUDING BUT NOT LIMITED TO THE FOLLOWING:

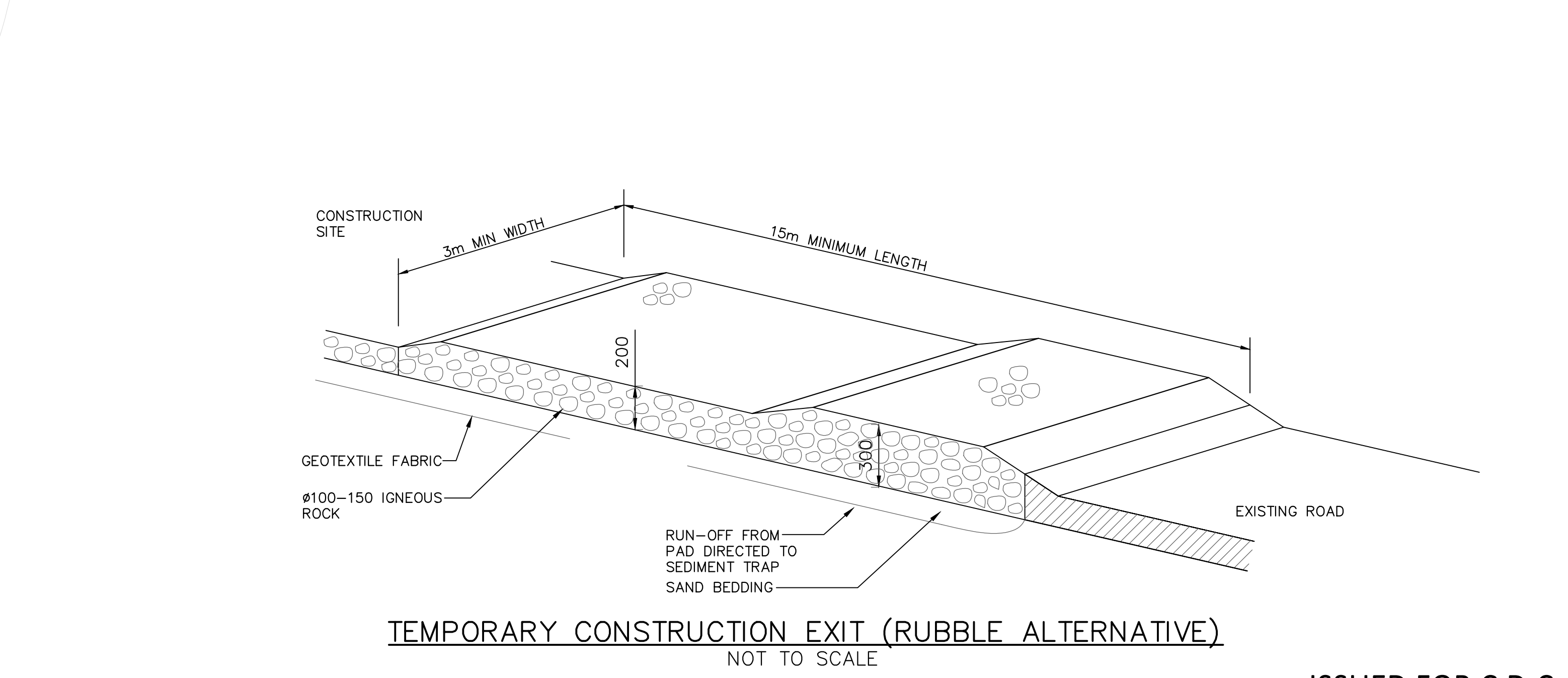
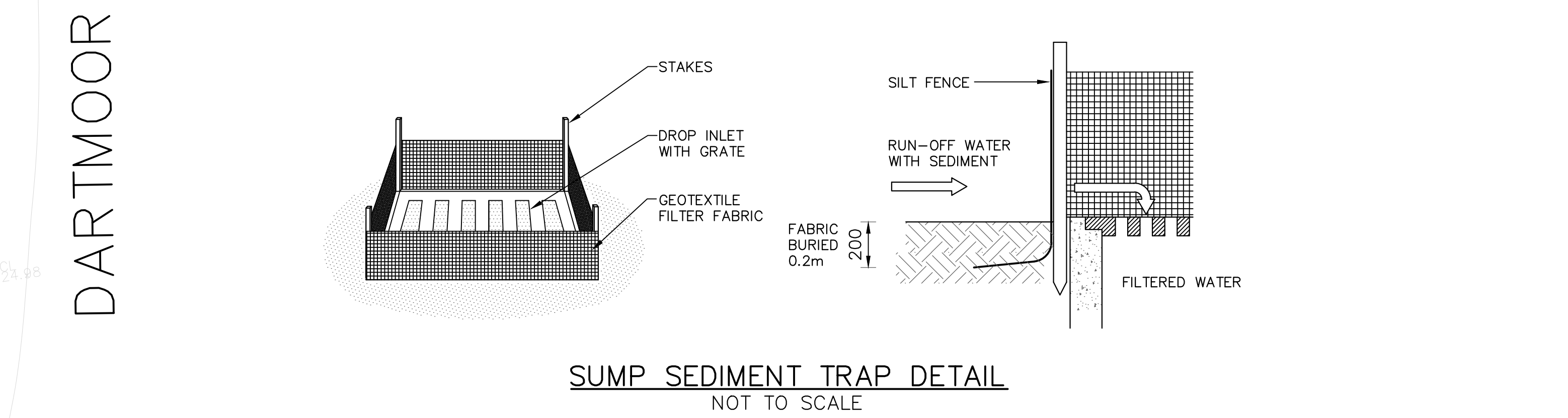
- DRAINS: TEMPORARY DRAINS AND CATCH DRAINS.
- SPREADER BANKS OR OTHER STRUCTURES: TO DISPERSE CONCENTRATED RUNOFF.
- SILT TRAPS: CONSTRUCTION AND MAINTENANCE OF SILT TRAPS TO PREVENT DISCHARGE OF SCOURED MATERIAL TO DOWNSTREAM AREAS.

AFTER RAIN, INSPECT, CLEAN, AND REPAIR IF REQUIRED, TEMPORARY EROSION & SEDIMENT CONTROL MEASURES.

REMOVE TEMPORARY EROSION & SEDIMENT CONTROL MEASURES WHEN THEY ARE NO LONGER REQUIRED.

COMPLY WITH THE REQUIREMENTS OF LANDCOM'S MANAGING URBAN STORMWATER - SOIL AND CONSTRUCTION 'THE BLUE BOOK' LATEST EDITION

THE EROSION & SEDIMENT CONTROL PLAN PROVIDED IS ONLY INDICATIVE. THE CONTRACTOR SHOULD PREPARE A DETAILED ESCP SUITABLE FOR THE SPECIFIC SITE CONDITIONS



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01 FOR CDC APPROVAL				A.N. A.N. 14/03/2026				APPROVAL TYPE: 1. PRELIMINARY 4. CONSTRUCTION CERTIFICATE 2. DEVELOPMENT APPLICATION 5. FOR TENDER 3. CONSTRUCTION DEVELOPMENT 6. OTHER				ADDRESS : 5 BURROWS AVENUE, CHESTER HILL PH: 0452489806 EMAIL: ahmad@nceeng.com																			
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